

Final sweep roll-up

Sweep status roll-up:

[OK] identifiability (57m) →
/home/ff235/dev/MLResearch/automl/automl_package/examples/probreg_identifiability_results
[OK] classreg_k (1m) →
/home/ff235/dev/MLResearch/automl/automl_package/examples/classreg_k_sweep_results
[OK] full_benchmark (63m) →
/home/ff235/dev/MLResearch/automl/automl_package/examples/full_benchmark_results
[OK] probreg_ablation (10m) →
/home/ff235/dev/MLResearch/automl/automl_package/examples/probreg_ablation_results
[OK] flex_nn_ablation (21m) →
/home/ff235/dev/MLResearch/automl/automl_package/examples/flex_nn_ablation_results
[OK] ablation_study (22m) →
/home/ff235/dev/MLResearch/automl/automl_package/examples/ablation_study_results
[OK] hpo (1m) →
/home/ff235/dev/MLResearch/automl/automl_package/examples/hpo_sweep_results

Each per-sweep page below links back to its own CSV/PDF artefacts.

Missing sweeps render a status-only page (no data to plot).

ProbReg identifiability (8-cell matrix)

Tests three orthogonal fixes (MDN NLL, CE_STOP_GRAD, anchored heads) across 4 toy datasets $\times k \in \{3, 5\} \times 3$ seeds. Cells A-H labelled in the sweep docstring. Headline = best cell per dataset by mean MSE.

Results dir: /home/ff235/dev/MLResearch/automl/automl_package/examples/probreg_identifiability_results

Per-sweep PDFs:

-
/home/ff235/dev/MLResearch/automl/automl_package/examples/probreg_identifiability_results/results_bimodal.pdf
-
/home/ff235/dev/MLResearch/automl/automl_package/examples/probreg_identifiability_results/results_exponential.p
-
/home/ff235/dev/MLResearch/automl/automl_package/examples/probreg_identifiability_results/results_heteroscedas
-
/home/ff235/dev/MLResearch/automl/automl_package/examples/probreg_identifiability_results/results_piecewise.pdf

Key findings:

- best cell on bimodal: F (MSE=2.305)
- best cell on exponential: A (MSE=0.503)
- best cell on heteroscedastic: C (MSE=1.775)
- best cell on piecewise: D (MSE=0.284)

Status: OK (57m)

ProbReg identifiability (8-cell matrix) — headline

dataset	cell	mse_mean	nll_own_mean
bimodal	F	2.3053	--
exponential	A	0.5025	1.0277
heteroscedastic	C	1.7752	1.6513
piecewise	D	0.2836	0.6308

ClassifierRegression k-sweep

$k \in \{2, 3, 5, 7, 10\}$ across toy datasets. Headline = argmin MSE per dataset.

Results dir: /home/ff235/dev/MLResearch/automl/automl_package/examples/classreg_k_sweep_results

Per-sweep PDFs:

- /home/ff235/dev/MLResearch/automl/automl_package/examples/classreg_k_sweep_results/k_sweep.pdf

Key findings:

- best k on bimodal: k=5 (MSE=2.953)
- best k on exponential: k=7 (MSE=2.800)
- best k on heteroscedastic: k=2 (MSE=1.826)
- best k on piecewise: k=7 (MSE=0.296)

Status: OK (1m)

ClassifierRegression k-sweep — headline

dataset	k	mse
bimodal	5	2.9529
exponential	7	2.7999
heteroscedastic	2	1.8262
piecewise	7	0.2957

Full benchmark (all models × all toy+UCI)

15 models on 4 toy + 4 UCI datasets. Metrics: MSE, NLL, CRPS, ECE, PICP@95, MPIW@95, sharpness. Human-readable report in REPORT.md alongside the PDF.

Results dir: /home/ff235/dev/MLResearch/automl/automl_package/examples/full_benchmark_results

Key findings:

- see /home/ff235/dev/MLResearch/automl/automl_package/examples/full_benchmark_results/REPORT.md

Status: OK (63m)

Full benchmark (all models x all toy tasks) headline

artifact	path
REPORT.md	/home/ff235/dev/MLResearch/automl/automl_package/examples/full_benchmark_results/REPORT.md

ProbReg ablation (Paper A primary table)

Cross-product over regression_strategy × loss_type × target_transform × opt_strategy × dynamic-k selection × n_classes_regularization. Headline = top-3 configs per dataset by MSE.

Results dir: /home/ff235/dev/MLResearch/automl/automl_package/examples/probreg_ablation_results

Key findings:

- b1_gravitational: best = single_head_final_output/soft_gating/nll (MSE=0.037)
- bimodal: best = separate_heads/soft_gating/nll (MSE=2.231)
- heteroscedastic: best = single_head_final_output/soft_gating/nll (MSE=1.612)

Status: OK (10m)

ProbReg ablation (Paper A primary table) — headline

	dataset	strategy	selection	regularization	loss	beta	target_transform	opt_strategy	mse	nll	crps	nll_mixture	mean_k	seconds	error
single_head	final_output	soft_gating		elbo	nll	0.5000	regression	regression_only	0.0374	-0.8772	0.0774	--	2.2978	85.1028	--
single_head	final_output		none	none	nll	0.5000	regression	regression_only	0.0656	-0.4768	0.1164	--	--	36.9436	--
	b1_separate_heads		none	none	nll	0.5000	regression	regression_only	0.1047	0.1031	0.1726	0.2616	--	48.7697	--
single_head	final_output	soft_gating		elbo	nll	0.5000	regression	regression_only	1.6121	1.3545	0.6335	--	2.0000	29.9464	--
single_head	final_output		ste	none	nll	0.5000	regression	regression_only	1.5574	1.3744	0.6281	--	--	17.2399	--
single_head	final_output	soft_gating		none	nll	0.5000	regression	regression_only	1.6250	1.3875	0.6331	--	5.3137	15.1313	--
	b1_separate_heads	soft_gating		none	nll	0.5000	regression	regression_only	2.2314	1.8307	0.8901	--	4.3058	18.0522	--
single_head	final_output	soft_gating		elbo	nll	0.5000	regression	regression_only	2.3187	1.8395	0.9164	--	2.0000	43.7820	--
single_head	final_output	soft_gating		none	nll	0.5000	regression	regression_only	2.3227	1.8413	0.9209	--	--	15.3680	--

FlexibleNN ablation (Paper B primary table)

max_hidden_layers × depth_regularization × layer_selection × weights × inference × uncertainty. Headline = top-3 per dataset.

Results dir: /home/ff235/dev/MLResearch/automl/automl_package/examples/flex_nn_ablation_results

Status: OK (21m)

FlexibleNN ablation (Paper B primary table) — headline

dataset	weights	max_layers	uq	layer_method	depth_reg	mse	mean_depth	seconds	error
b3_two_phase	shared	5	constant	soft_gating	cost_aware_elbo	0.0823	1.7718	127.3997	nan
b3_two_phase	shared	5	constant	gumbel_softmax	depth_penalty	0.0848	1.4756	103.8324	nan
b3_two_phase	shared	5	constant	soft_gating	elbo	0.0850	2.0893	141.3193	nan
piecewise	shared	5	constant	reinforce	none	0.2603	3.9833	6.4632	nan
piecewise	shared	5	constant	soft_gating	elbo	0.2610	2.0057	5.4658	nan
piecewise	shared	5	constant	reinforce	depth_penalty	0.2625	3.5750	4.7796	nan

Ablation study (smaller probe)

Compact ablation over FlexNN and ProbReg configurations on heteroscedastic + exponential.

Results dir: /home/ff235/dev/MLResearch/automl/automl_package/examples/ablation_study_results

Status: OK (22m)

HPO sweep (Optuna, tuned UCI baselines)

N=50 Optuna trials per (model, dataset). Default dataset: uci-yacht. ALL_DATASETS=1 adds uci-california (3k subsample).

Results dir: /home/ff235/dev/MLResearch/automl/automl_package/examples/hpo_sweep_results

Key findings:

- HPO winner on uci-yacht: XGB+HPO (MSE=0.323)

Status: OK (1m)

HPO sweep (Optuna, tuned UCI baselines) — headline

dataset	model	mse	nll
uci-yacht	XGB+HPO	0.3227	1.8958
uci-yacht	LGBM+HPO	8.7965	2.5429
uci-yacht	ProbReg+HPO	206.0500	2.5055
uci-yacht	NN+HPO	87.8980	2.1801

bimodal — metrics summary (mean ± std across 3 seeds)

k	cell	mse_mean	mse_std	nll_gaussian_mean	nll_mdn_mean	max_p_mid_mean	anchor_error_mean
3	A	2.372211	0.03954392533626387	1.853426333333332	nan	nan	3.672133333333335
3	B	2.3966766666666666	0.0792268646377812	1.85843	nan	nan	0.0302
3	C	2.5903806666666664	0.2511624399832372	1.9235389999999999	nan	nan	0.741733333333334
3	ClassReg	3.3498486666666665	0.08319120569106642	2.029598	nan	nan	nan
3	D	2.589456333333333	0.2116922536852338	1.9214149999999999	nan	nan	0.10976666666666667
3	E	2.625681333333333	0.09815341866859938	1.9855859999999999	nan	nan	5.3599
3	F	2.3053386666666666	0.0394430017282322	1.851343	nan	nan	1.8006666666666666
3	G	2.727553333333333	0.24955274903381316	2.0033523333333334	nan	nan	0.5288666666666667
3	H	2.5990103333333336	0.19341639554425932	1.9411066666666665	nan	nan	0.07136666666666666
5	A	2.329315	0.050133762645546816	1.8440906666666665	nan	0.3698666666666666	3.8042
5	B	2.3791663333333335	0.05525763078827501	1.8562156666666667	nan	0.2889	0.22260000000000002
5	C	2.5424913333333334	0.08715552169732763	1.9107139999999998	nan	0.7461000000000001	0.7197
5	ClassReg	3.3931186666666666	0.09228135069088078	2.0324596666666666	nan	nan	nan
5	D	2.677008333333333	0.047585087131719536	1.925709333333333	nan	0.7545333333333334	0.3157333333333337
5	E	2.721929	0.05214956403461085	2.0808999999999997	nan	0.167	4.591533333333335
5	F	2.3822423333333336	0.1476211655093311	1.8658236666666668	nan	0.2097333333333333	1.3774
5	G	2.712297	0.12296789588750394	2.0072313333333334	nan	0.7814333333333333	0.3554
5	H	2.788091	0.14152603764678795	2.0463146666666667	nan	0.7443	0.3478333333333334

bimodal k=3 — $h_i(p_i)$ vs p_i (seed 42)

ClassReg k=3

Cell A k=3

Cell B k=3

Cell C k=3

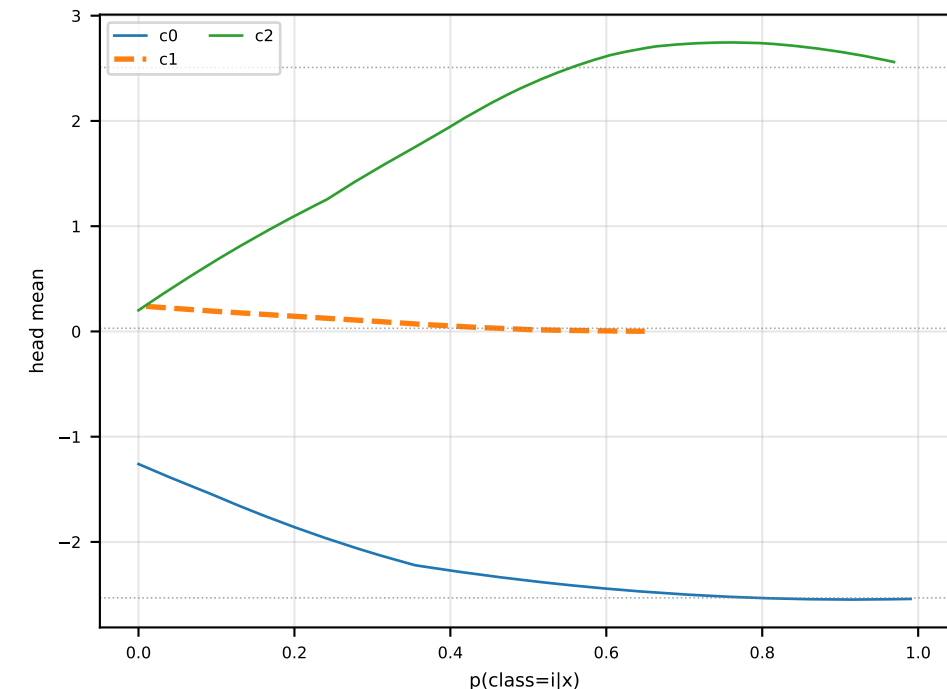
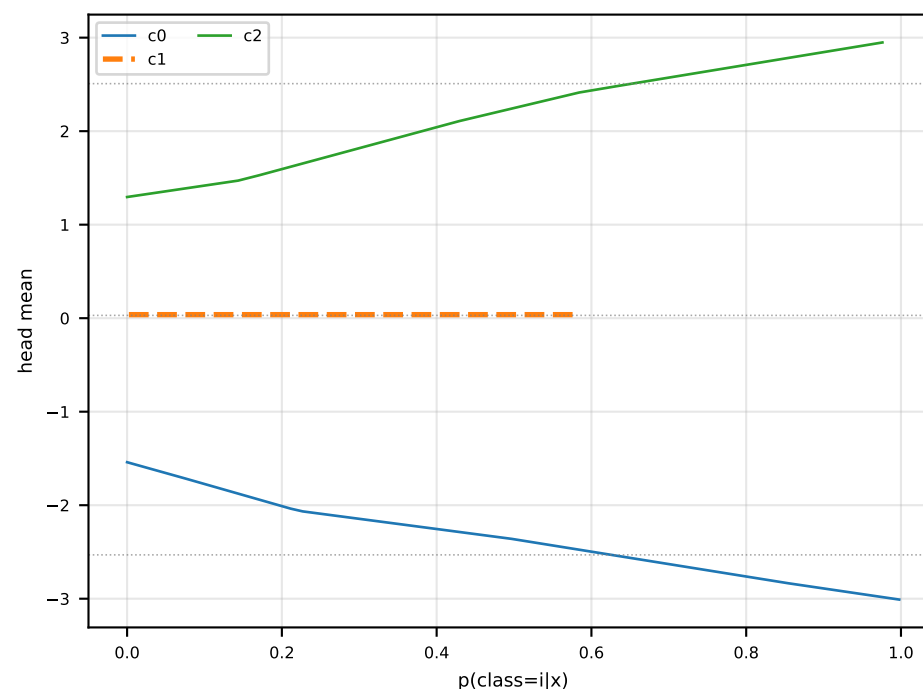
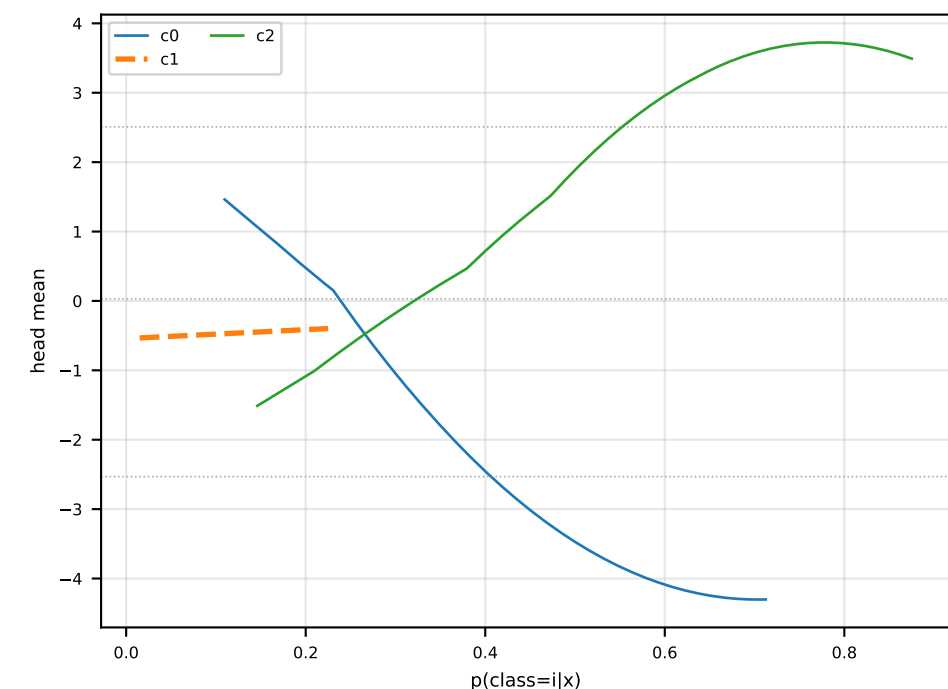
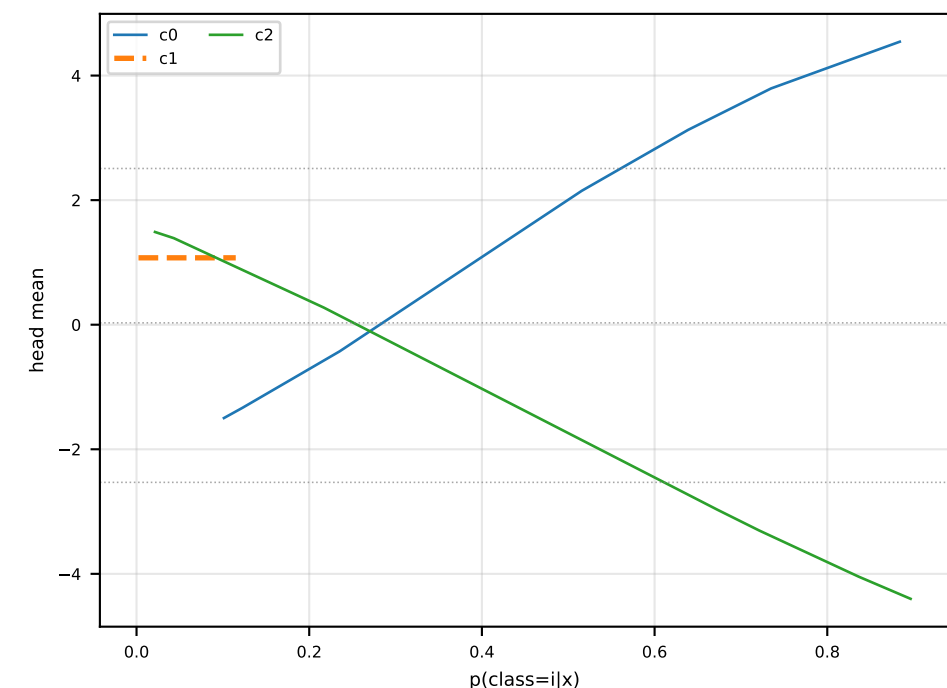
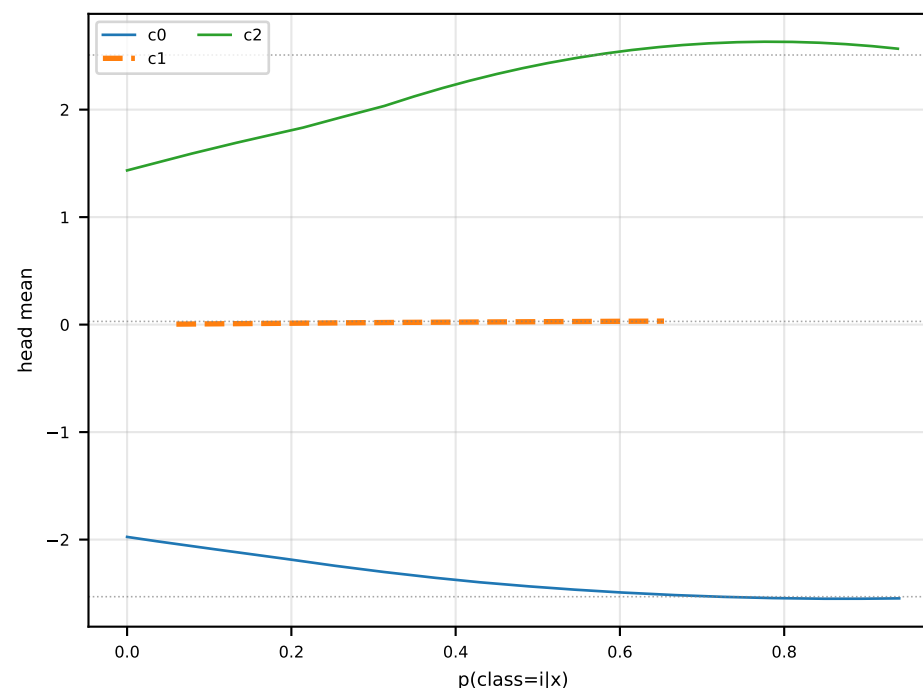
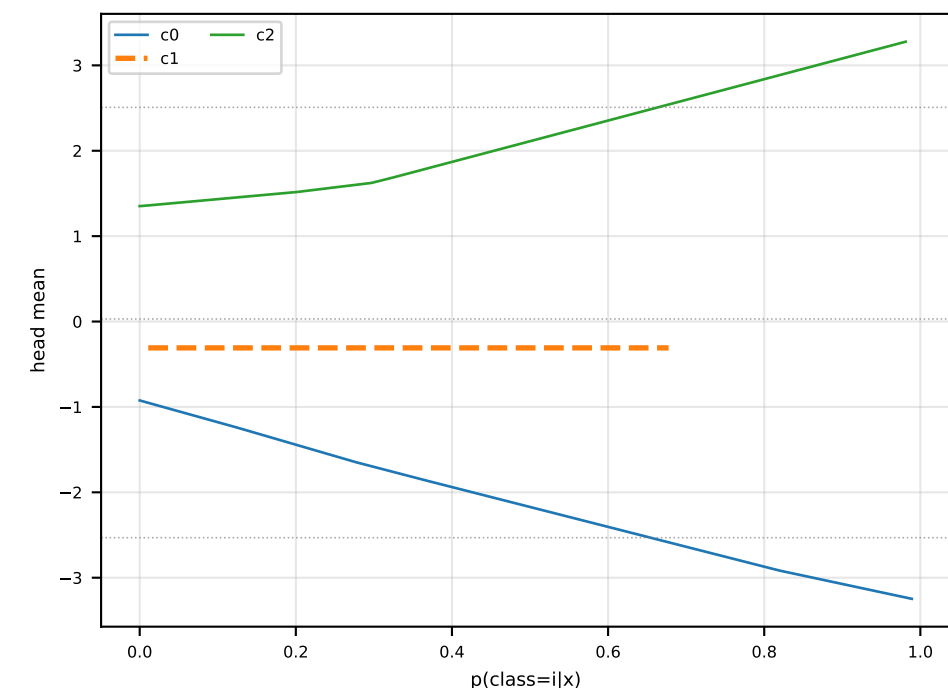
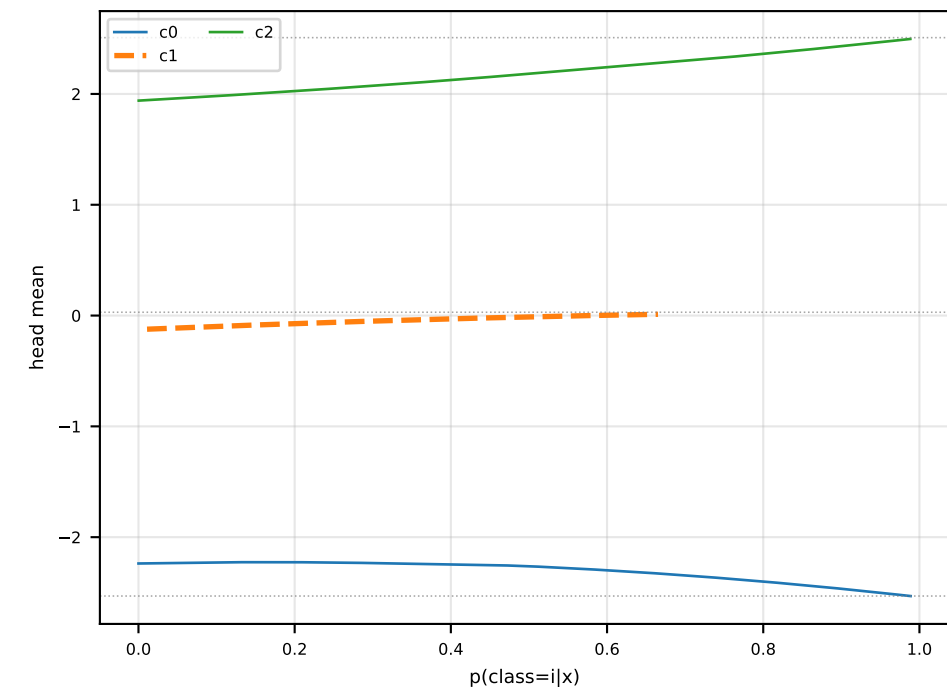
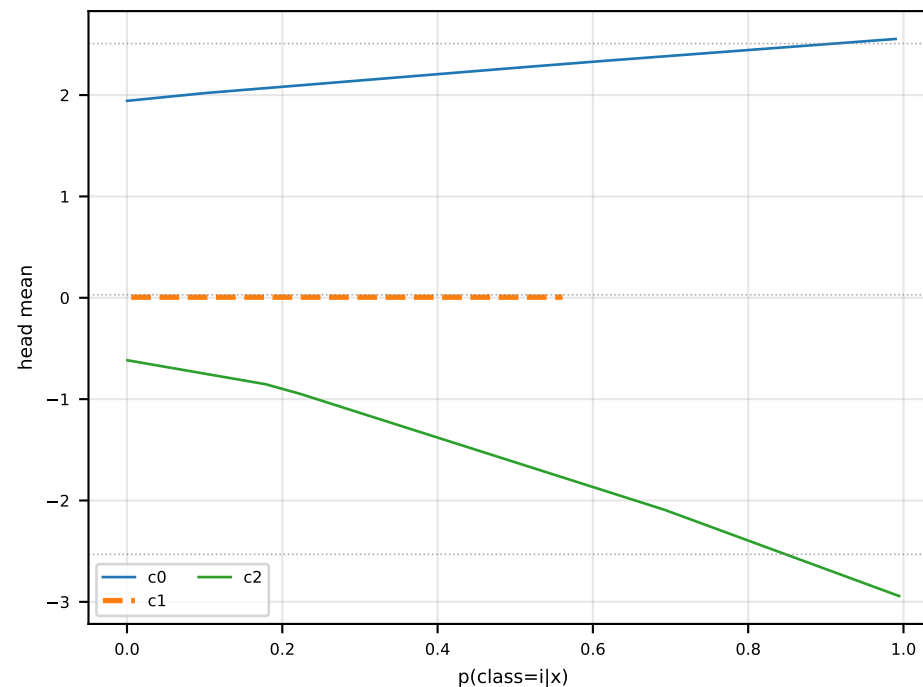
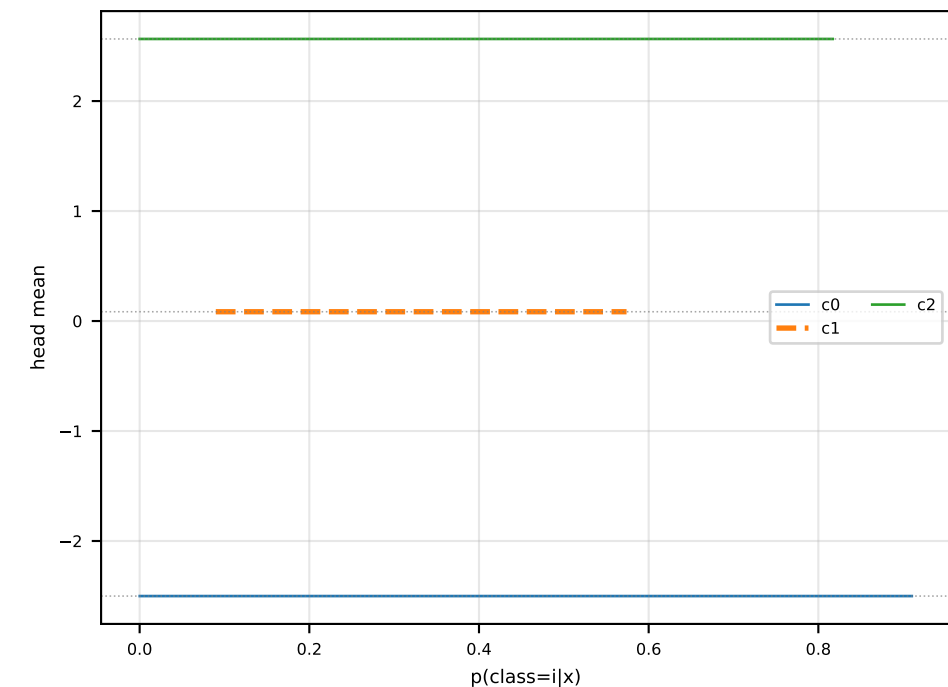
Cell D k=3

Cell E k=3

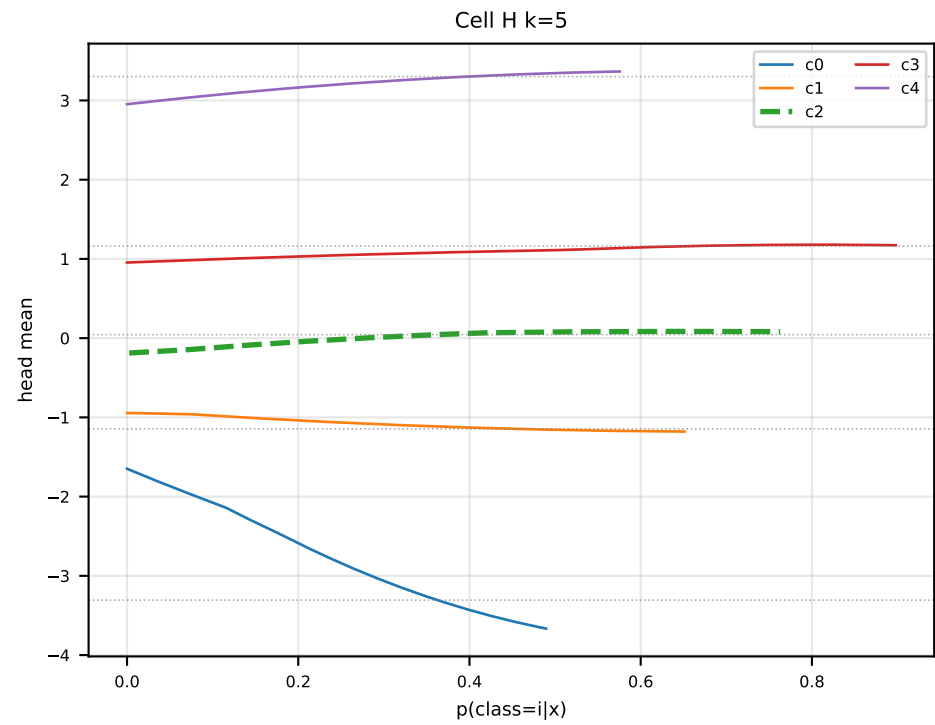
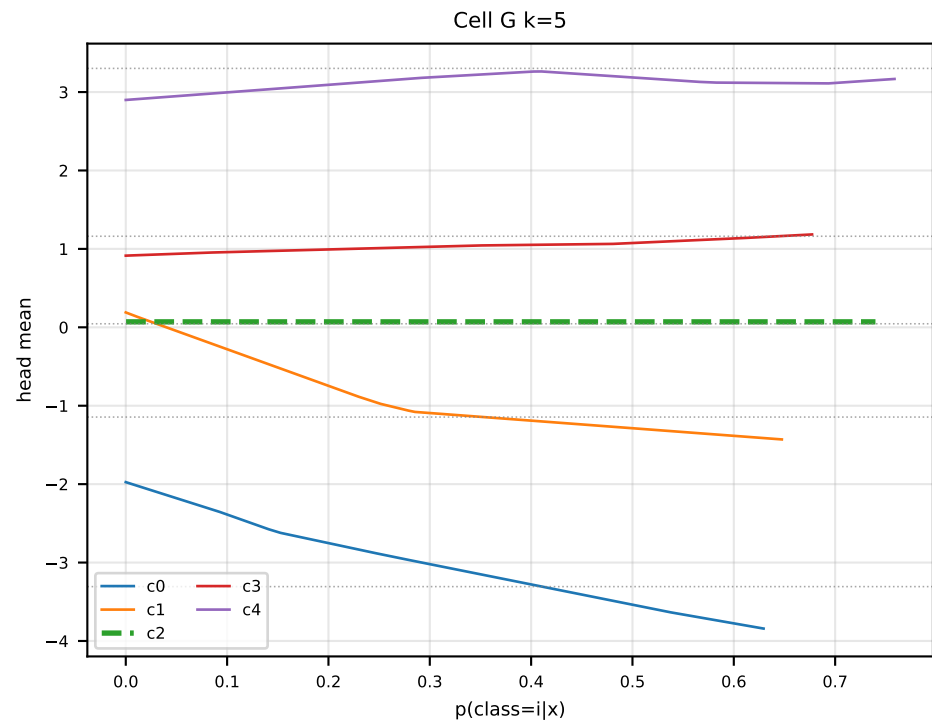
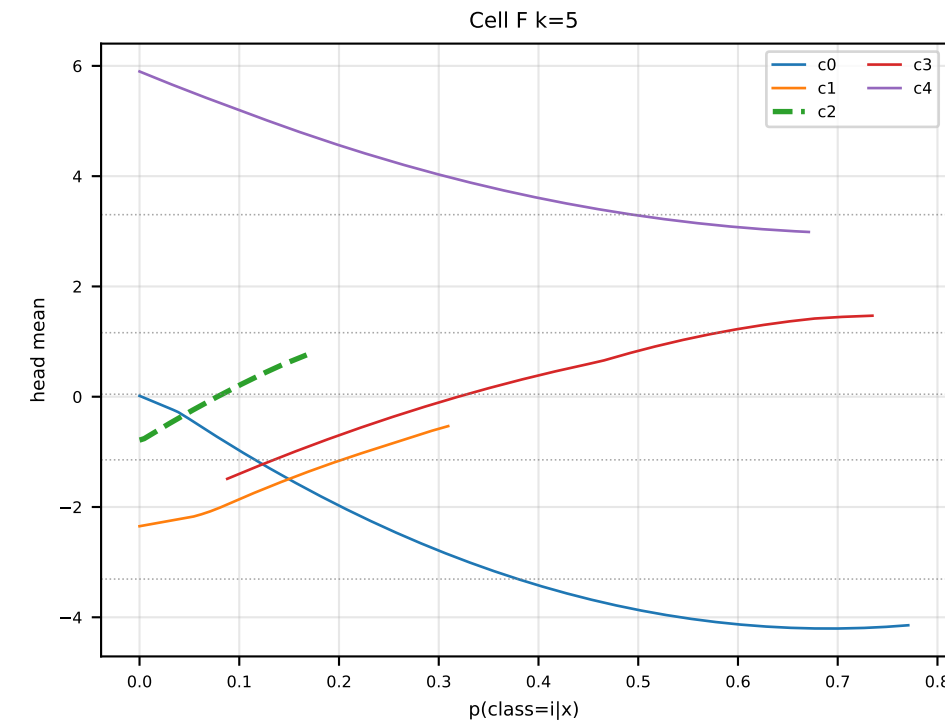
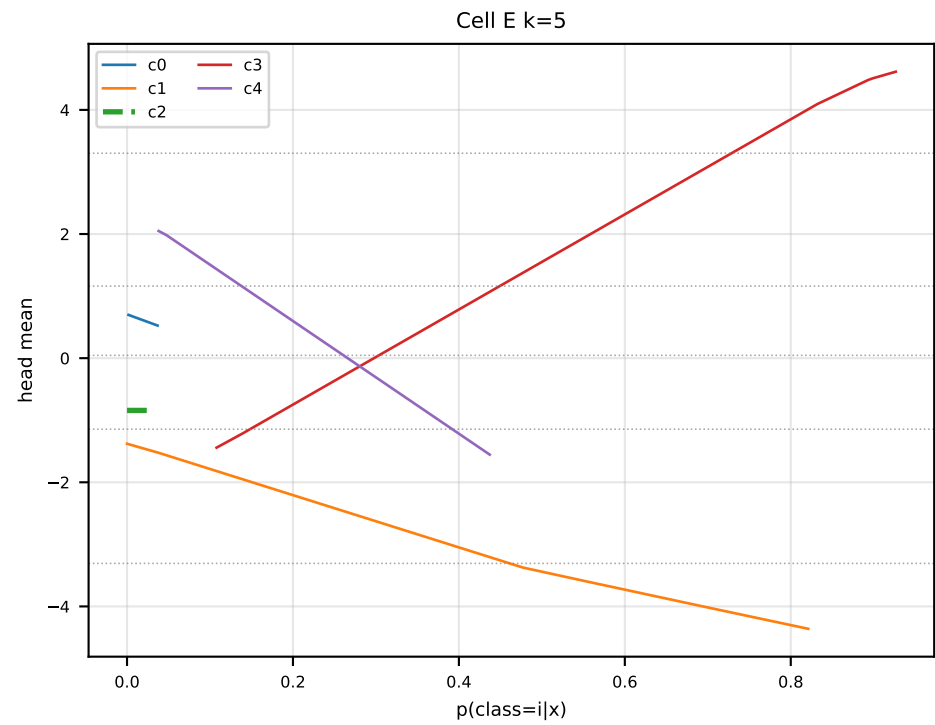
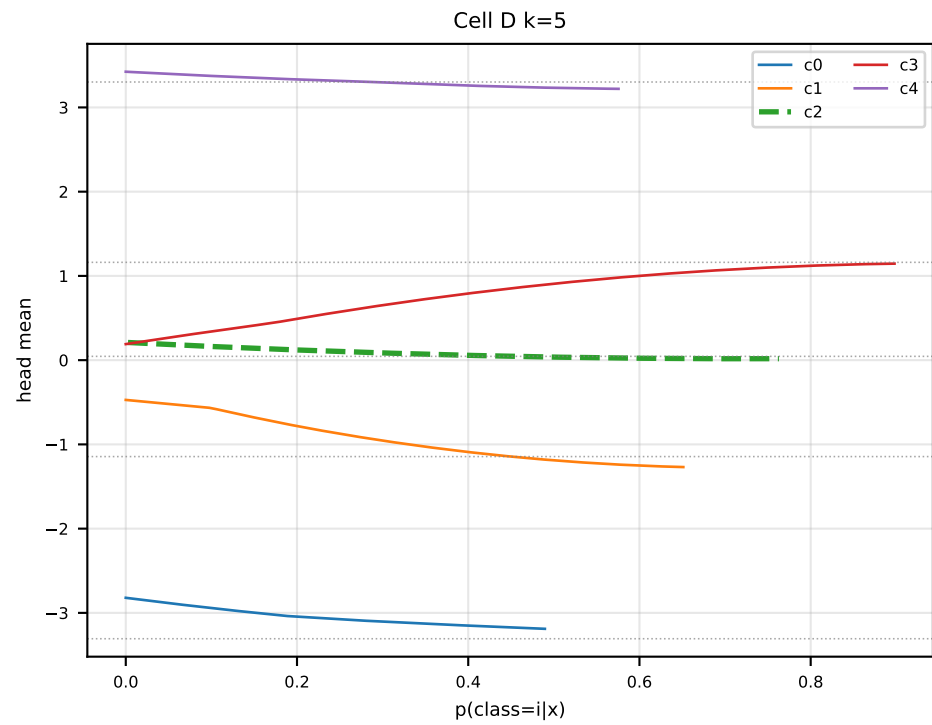
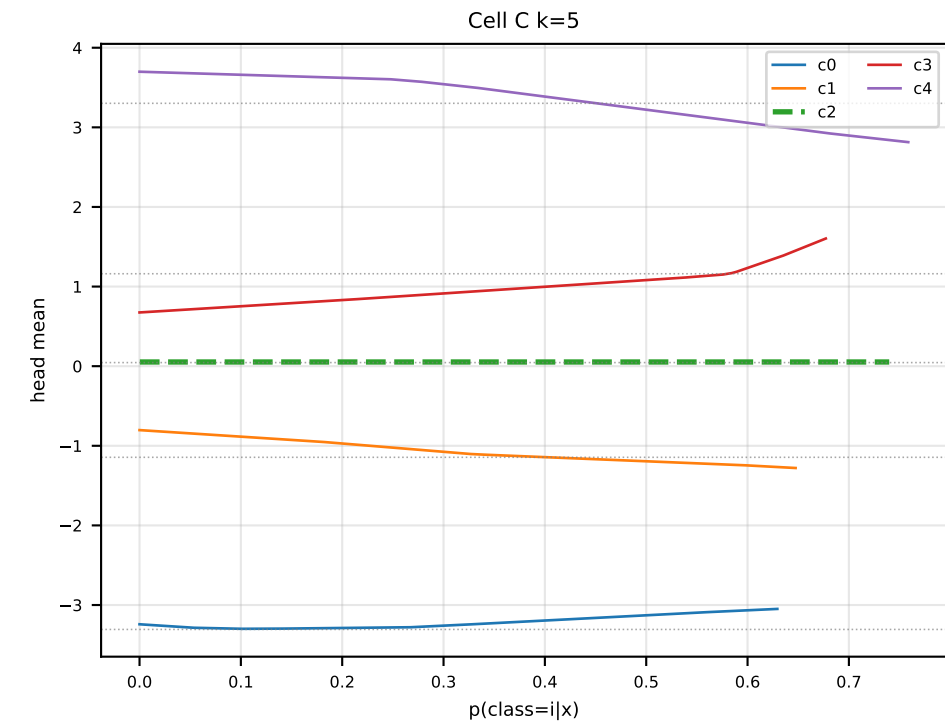
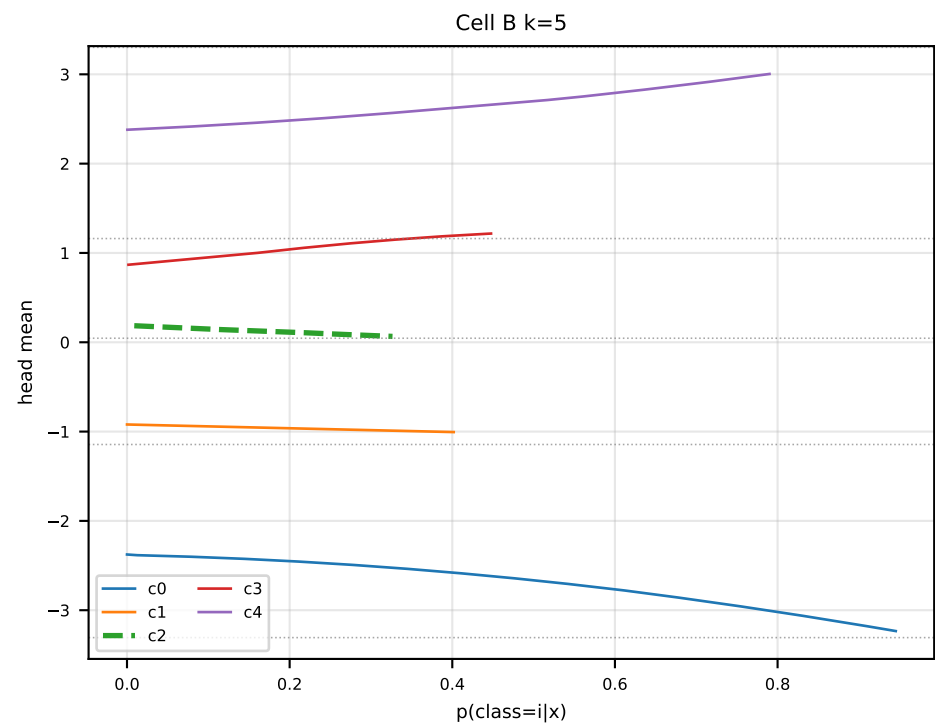
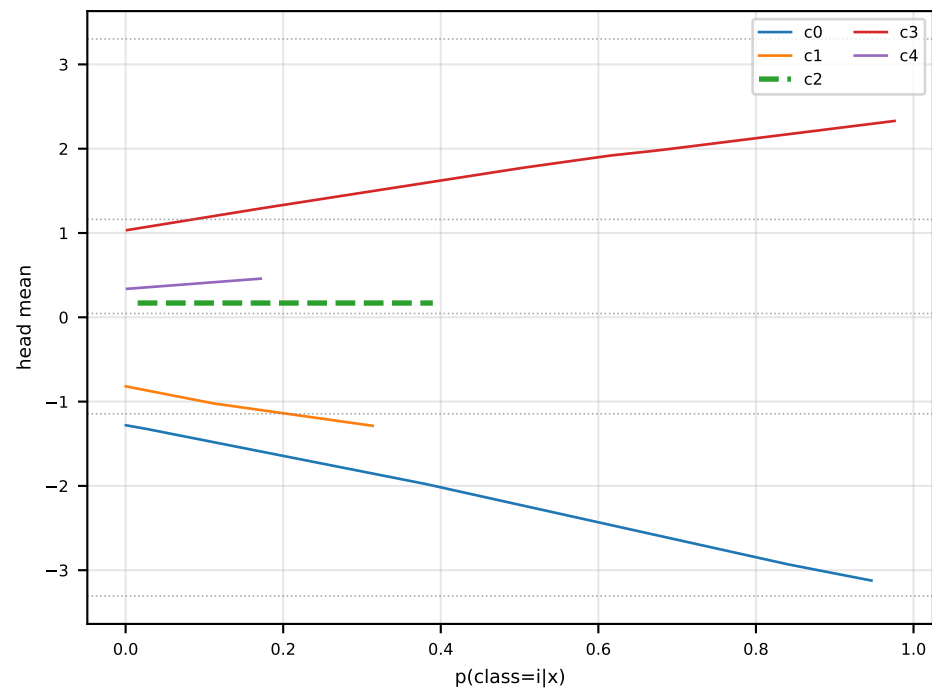
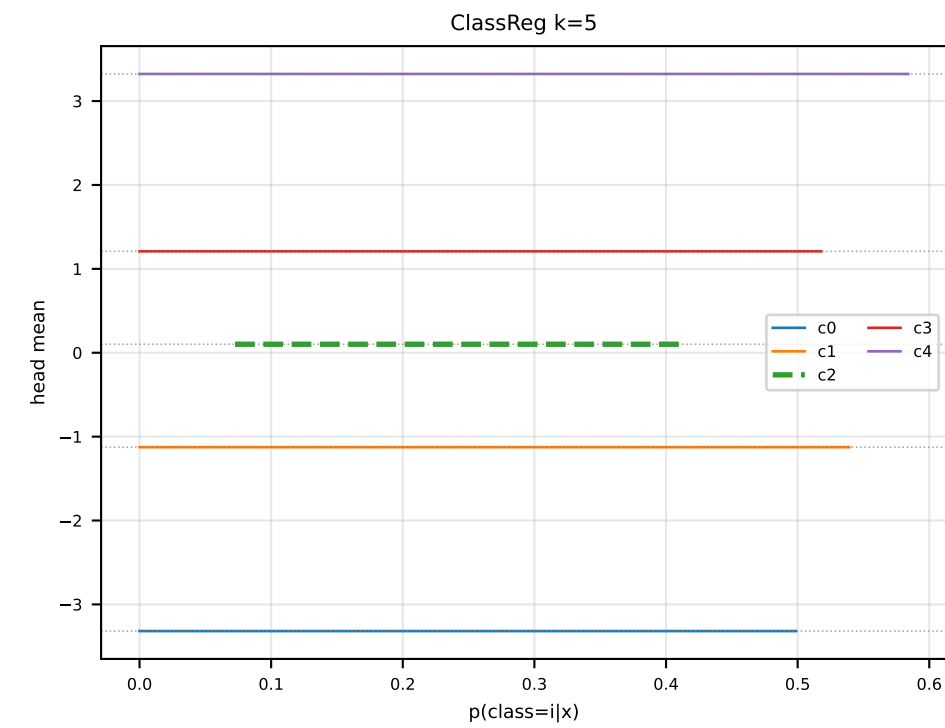
Cell F k=3

Cell G k=3

Cell H k=3

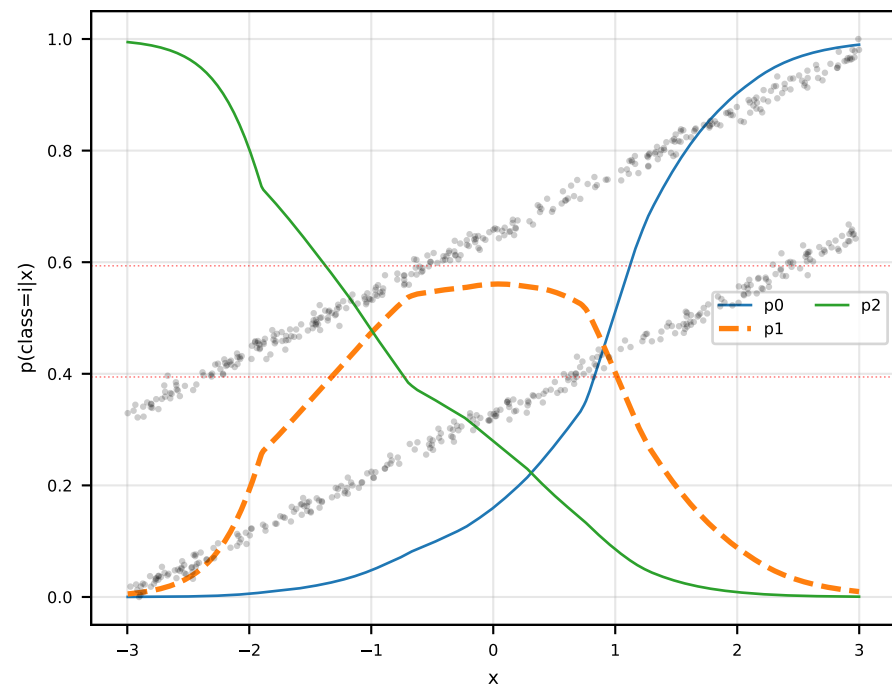


bimodal k=5 — $h_i(p_i)$ vs p_i (seed 42)

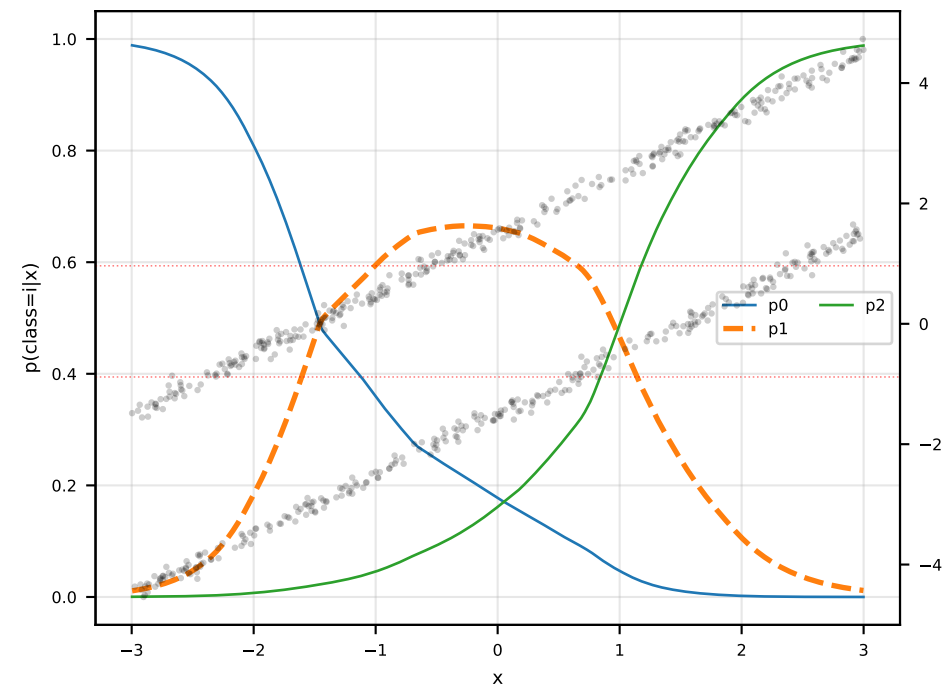


bimodal k=3 — $p_i(x)$ vs x (seed 42)

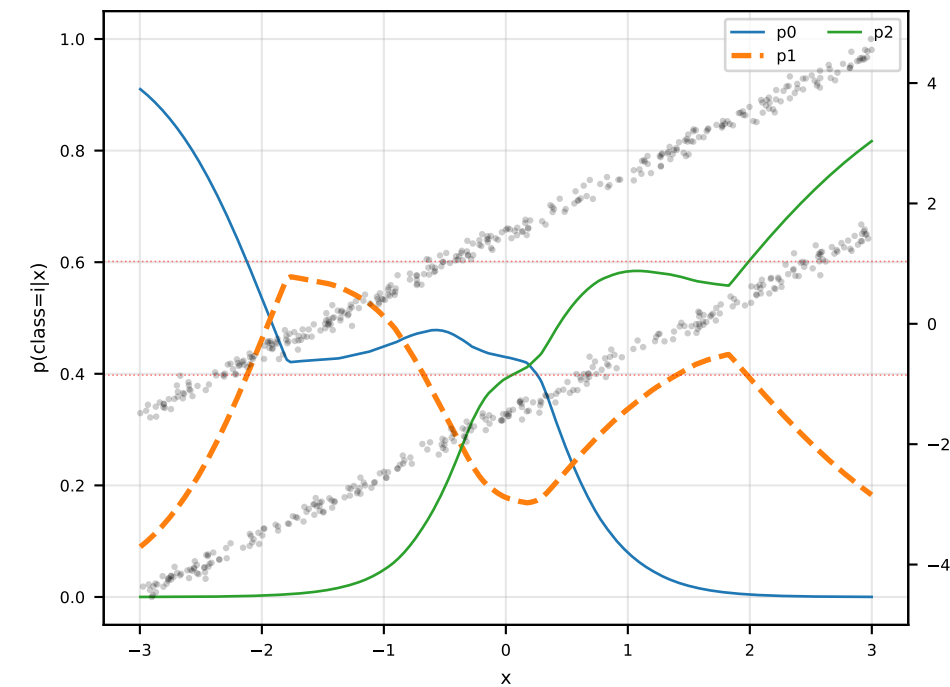
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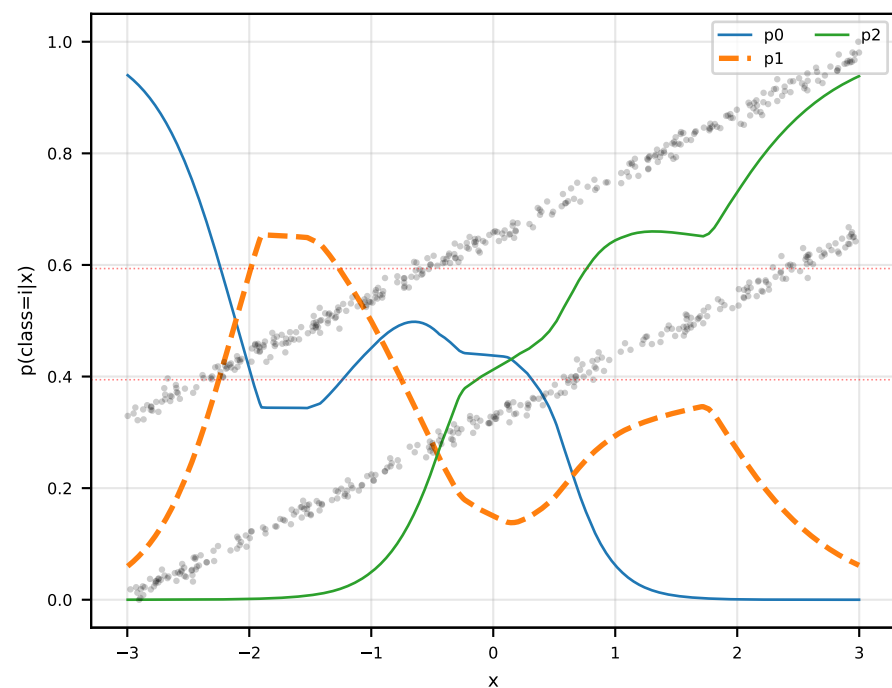
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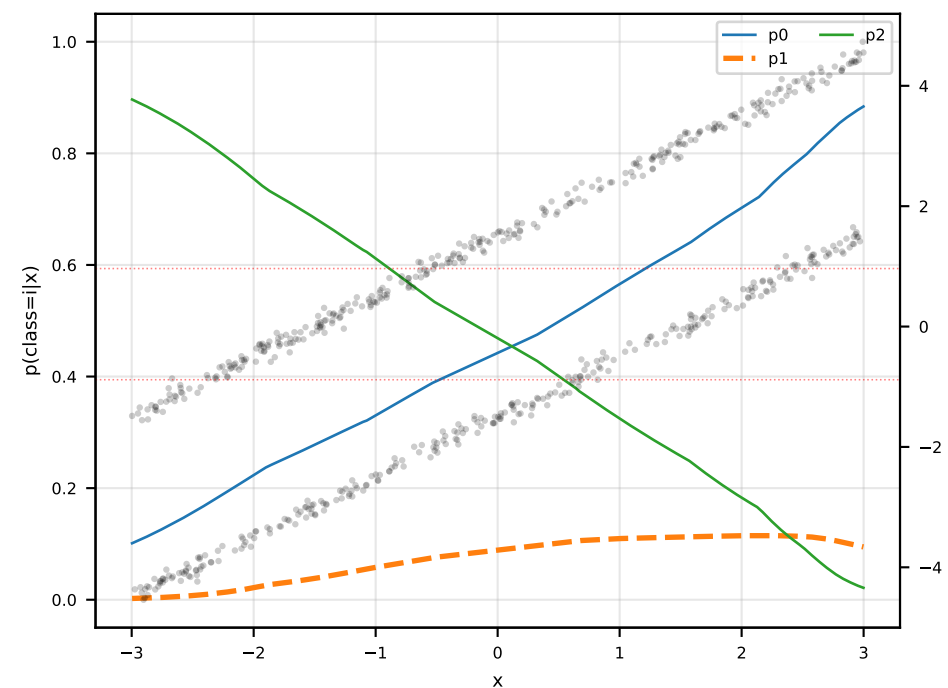
ClassReg k=3



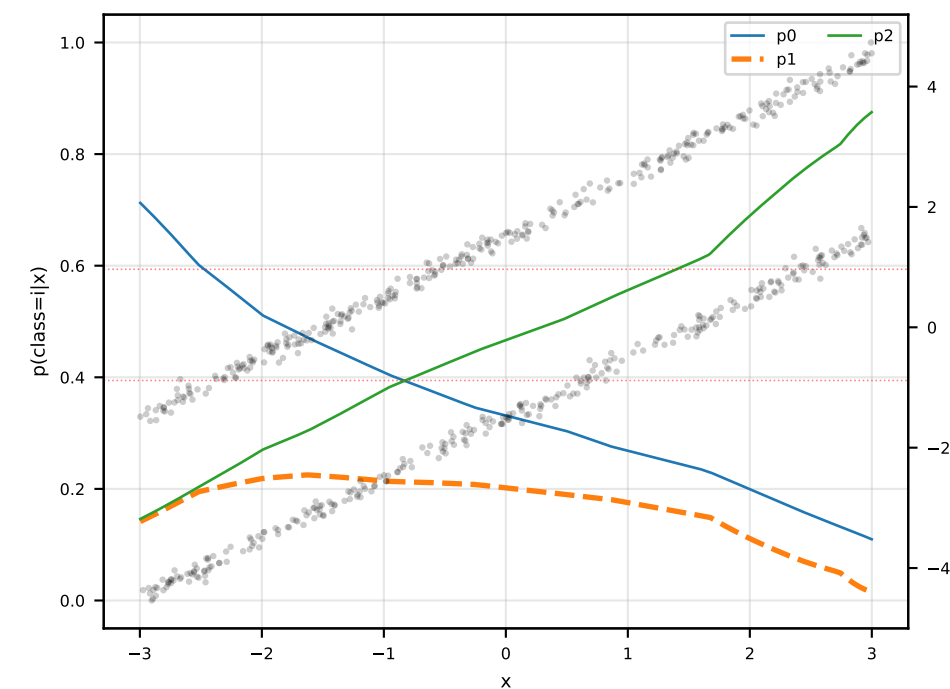
Cell D k=3



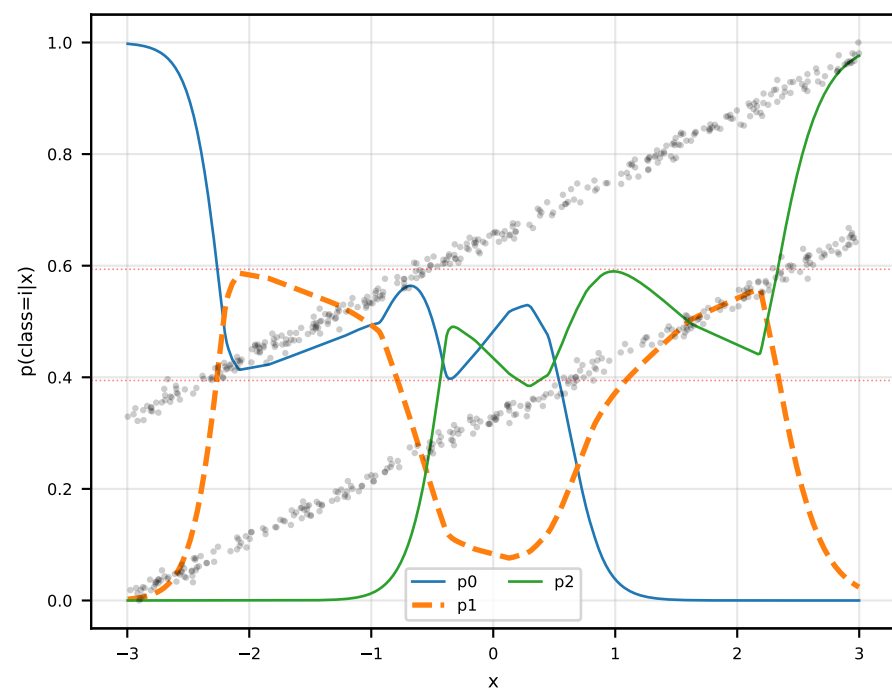
Cell E k=3



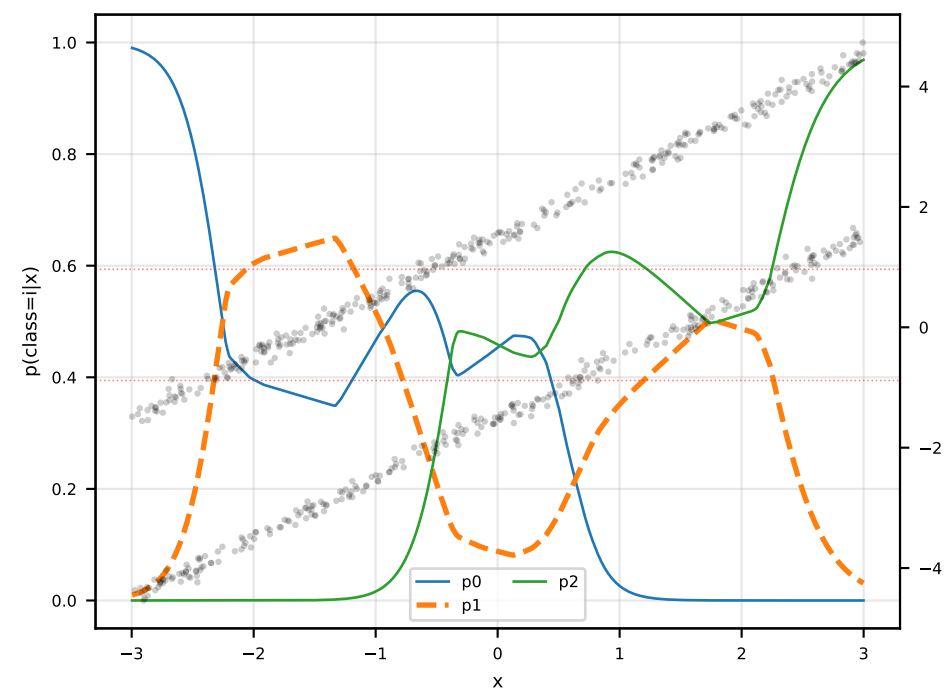
Cell F k=3



Cell G k=3

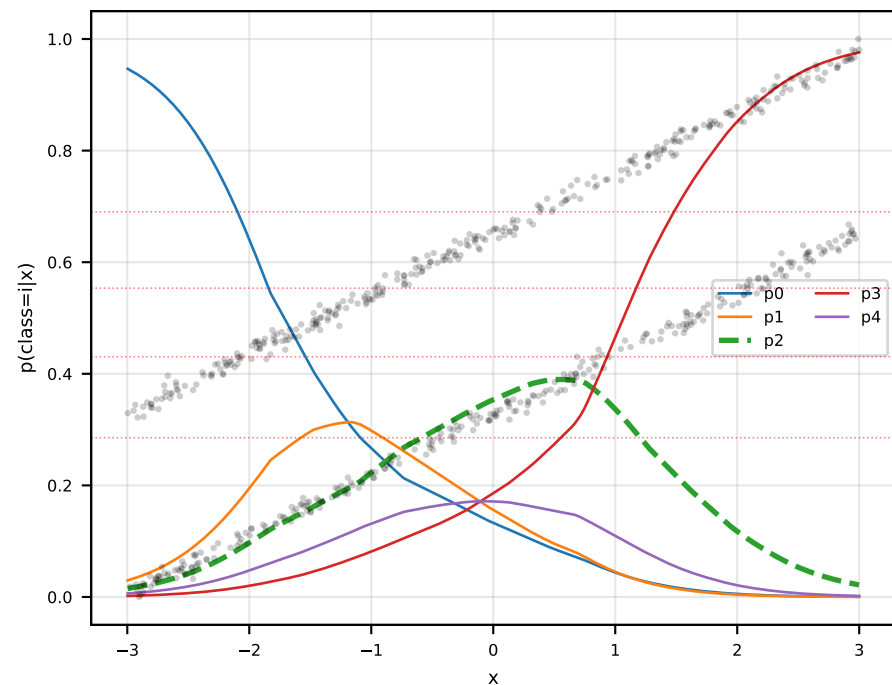


Cell H k=3

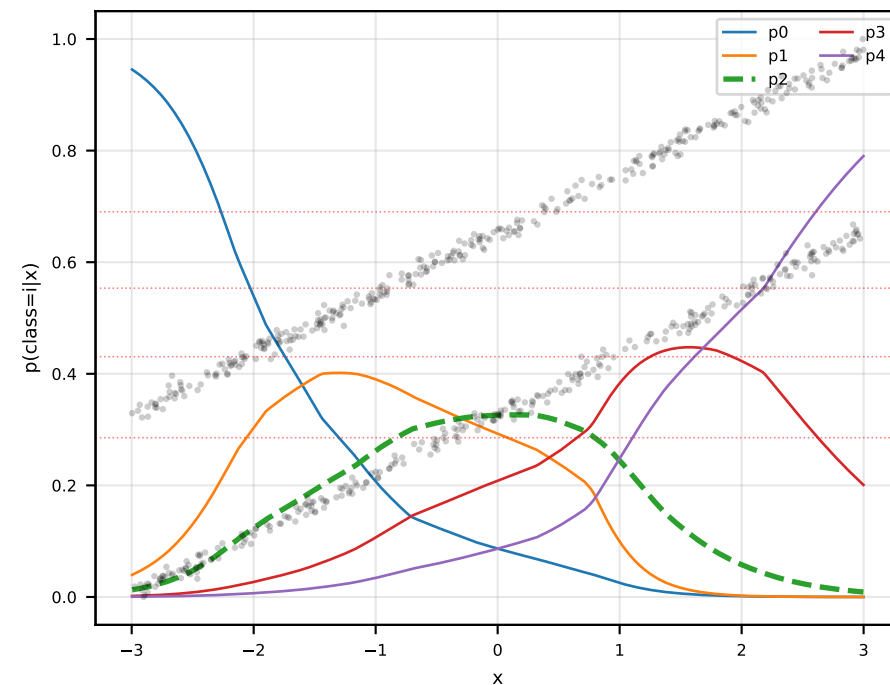


bimodal k=5 — $p_i(x)$ vs x (seed 42)

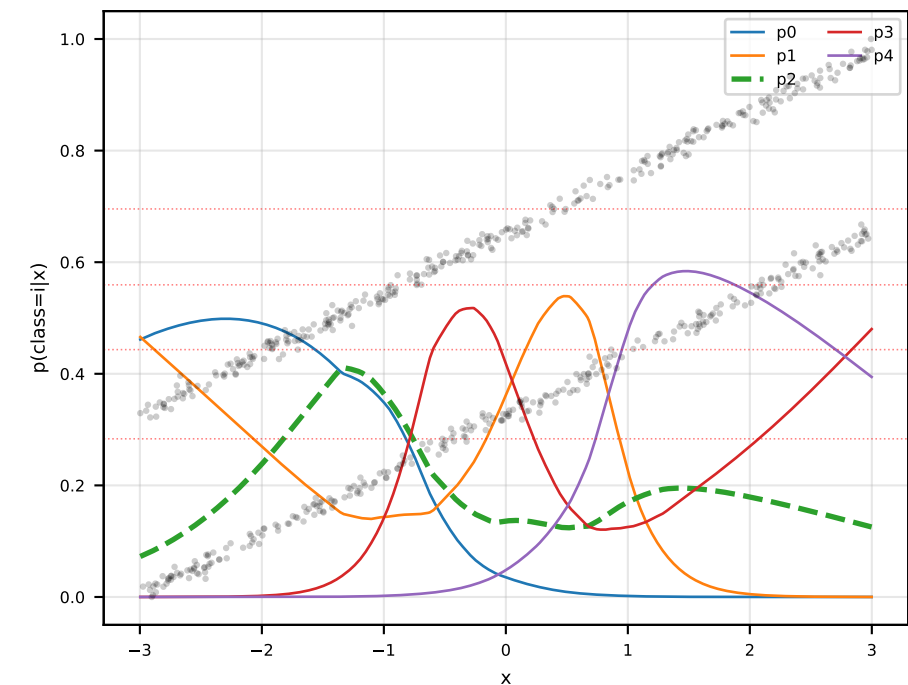
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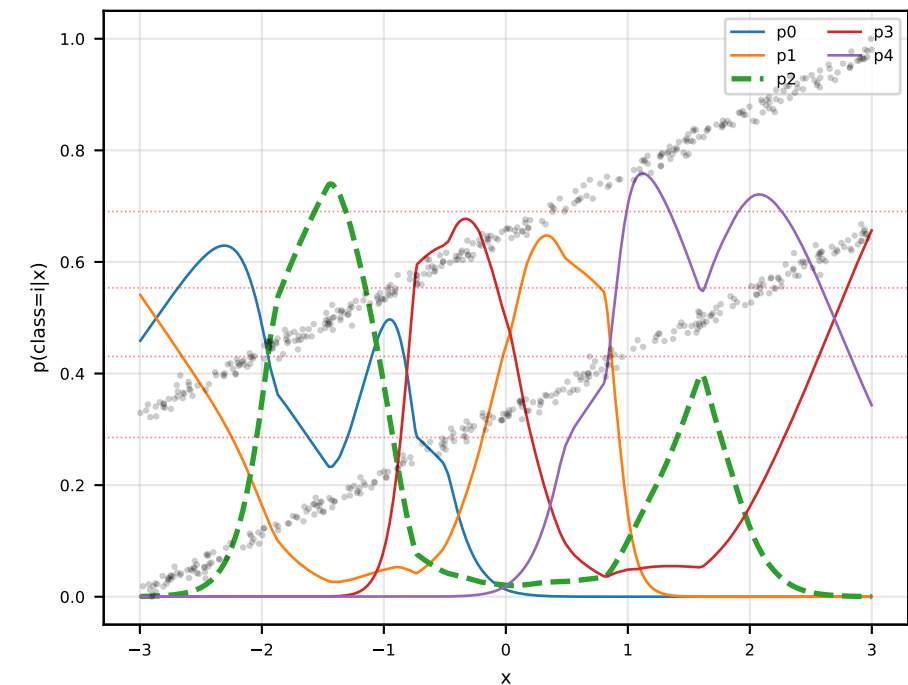
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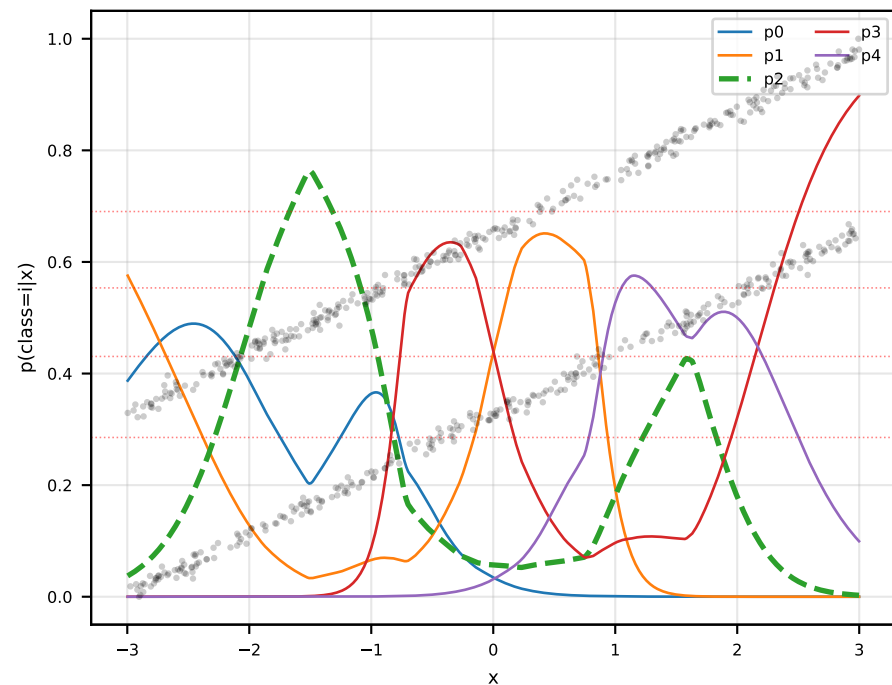
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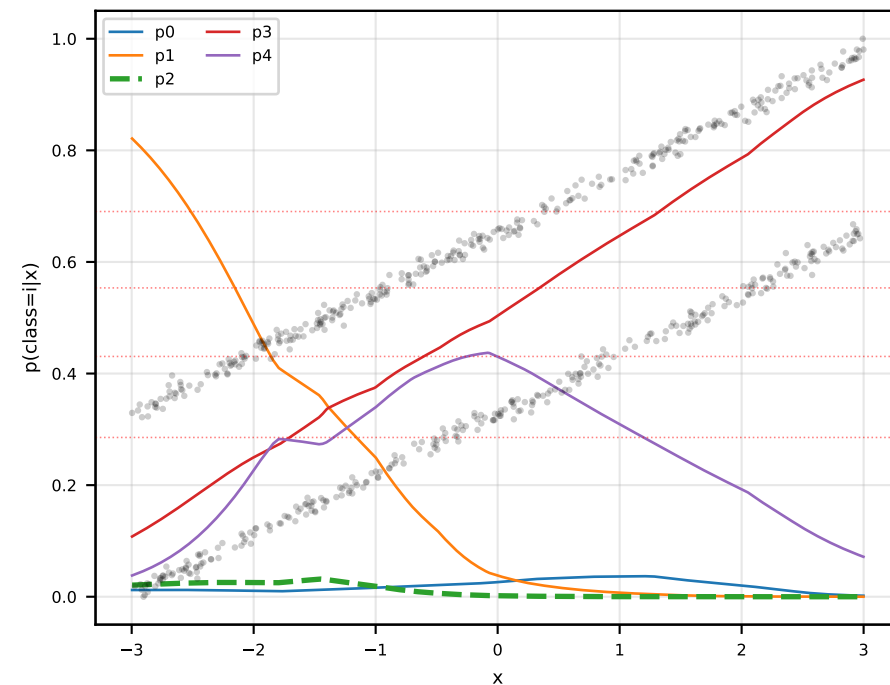
Cell C k=5



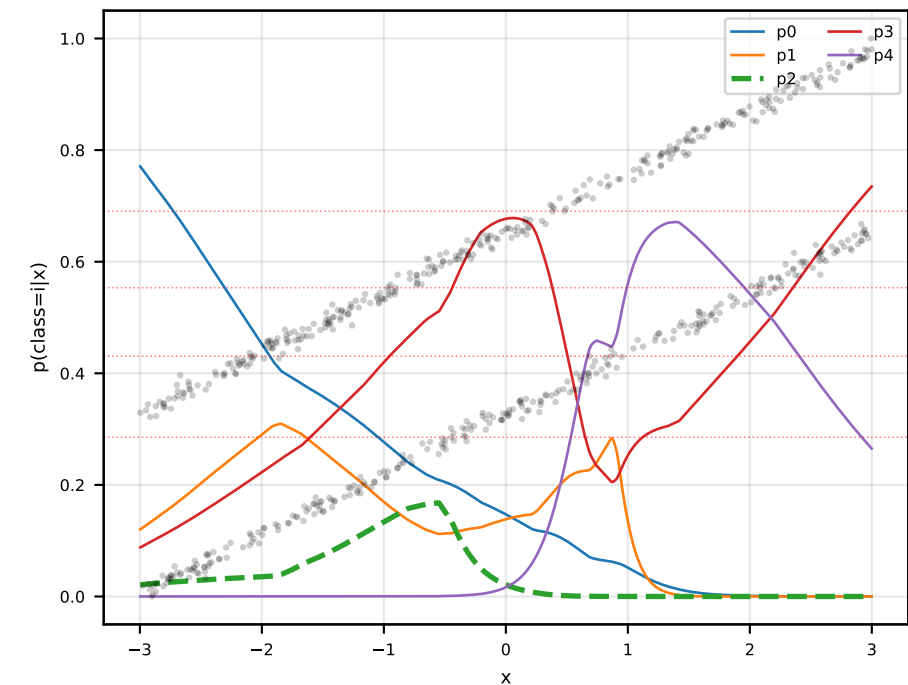
Cell D k=5



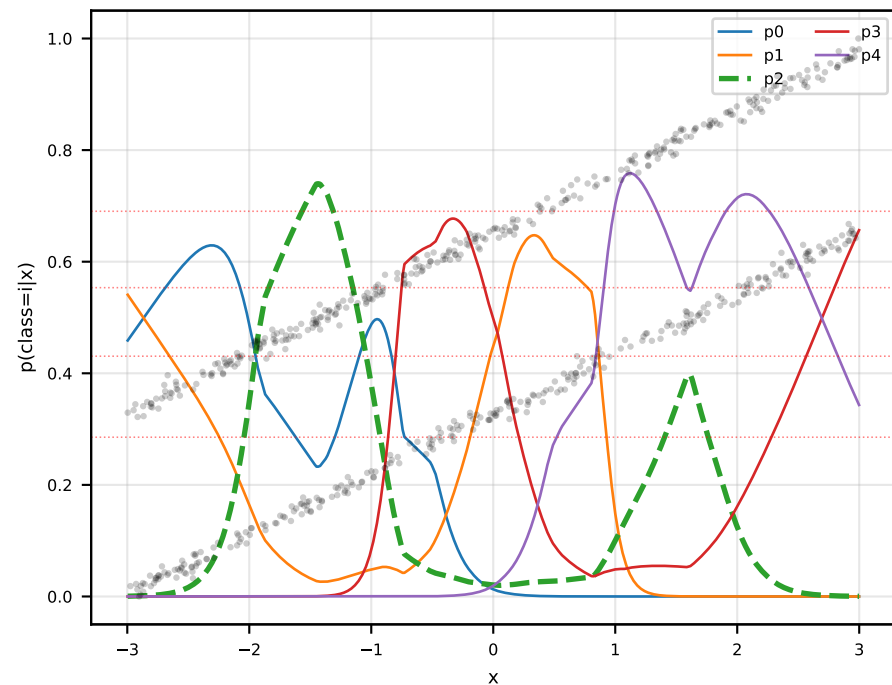
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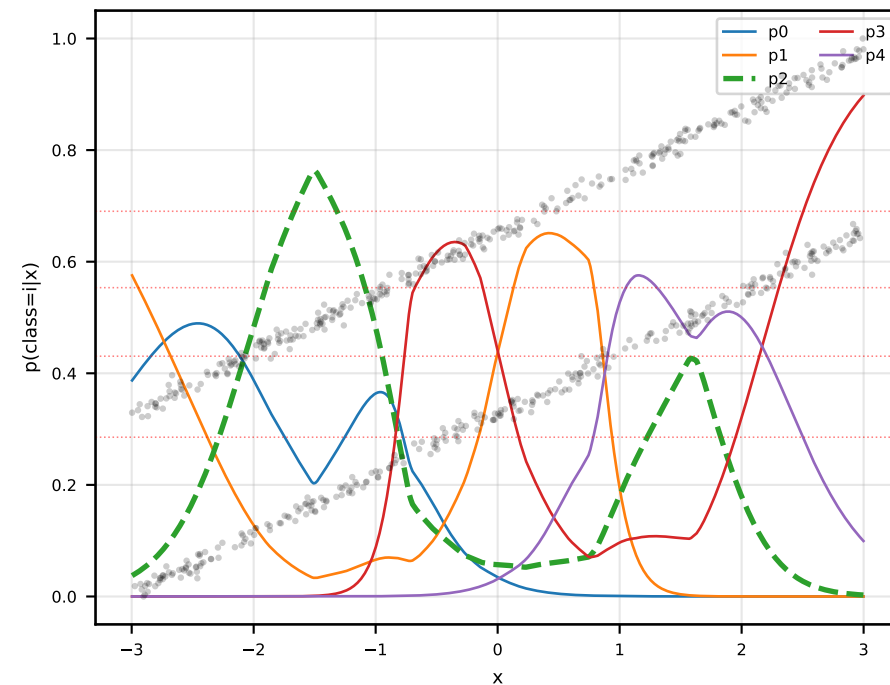
Cell F k=5



Cell G k=5

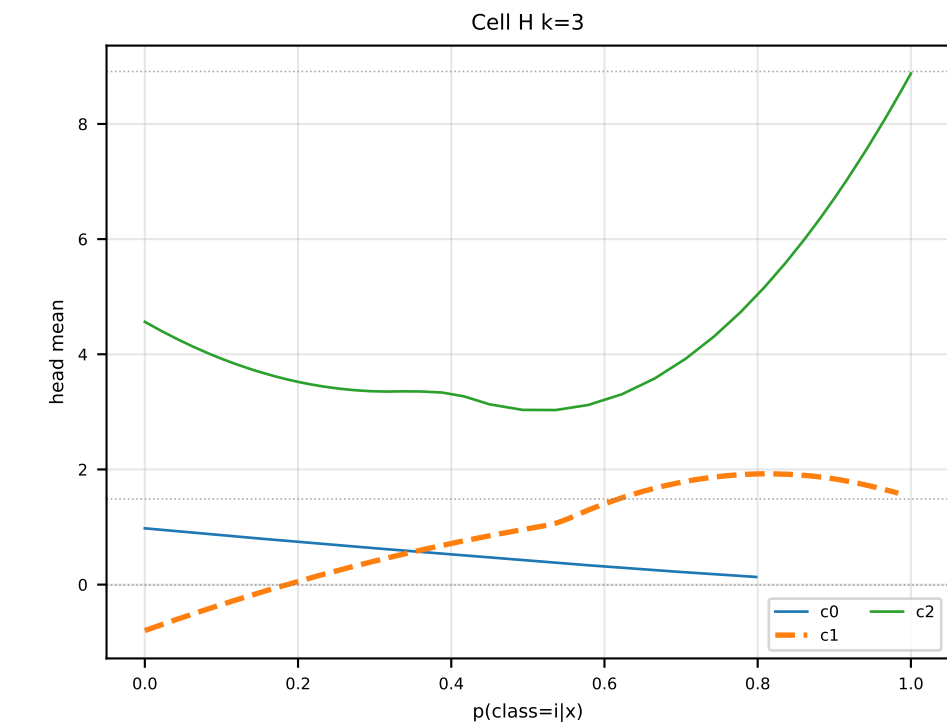
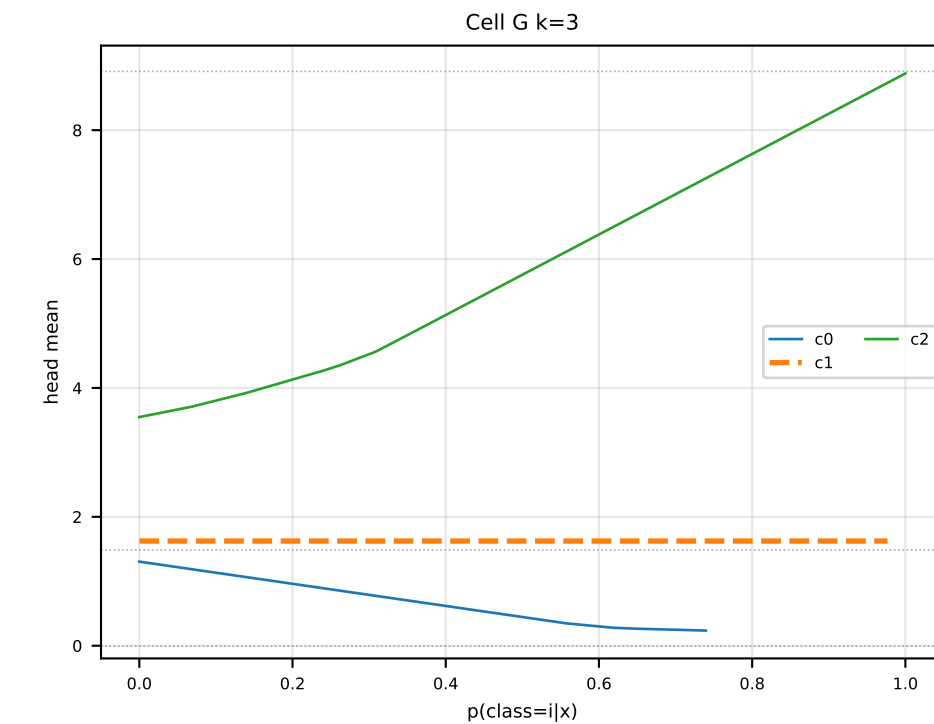
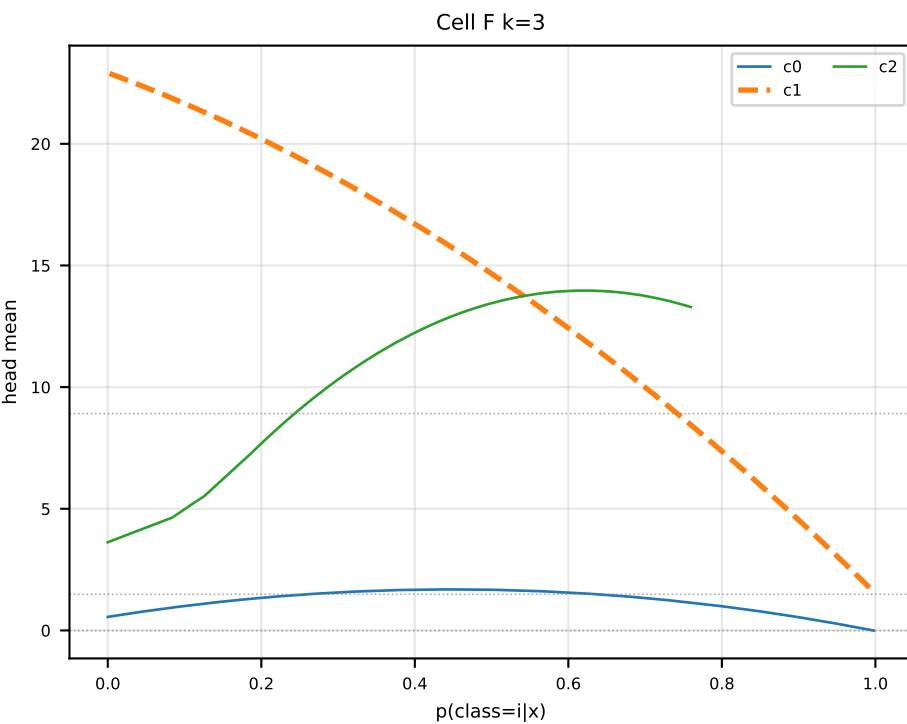
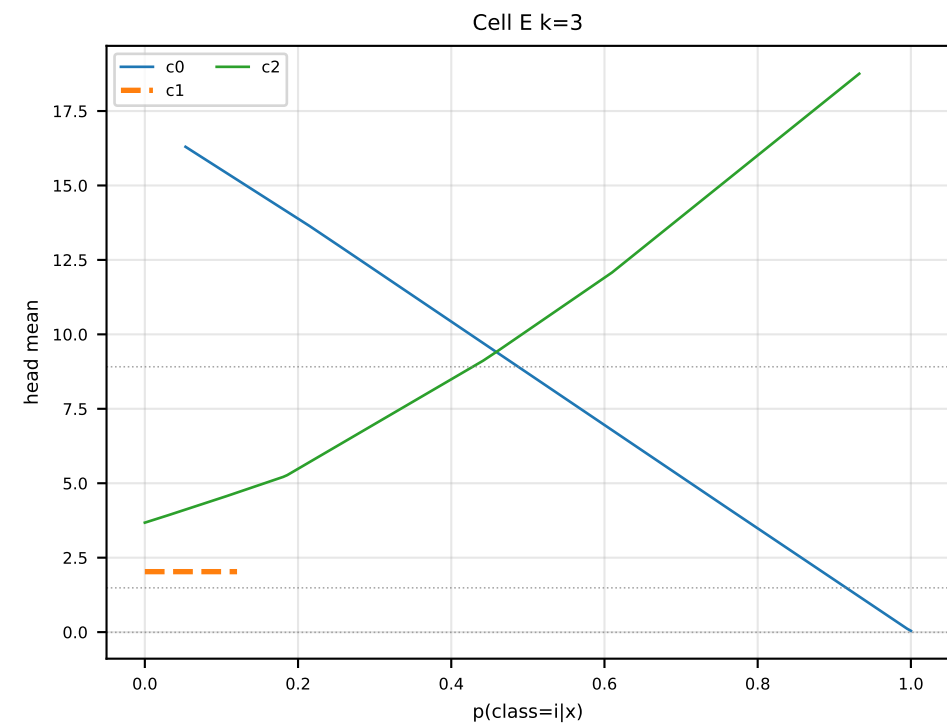
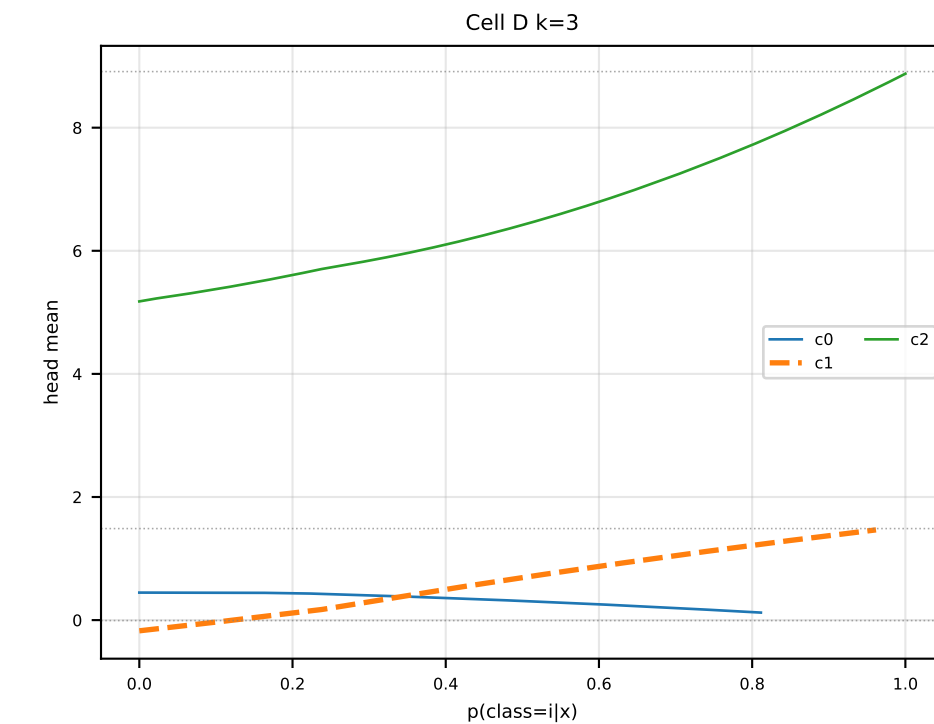
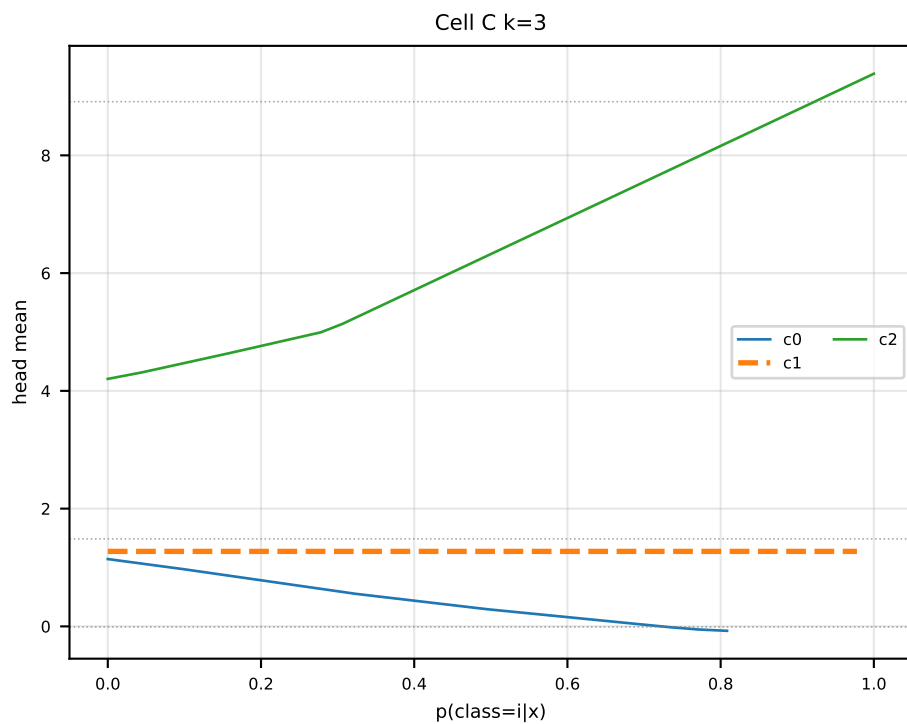
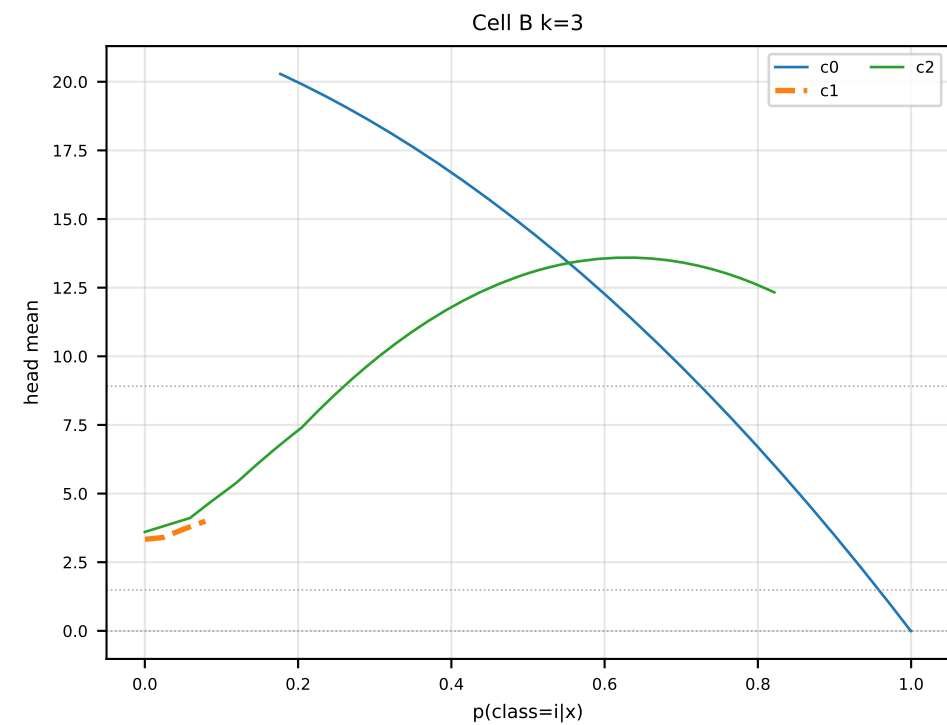
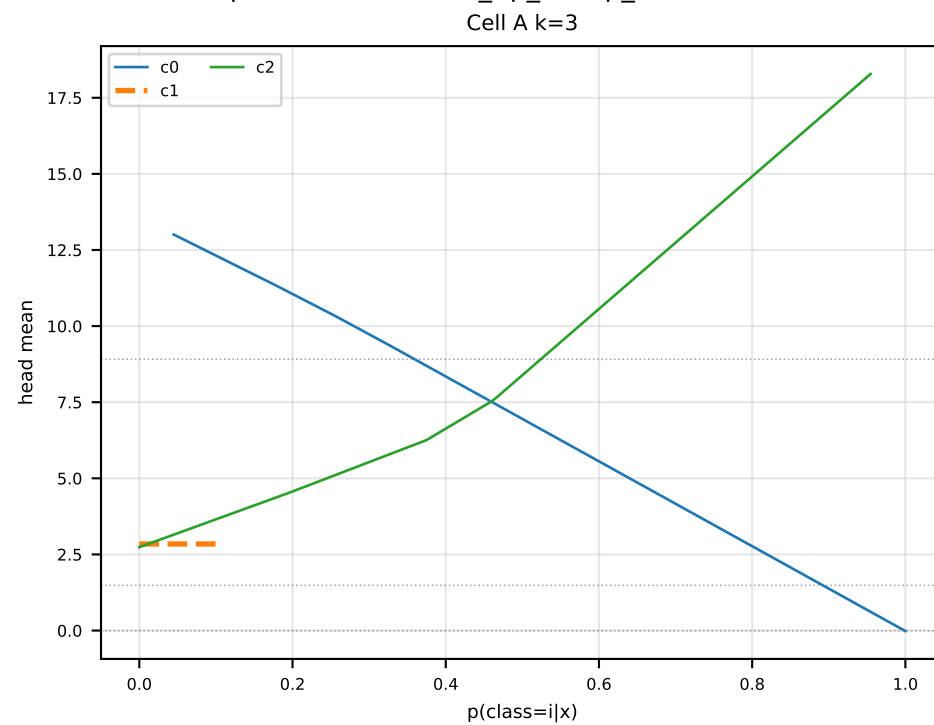
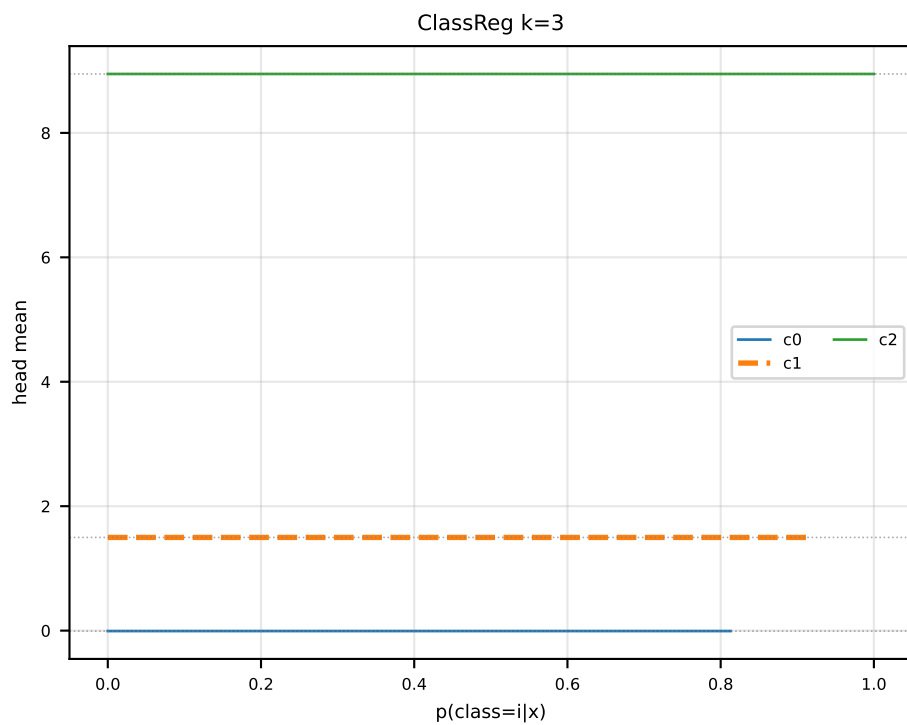


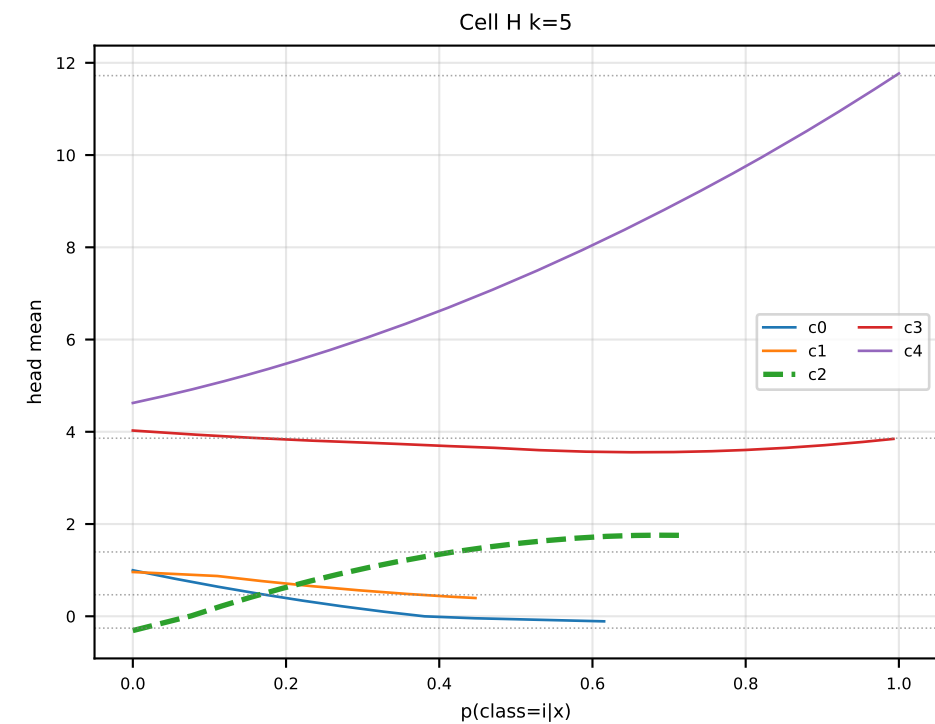
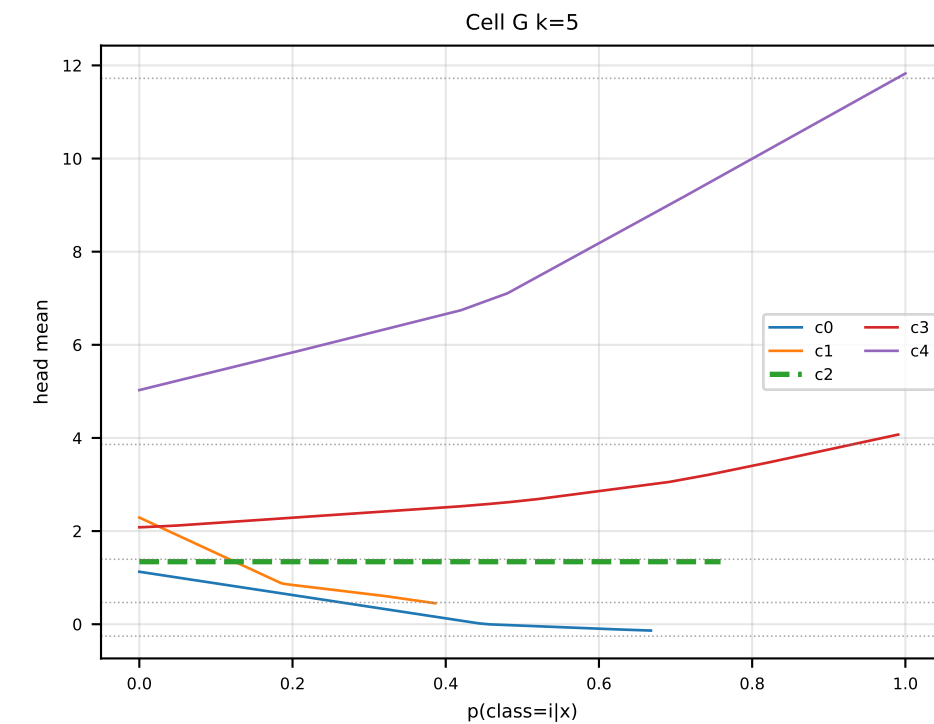
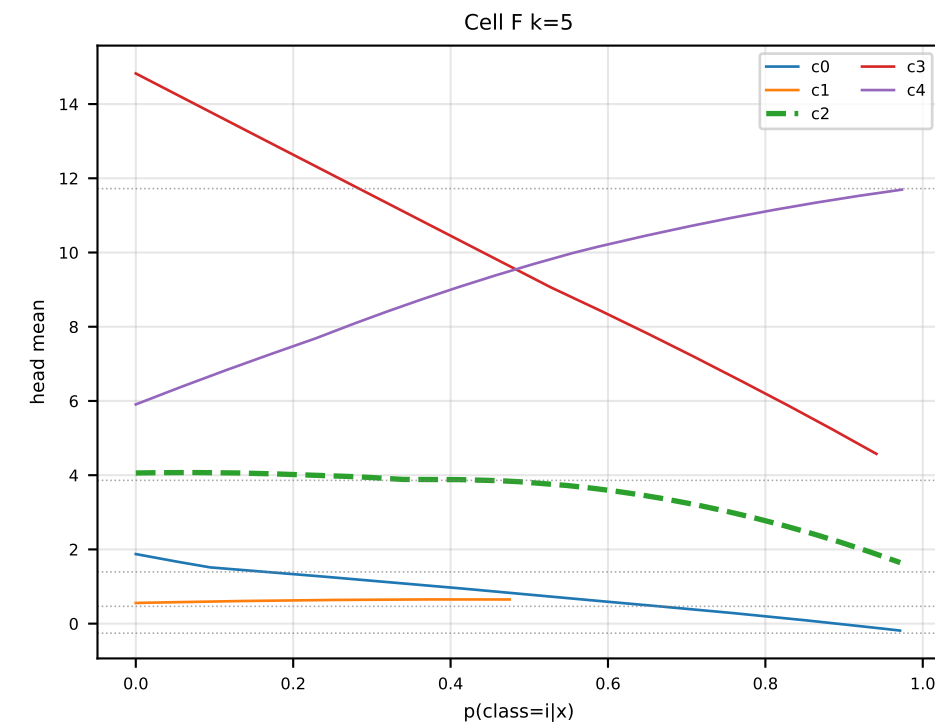
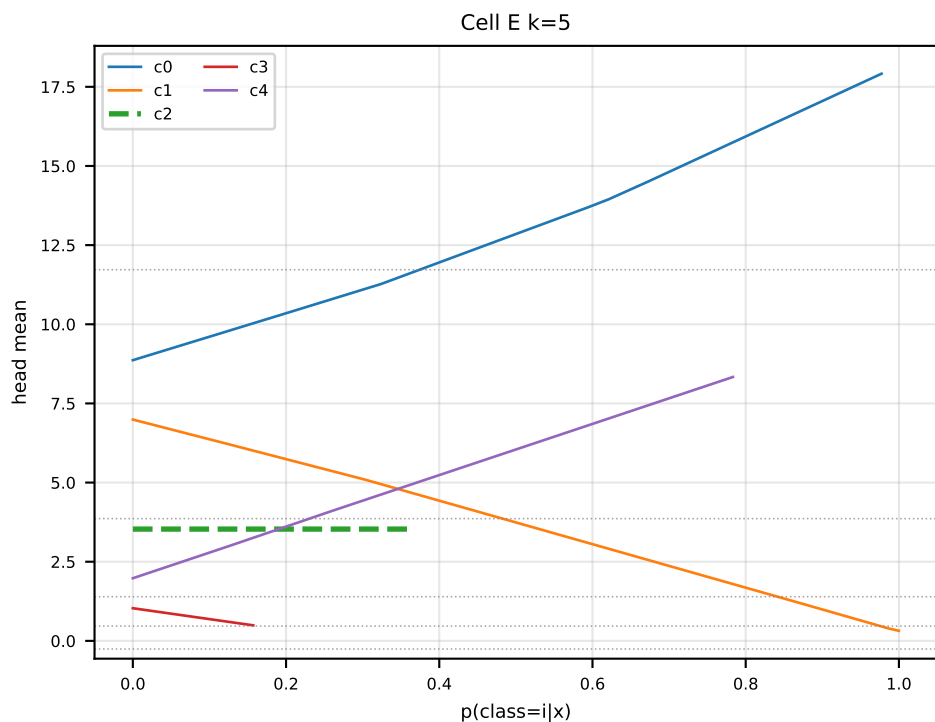
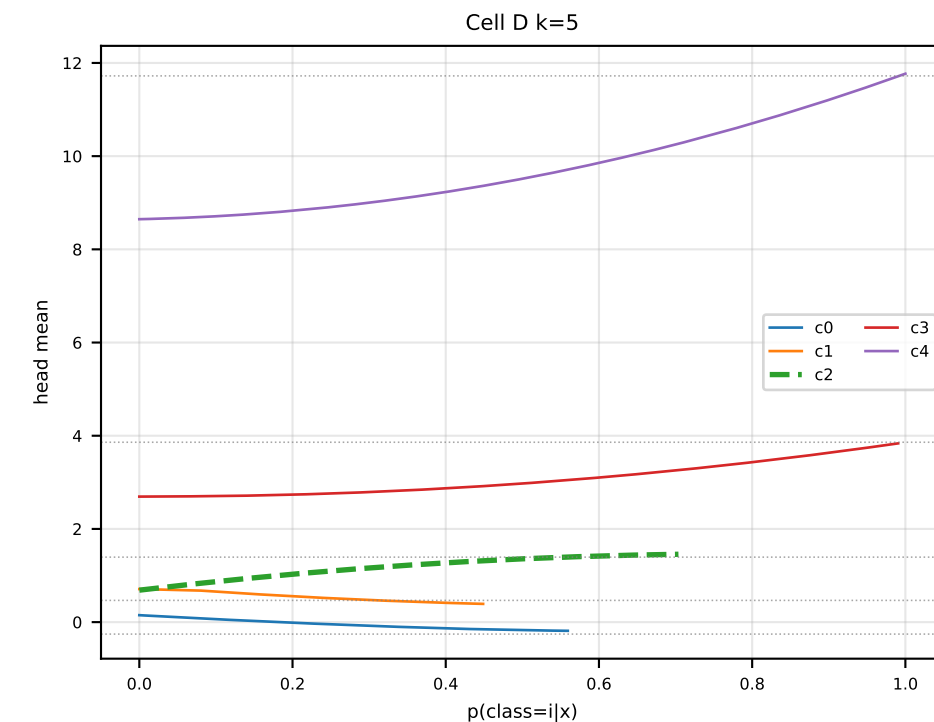
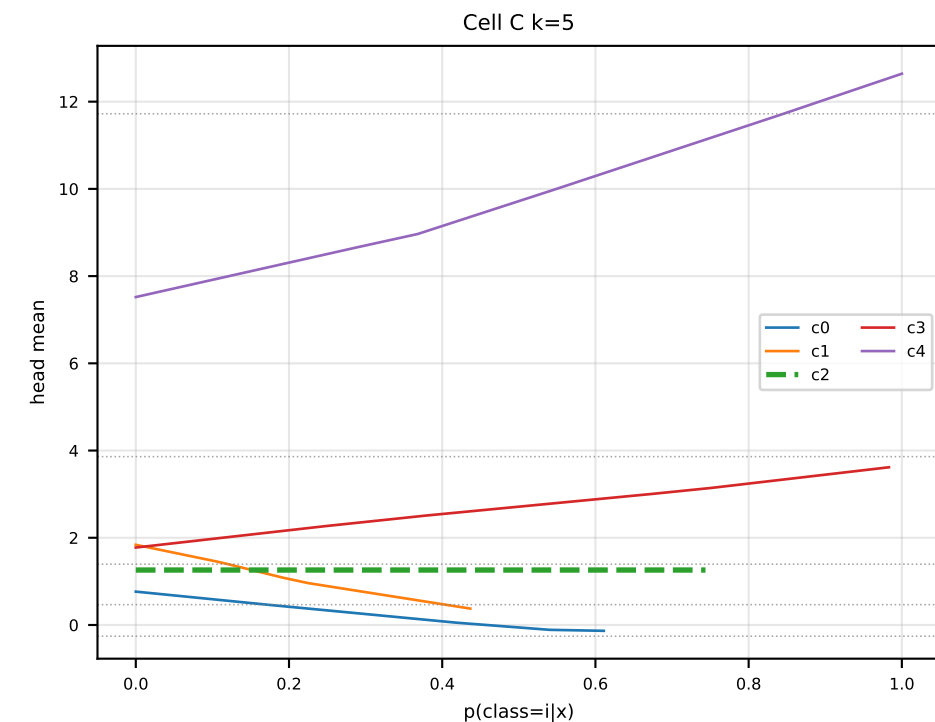
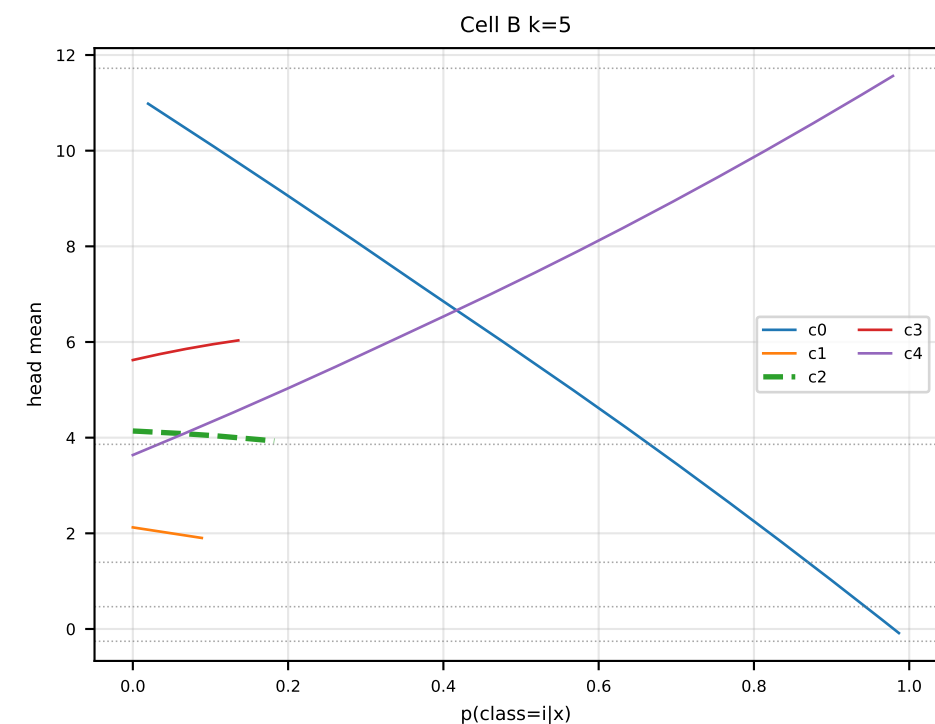
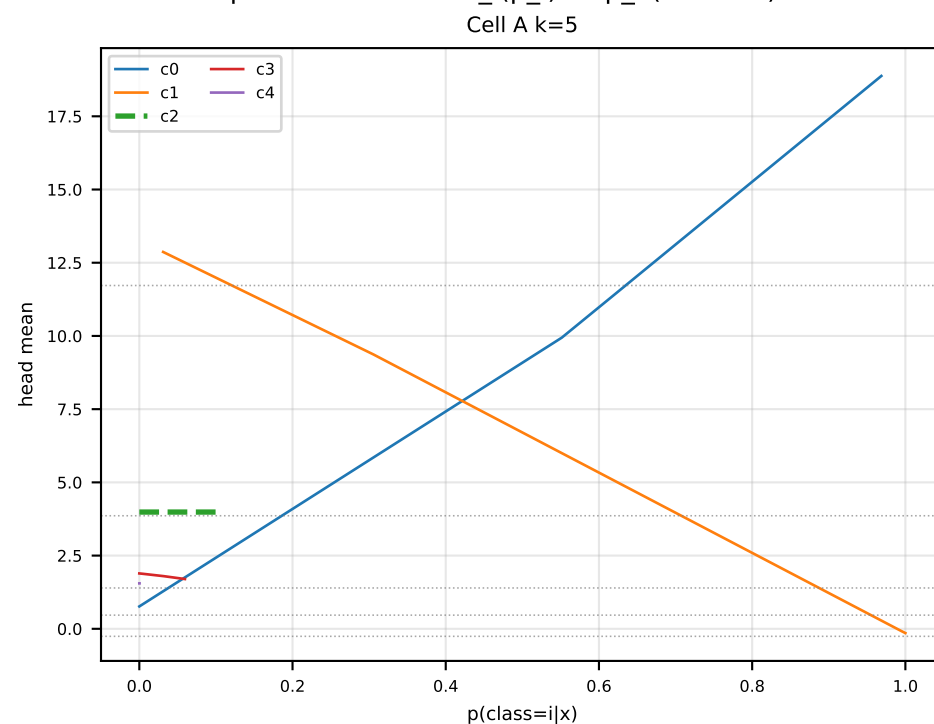
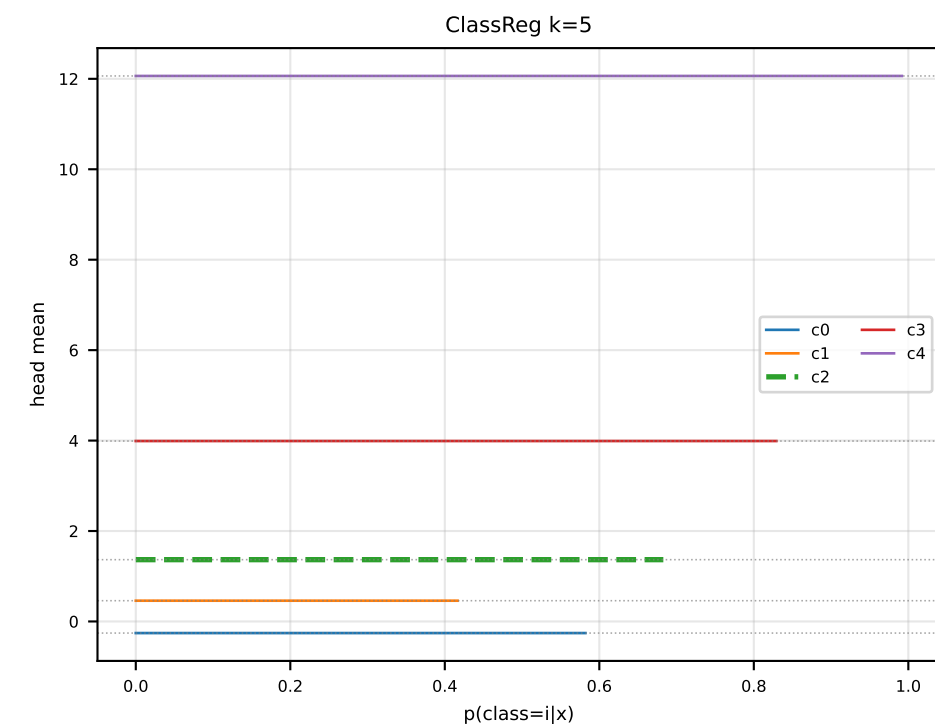
Cell H k=5

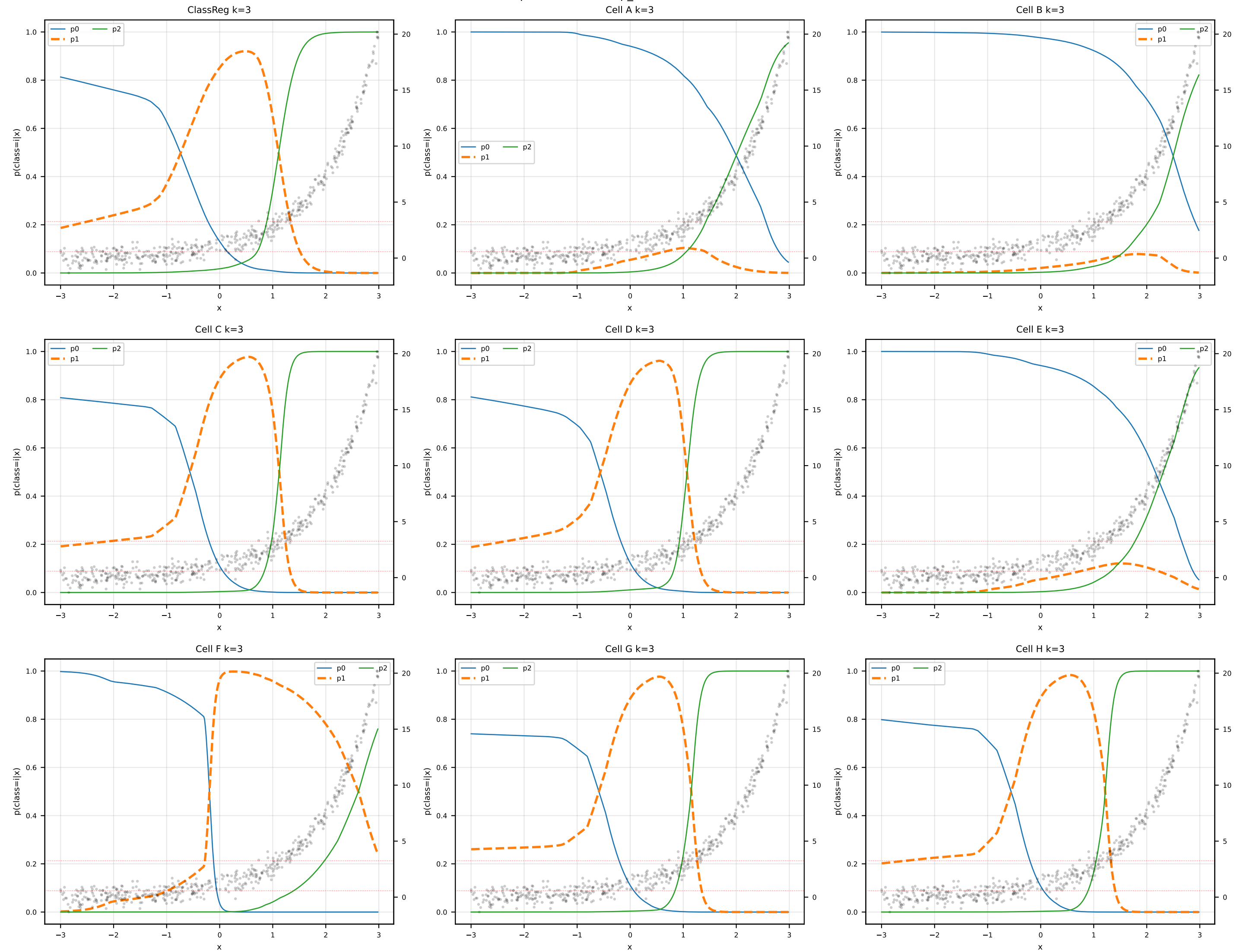


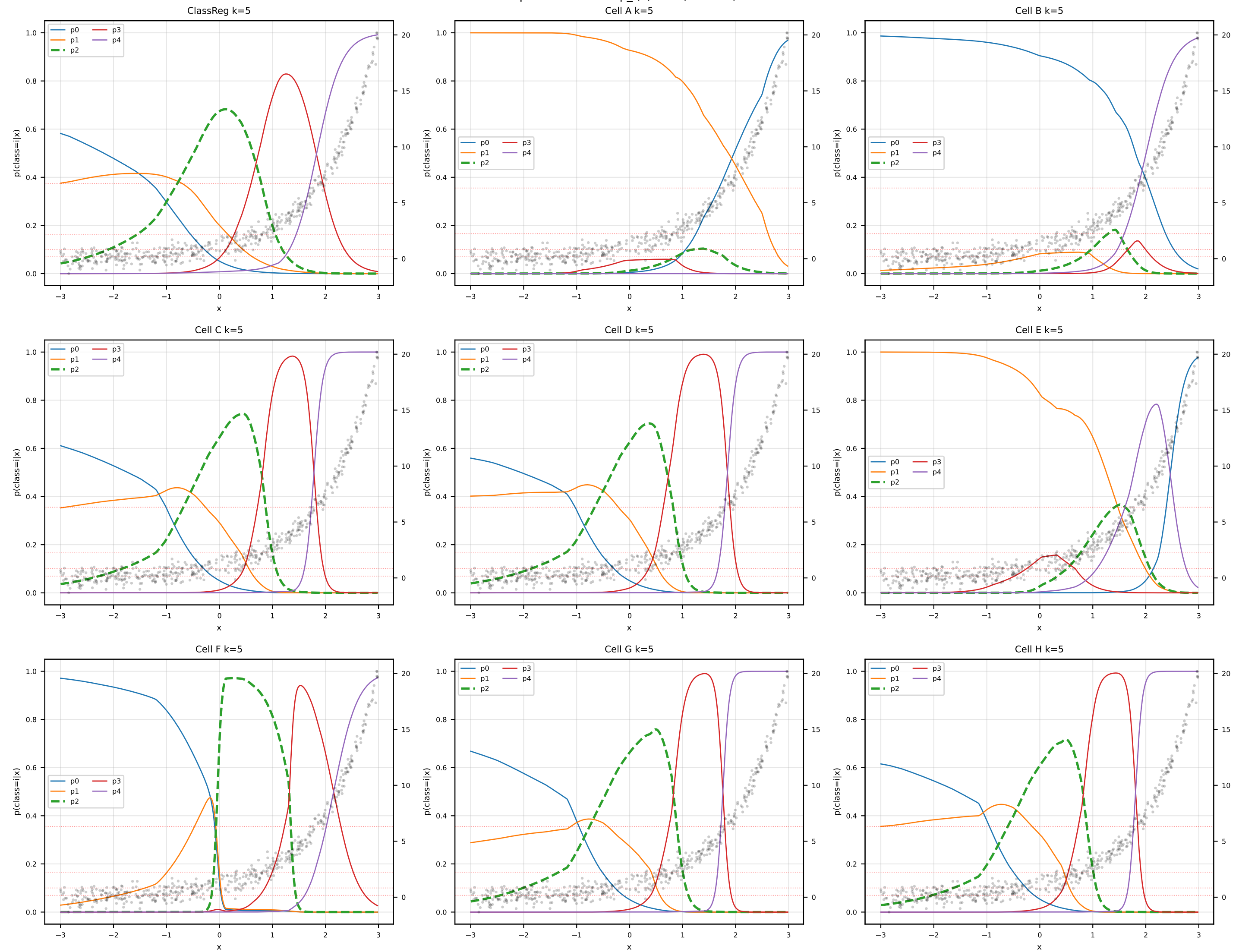
exponential — metrics summary (mean ± std across 3 seeds)

k	cell	mse_mean	mse_std	nll_gaussian_mean	nll_mdn_mean	max_p_mid_mean	anchor_error_mean
3	A	0.5687953333333333	0.1164936489785316	1.0646226666666667	nan	nan	8.5989
3	B	0.7987786666666666	0.44334902623253086	0.9735973333333333	nan	nan	4.201
3	C	5.768589333333334	0.5993505556161043	1.603808	nan	nan	0.4475666666666667
3	ClassReg	2.838287	0.42979332207120235	1.9523620000000002	nan	nan	nan
3	D	5.488702	0.6121256859567326	1.561203	nan	nan	0.1771666666666667
3	E	0.50705	0.1325158415775261	1.0414763333333334	nan	nan	9.392866666666668
3	F	2.9441539999999997	0.3106504228019014	1.8477033333333335	nan	nan	3.0005333333333333
3	G	5.691092	0.5435694283934659	1.682448	nan	nan	0.22516666666666665
3	H	5.781742333333334	0.632757540421553	1.7101063333333333	nan	nan	0.1872
5	A	0.5025430000000001	0.18931254758995766	1.0276646666666667	nan	0.08336666666666666	17.789033333333332
5	B	1.8360786666666666	0.36638167222902024	1.1334916666666668	nan	0.6105	1.2775
5	C	1.8758186666666665	0.6978525937533893	1.3580196666666666	nan	0.7258	1.1170333333333333
5	ClassReg	2.5709883333333337	0.216304402924058	1.9008983333333334	nan	nan	nan
5	D	1.8600806666666667	0.43002099507620956	1.2343356666666667	nan	0.7402333333333333	0.1509333333333334
5	E	0.9950536666666667	0.542757741156525	1.2203126666666666	nan	0.21383333333333332	16.7748
5	F	2.2948316666666666	0.3157397757035583	1.4142213333333333	nan	0.9683666666666667	1.1041
5	G	2.263089	0.6550193695616946	1.4064033333333334	nan	0.7460333333333334	0.24926666666666666
5	H	2.126559	0.4987137167113012	1.4436253333333333	nan	0.7423666666666667	0.3268

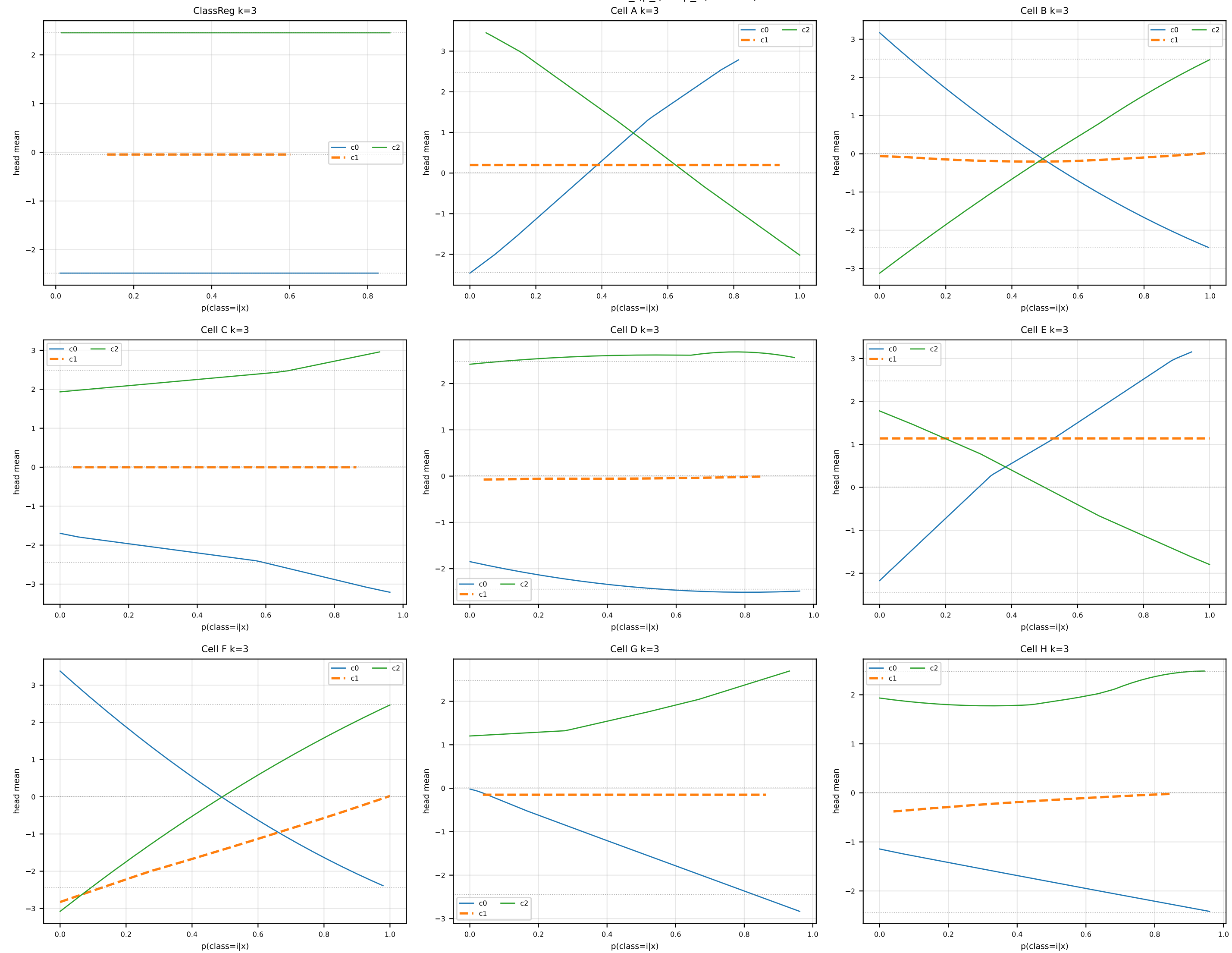
exponential k=3 — $h_i(p_i)$ vs p_i (seed 42)

exponential k=5 — $h_i(p_i)$ vs p_i (seed 42)

exponential k=3 — $p_i(x)$ vs x (seed 42)

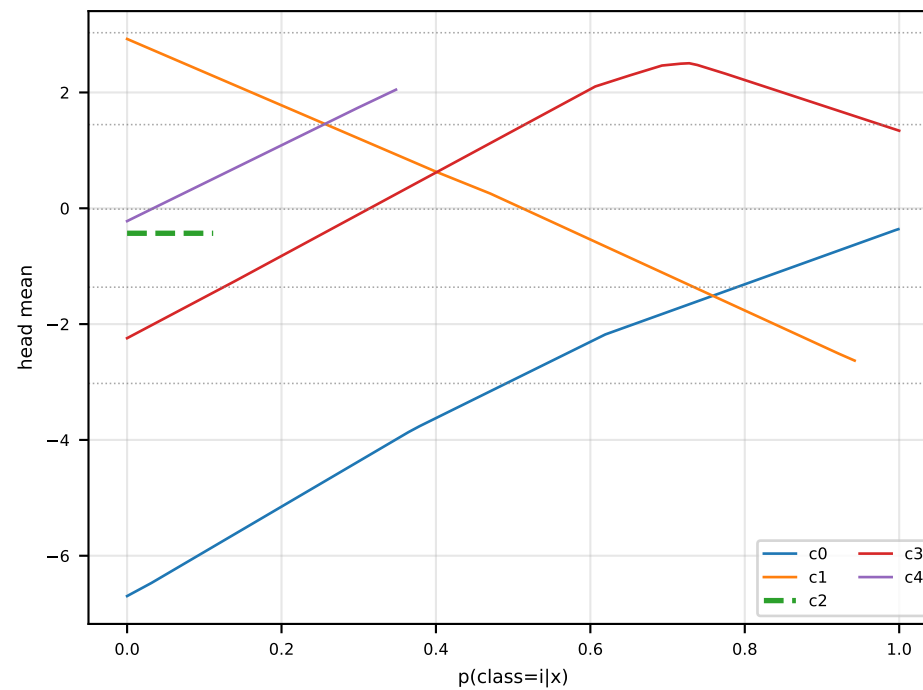
exponential k=5 — $p_i(x)$ vs x (seed 42)

k	cell	mse_mean	mse_std	nll_gaussian_mean	nll_mdn_mean	max_p_mid_mean	anchor_error_mean
3	A	2.1903023333333334	0.38671572660202647	1.646884	nan	nan	3.9851000000000005
3	B	1.9559313333333332	0.2449538704946982	1.4837796666666667	nan	nan	0.05876666666666667
3	C	1.7751906666666668	0.06038251118770513	1.651345	nan	nan	0.5993666666666667
3	ClassReg	2.013271	0.18900453744553355	1.769283	nan	nan	nan
3	D	1.958037	0.12793509100711964	1.672064	nan	nan	0.10753333333333333
3	E	2.4128613333333333	0.15806193606726868	1.7544156666666666	nan	nan	3.5800666666666667
3	F	2.1938786666666665	0.716708018331268	1.6017719999999998	nan	nan	0.07096666666666666
3	G	1.834486	0.15633366317911185	1.6943063333333335	nan	nan	0.2992333333333333
3	H	1.9714386666666668	0.08402006461157563	1.745126	nan	nan	0.0892
5	A	2.1235673333333334	0.20243963987404567	1.5864846666666665	nan	0.3549	3.3711
5	B	1.8033556666666666	0.14091237650516475	1.4859010000000001	nan	0.9584666666666667	0.7393333333333333
5	C	1.9975543333333334	0.20320489363283875	1.6694363333333335	nan	0.5942	0.35400000000000004
5	ClassReg	2.0762883333333333	0.1183292425790571	1.7861713333333336	nan	nan	nan
5	D	1.8721733333333332	0.16457596792160553	1.6462673333333333	nan	0.5881666666666666	0.26886666666666664
5	E	2.0797266666666667	0.049523288474548295	1.684366	nan	0.8468666666666667	3.2797666666666667
5	F	2.1017713333333333	0.11884418709526076	1.6119886666666667	nan	0.9883333333333333	0.9706333333333333
5	G	1.859609	0.13063342119075058	1.650205	nan	0.5943999999999999	0.42566666666666664
5	H	1.9629833333333333	0.2551410200425116	1.6802013333333334	nan	0.5531666666666667	0.20719999999999997

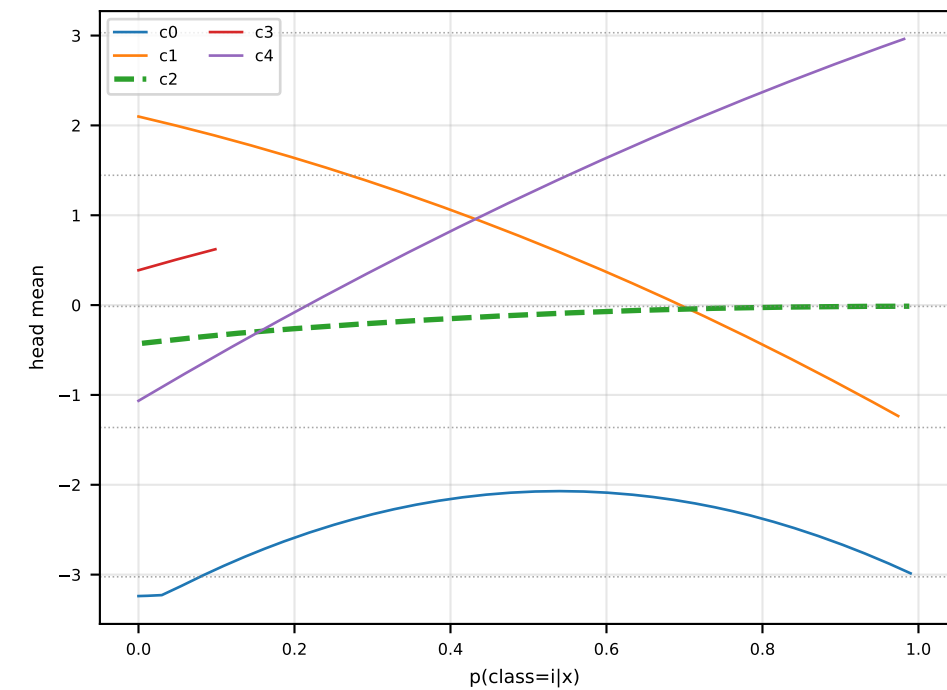
heteroscedastic k=3 — $h_i(p_i)$ vs p_i (seed 42)

heteroscedastic k=5 — $h_i(p_i)$ vs p_i (seed 42)

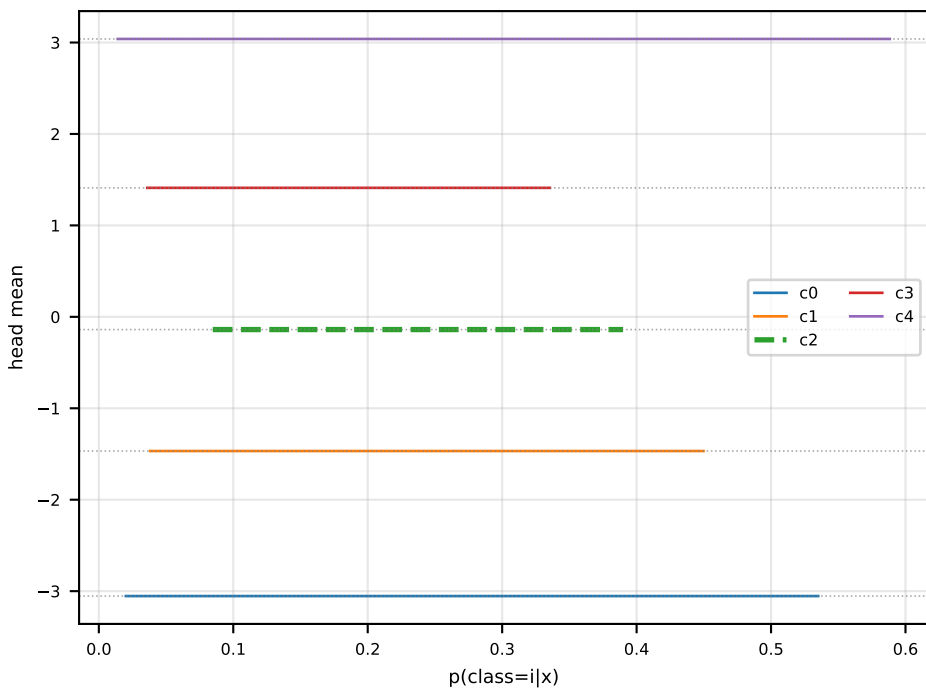
Cell A k=5



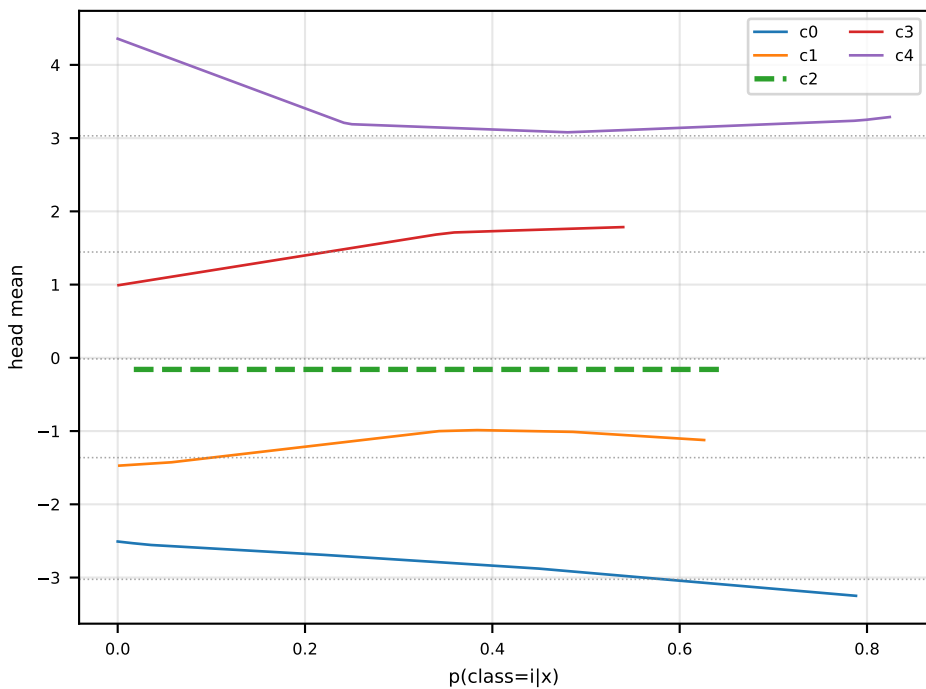
Cell B k=5



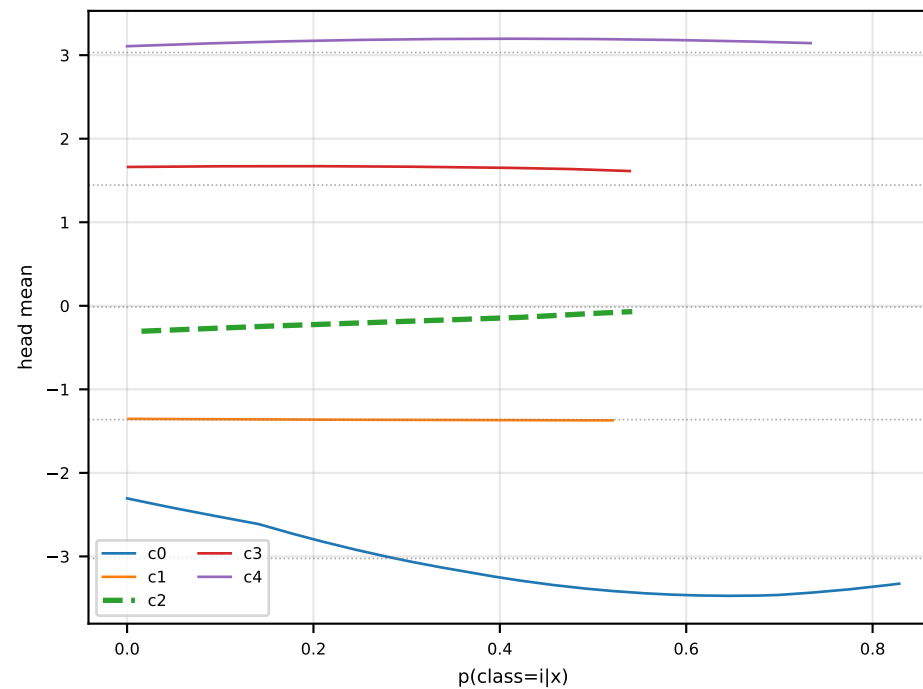
ClassReg k=5



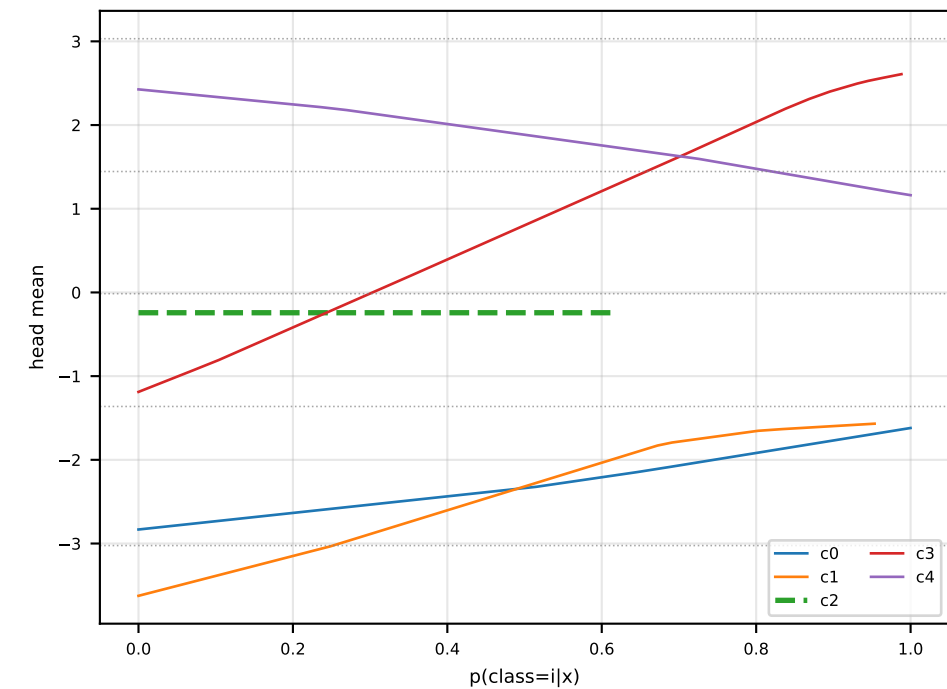
Cell C k=5



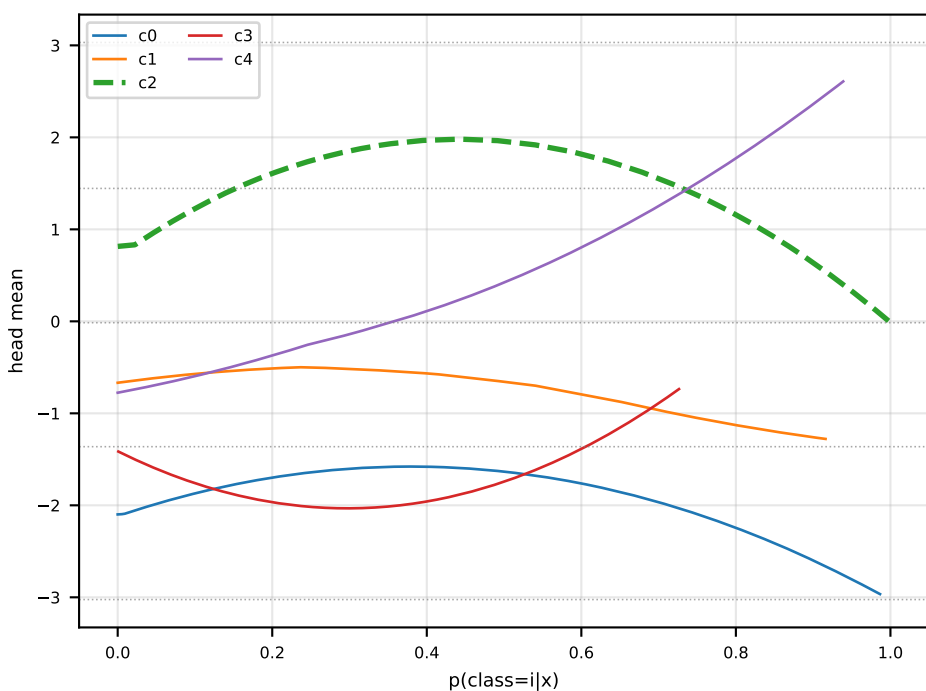
Cell D k=5



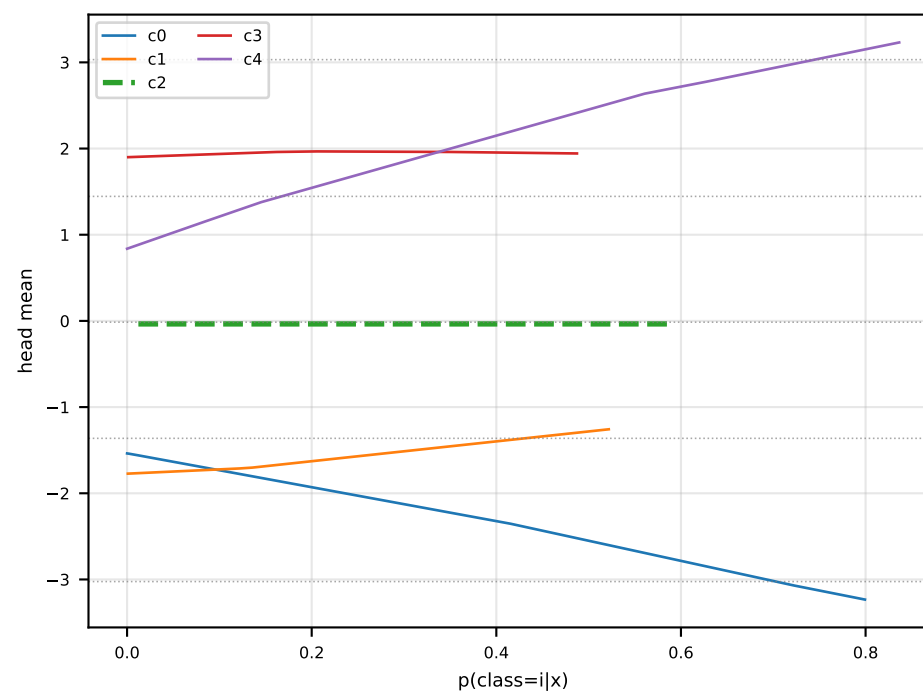
Cell E k=5



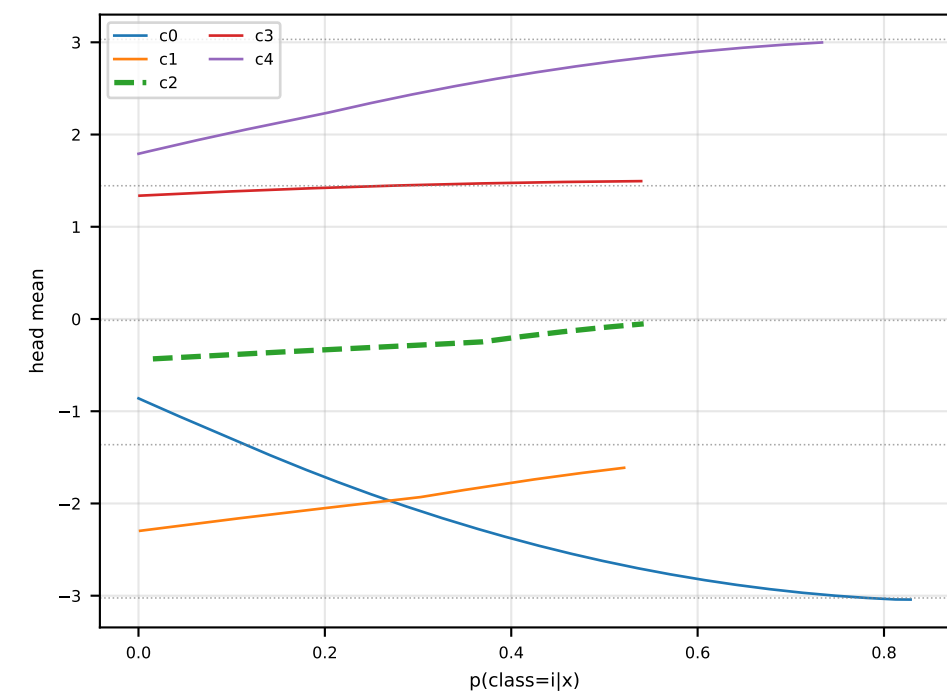
Cell F k=5

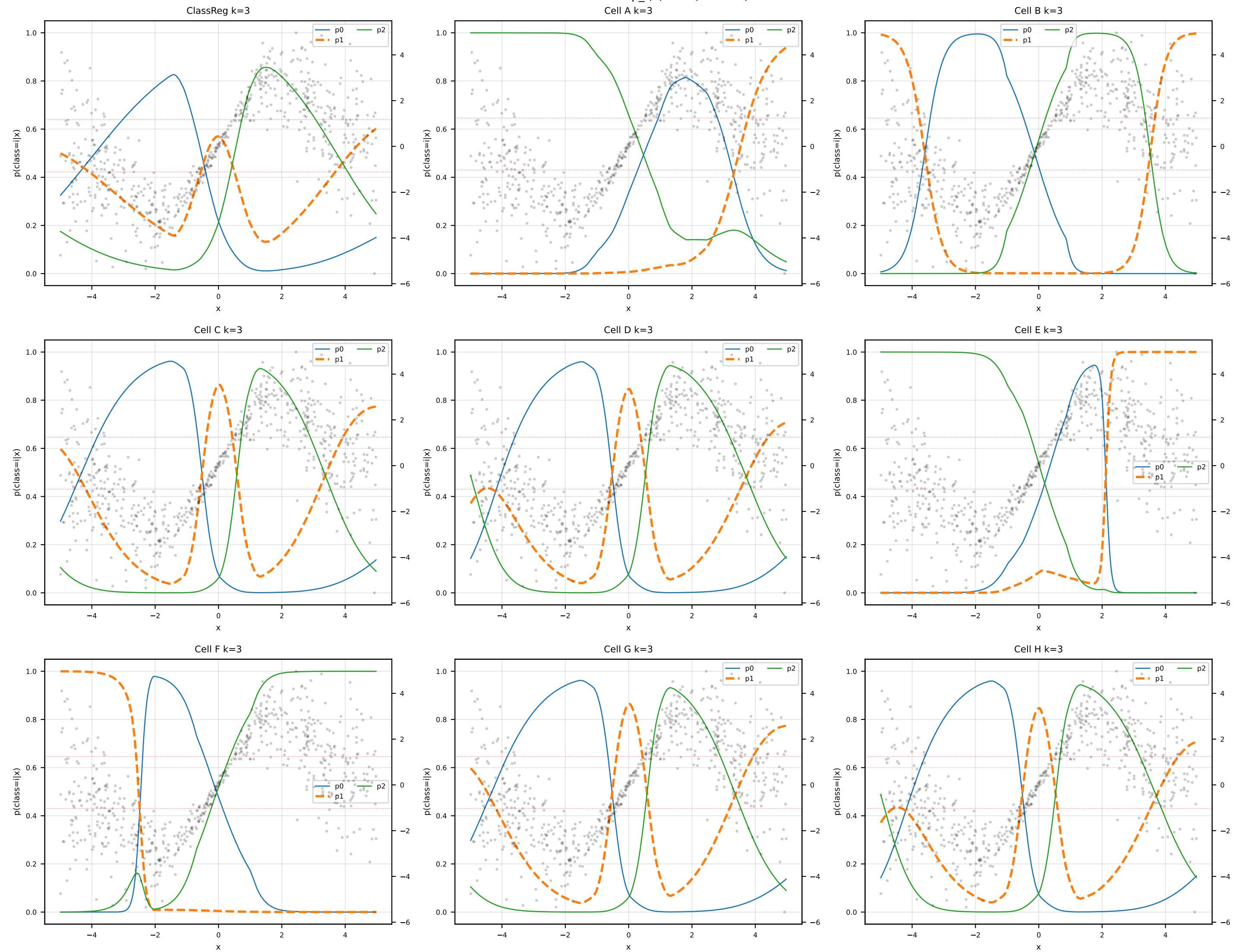


Cell G k=5



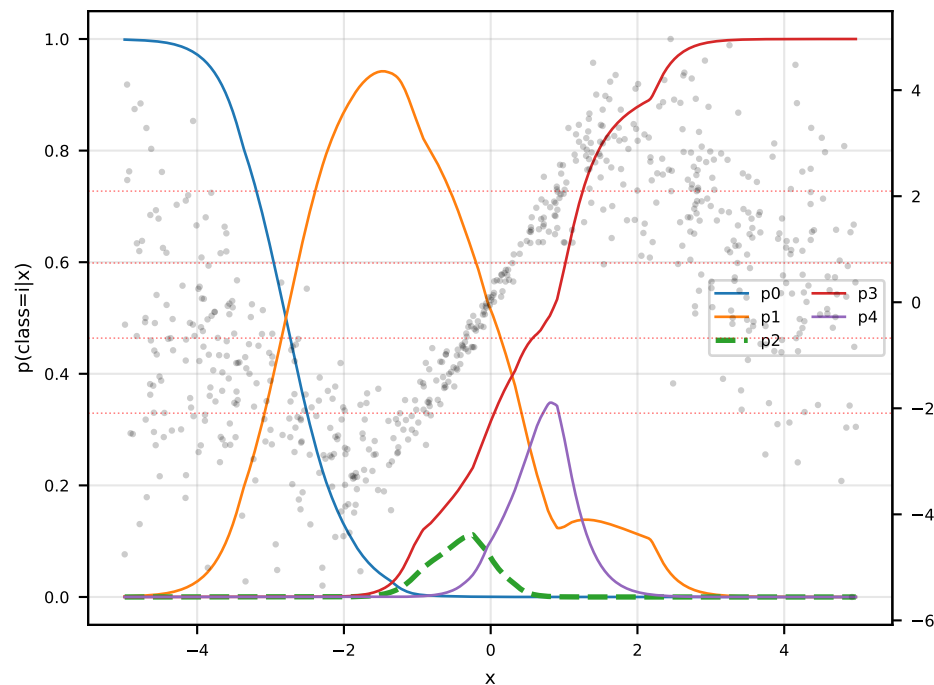
Cell H k=5



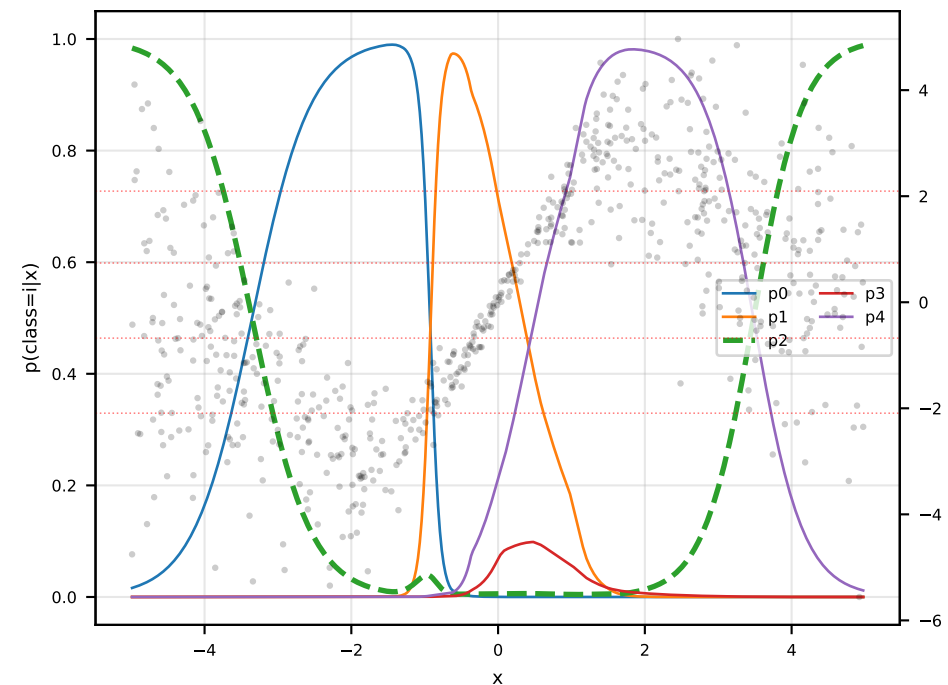
heteroscedastic k=3 — $p_i(x)$ vs x (seed 42)

heteroscedastic k=5 — $p_i(x)$ vs x (seed 42)

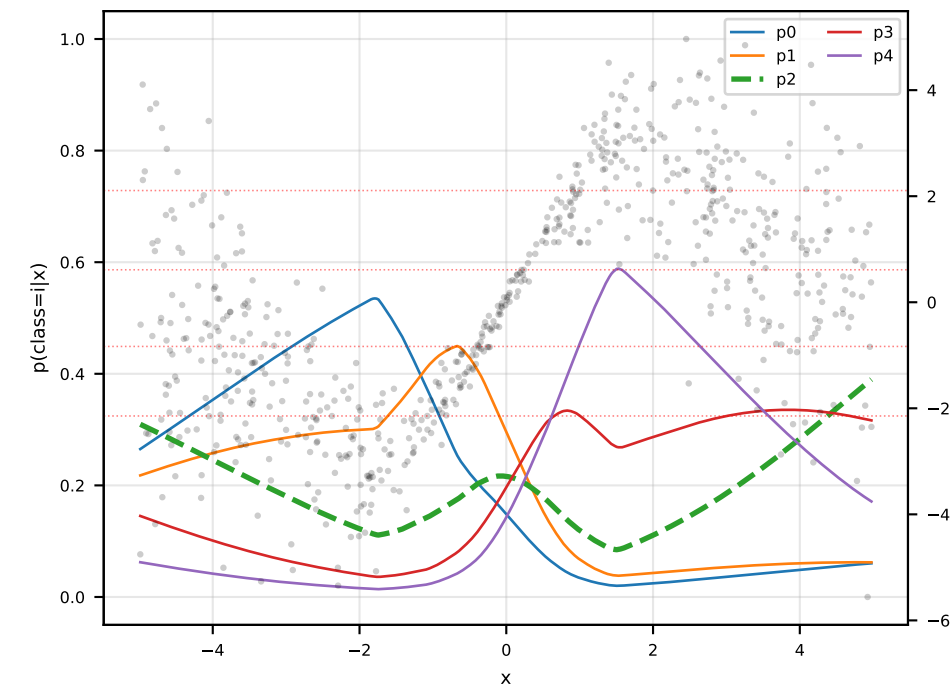
Cell A k=5



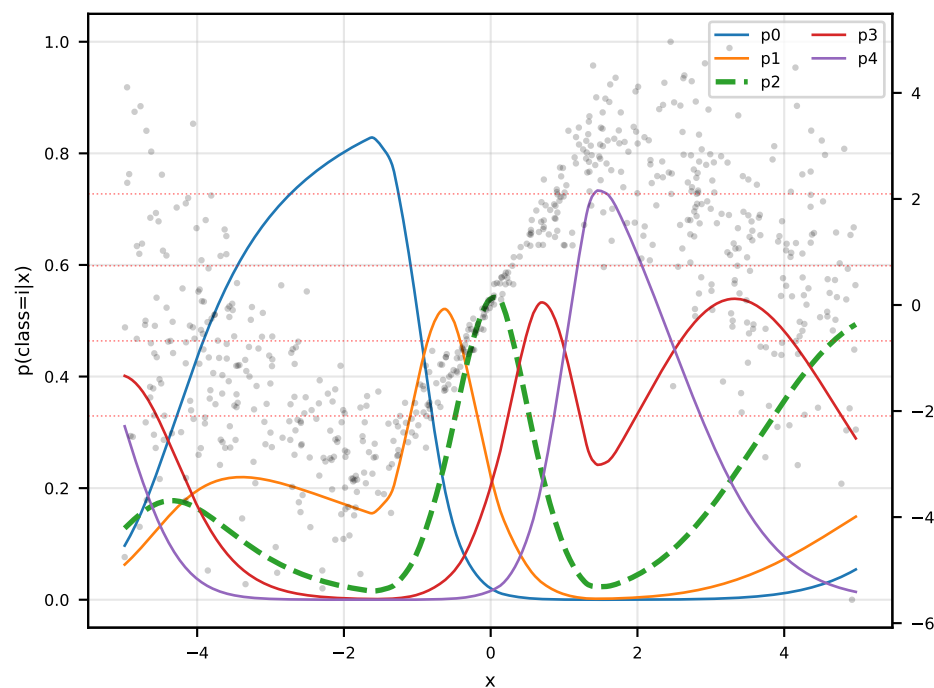
Cell B k=5



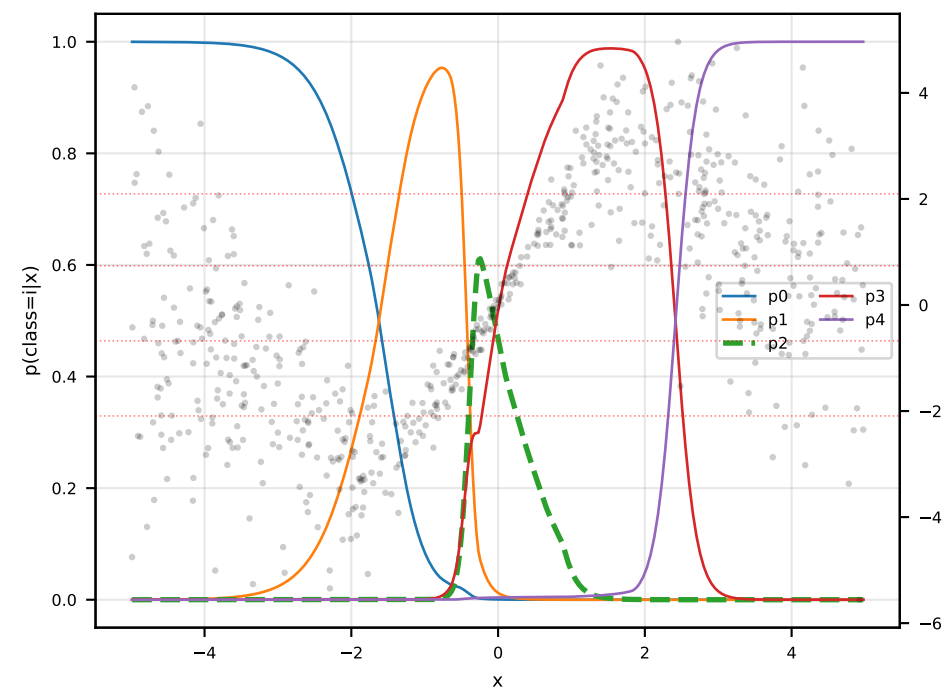
ClassReg k=5



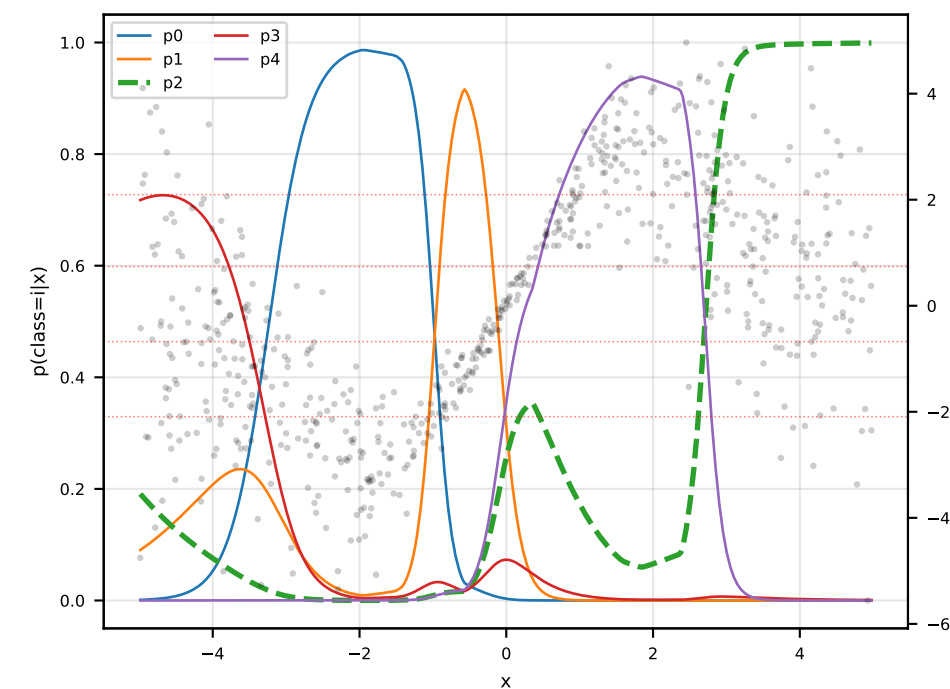
Cell D k=5



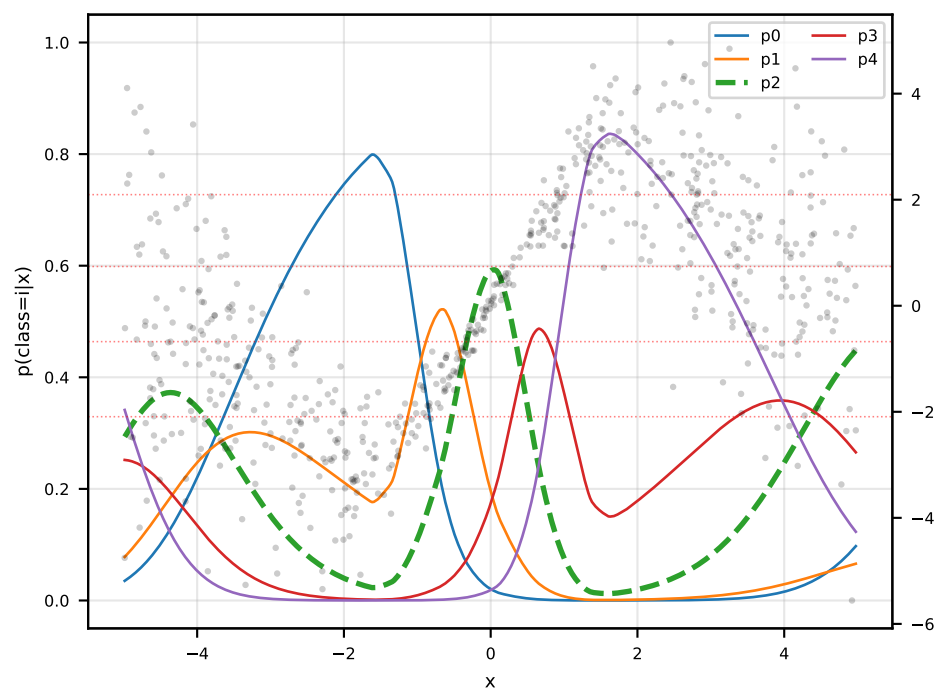
Cell E k=5



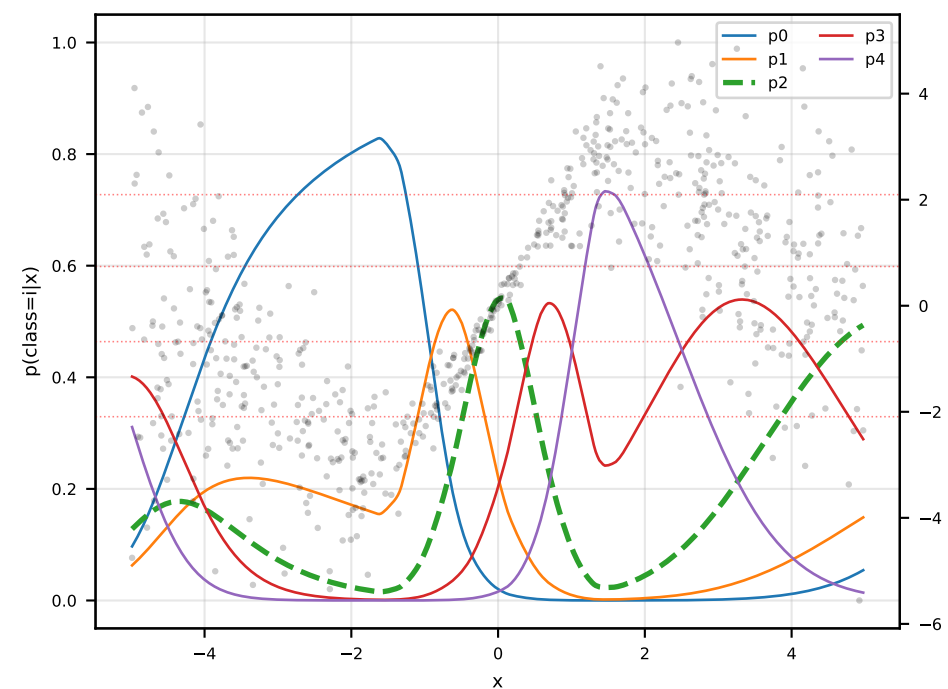
Cell F k=5



Cell G k=5



Cell H k=5

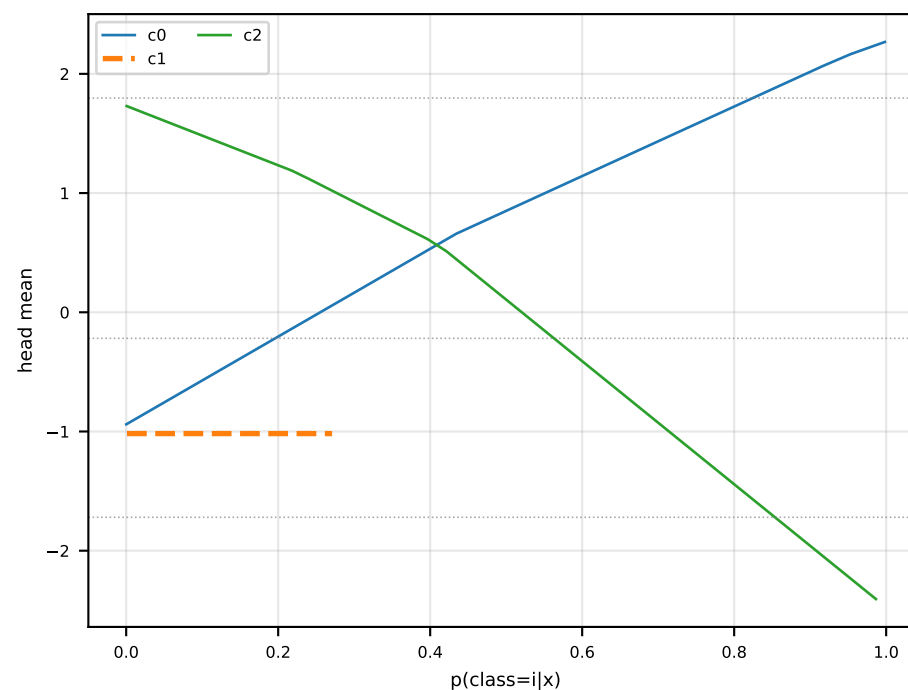


piecewise — metrics summary (mean ± std across 3 seeds)

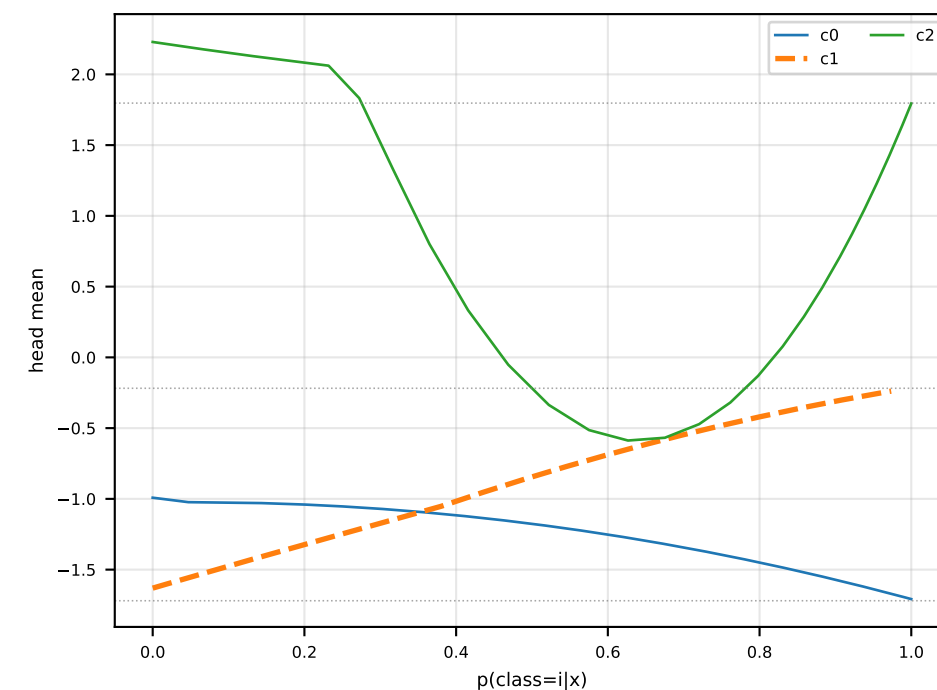
k	cell	mse_mean	mse_std	nll_gaussian_mean	nll_mdn_mean	max_p_mid_mean	anchor_error_mean
3	A	0.334195	0.08858017021884751	0.6207699999999999	nan	nan	3.3631666666666664
3	B	0.3210456666666667	0.04737747931595067	0.664899	nan	nan	0.01833333333333333
3	C	0.329368	0.01439650124162119	0.789081	nan	nan	0.4268
3	ClassReg	0.37243166666666666	0.051183614236719674	0.9262990000000001	nan	nan	nan
3	D	0.4140376666666667	0.11946754138398151	0.8255016666666667	nan	nan	0.06983333333333334
3	E	0.37782866666666665	0.15442321953104501	0.7101853333333333	nan	nan	3.3459000000000003
3	F	0.33151400000000003	0.04251007372376573	0.716837	nan	nan	0.061
3	G	0.42820400000000003	0.13862678354849037	0.8750326666666667	nan	nan	0.23806666666666665
3	H	0.351167	0.053238140397651006	0.8254376666666667	nan	nan	0.055566666666666674
5	A	0.2927406666666667	0.049592727212095664	0.5318516666666667	nan	0.05029999999999999	2.3823666666666665
5	B	0.297548	0.049869105666735226	0.5989140000000001	nan	0.6693000000000001	0.1481
5	C	0.31121899999999997	0.02997303074432078	0.624079	nan	0.9265333333333333	0.35189999999999994
5	ClassReg	0.3521156666666667	0.06201382285695127	0.8985816666666667	nan	nan	nan
5	D	0.28364799999999996	0.02038935987224709	0.6308373333333334	nan	0.8926666666666666	0.18613333333333335
5	E	0.2923926666666667	0.04832392207109573	0.572597	nan	0.7284	2.496233333333333
5	F	0.29788	0.030505105326813733	0.6237883333333333	nan	0.9438333333333334	0.20653333333333335
5	G	0.292373	0.022663308033029933	0.7099643333333333	nan	0.9226666666666666	0.3469
5	H	0.297786	0.05386368720204734	0.6334719999999999	nan	0.9465	0.08493333333333335

piecewise k=3 — $h_i(p_i)$ vs p_i (seed 42)

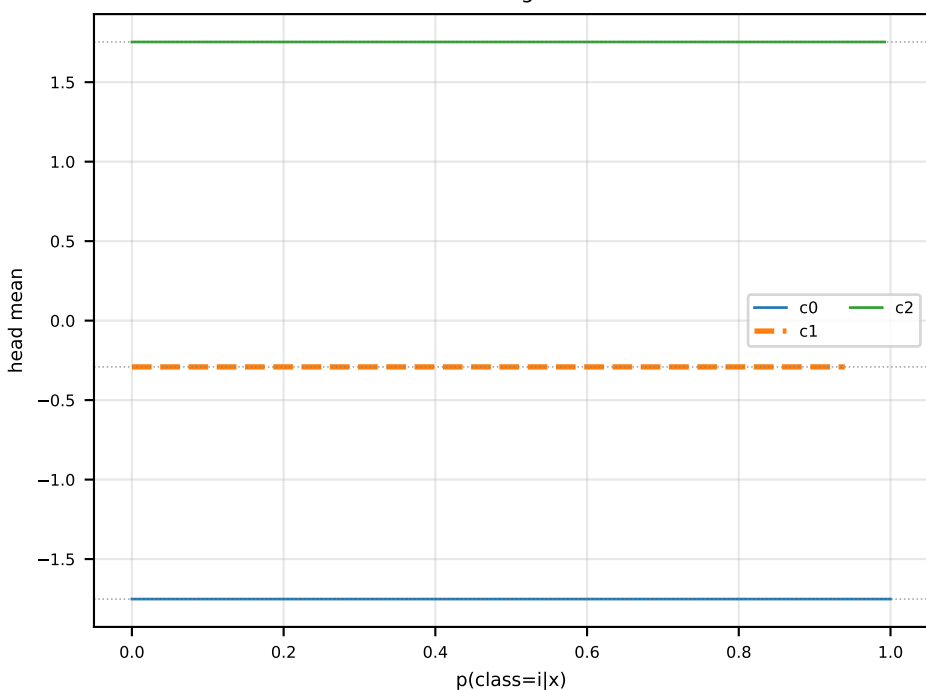
Cell A k=3



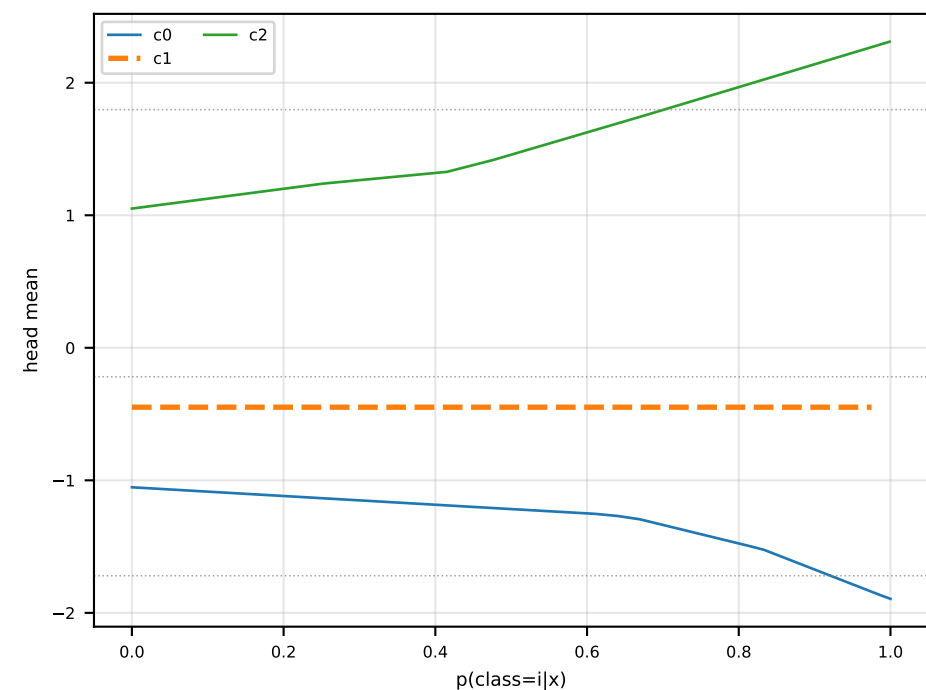
Cell B k=3



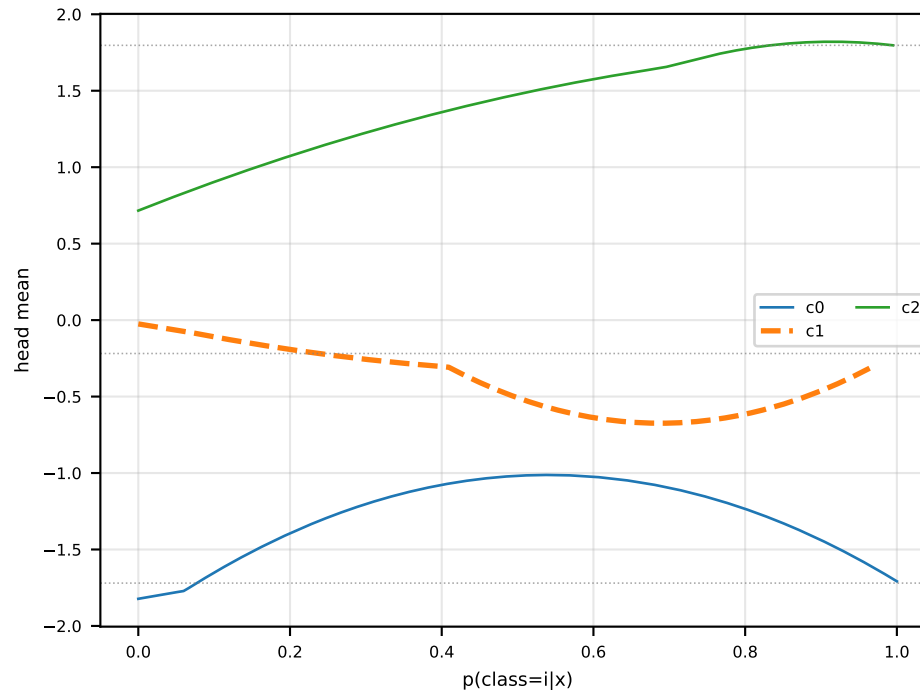
ClassReg k=3



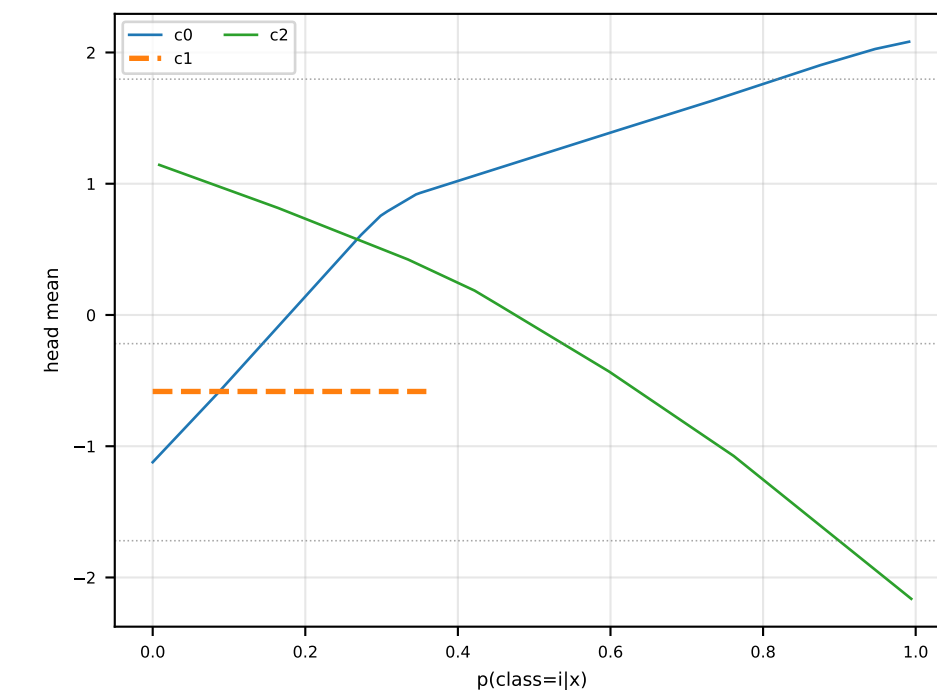
Cell C k=3



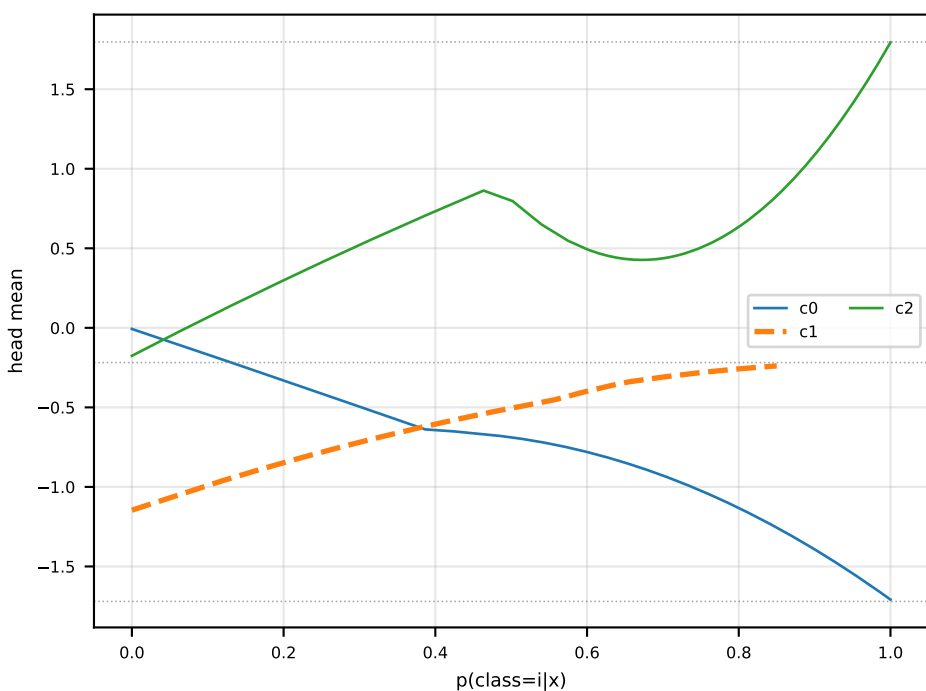
Cell D k=3



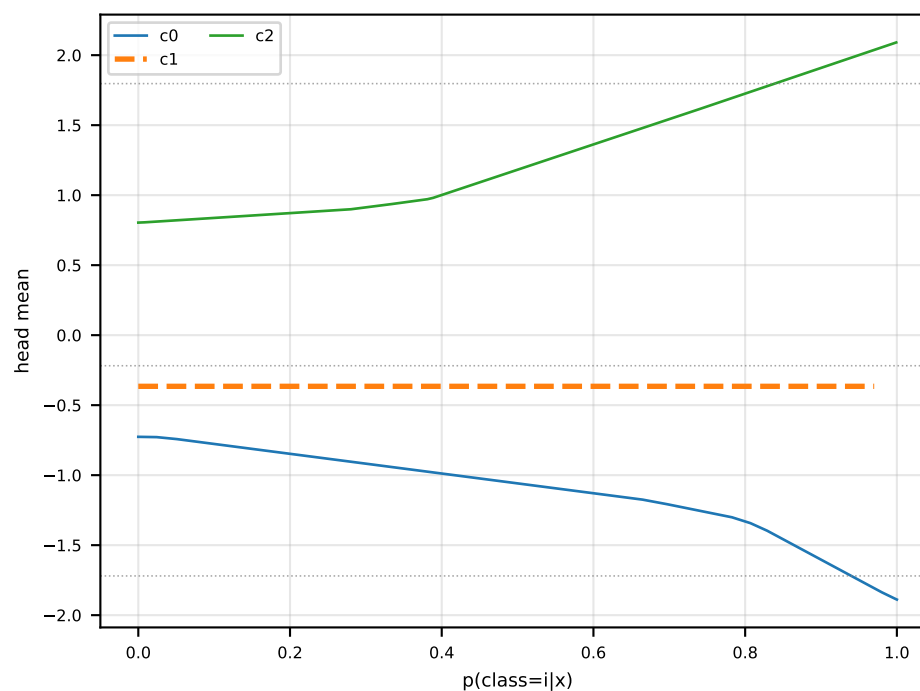
Cell E k=3



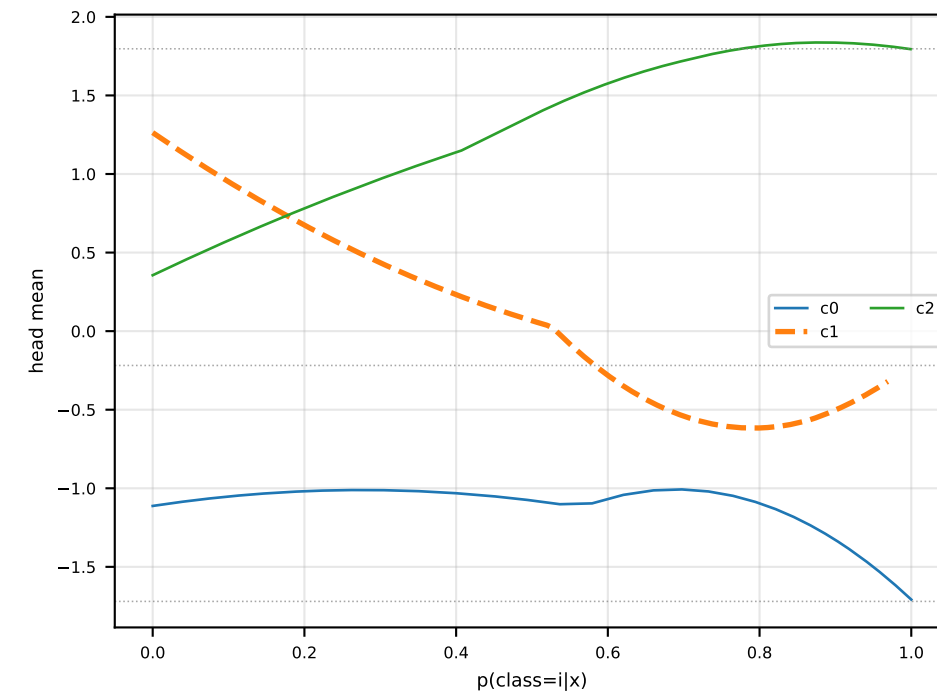
Cell F k=3



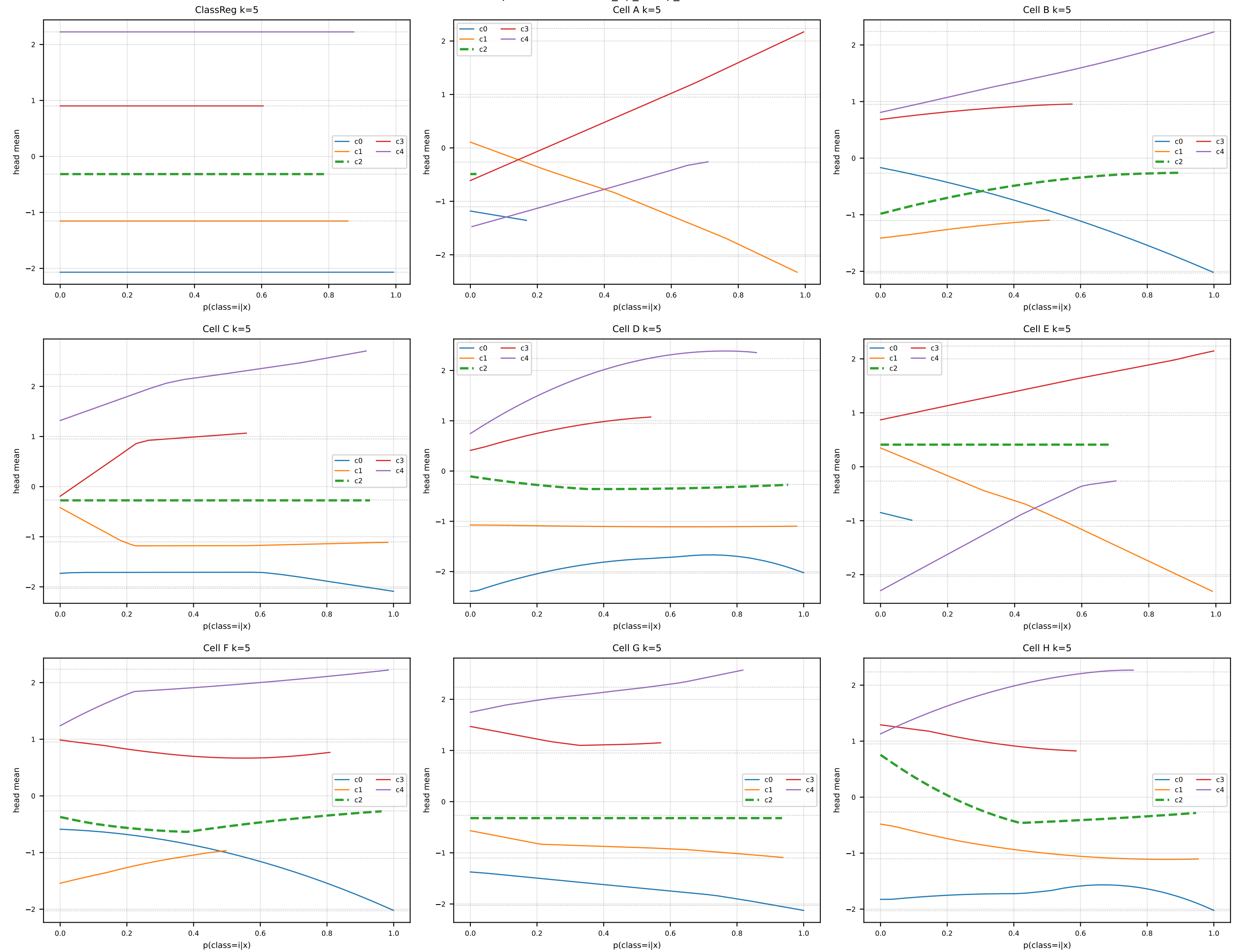
Cell G k=3



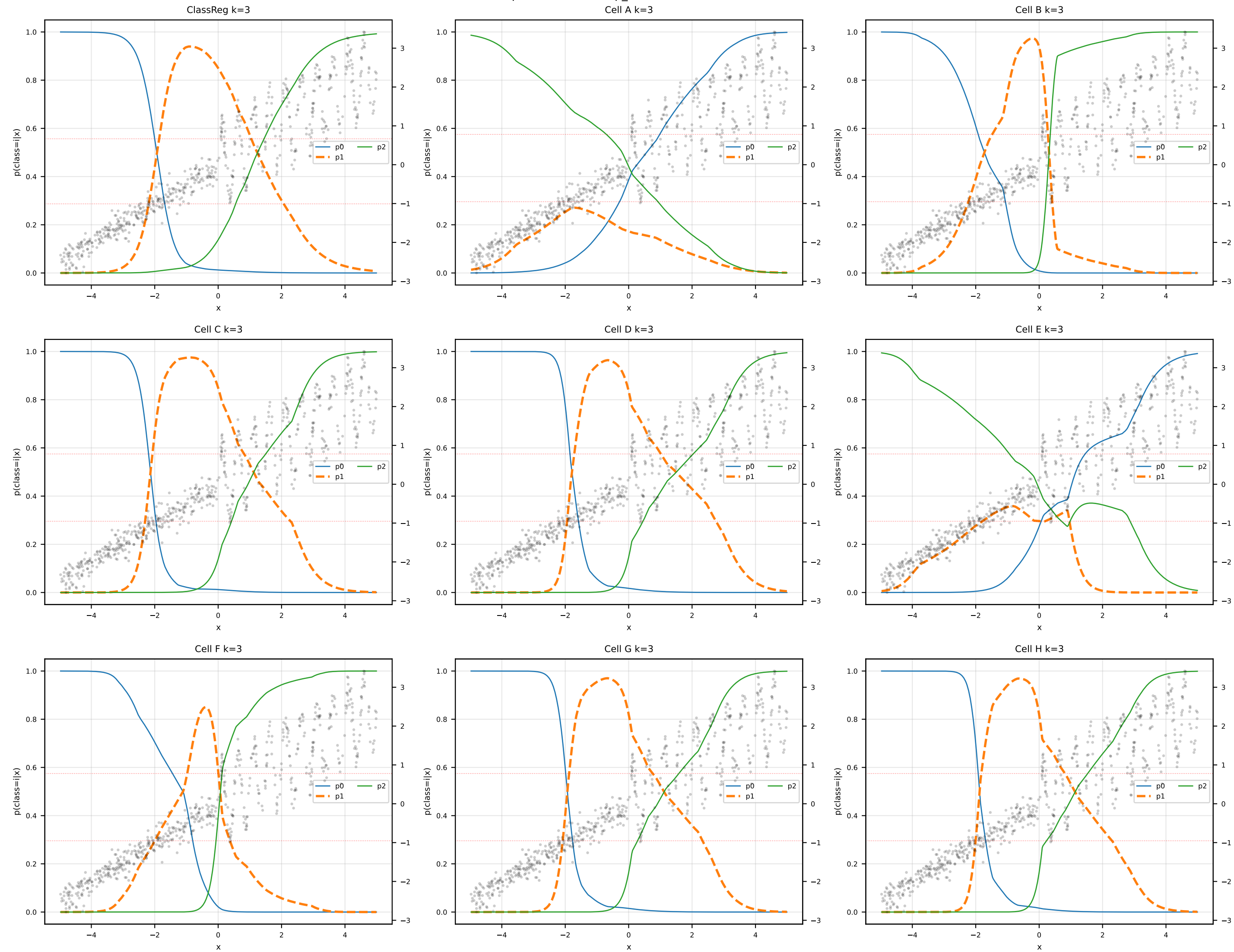
Cell H k=3



piecewise k=5 — $h_i(p_i)$ vs p_i (seed 42)

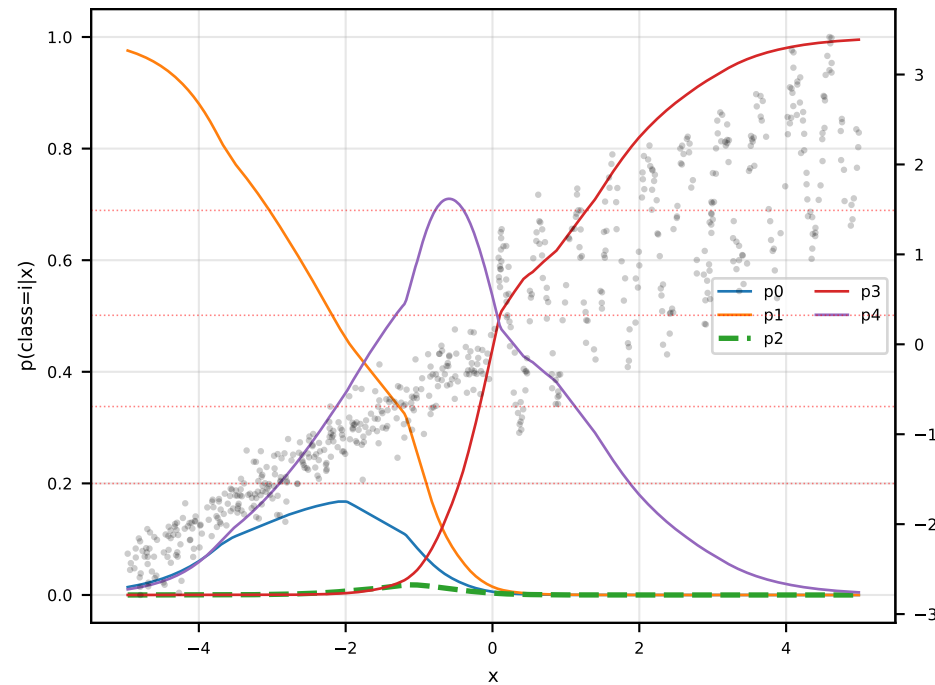


piecewise k=3 — $p_i(x)$ vs x (seed 42)

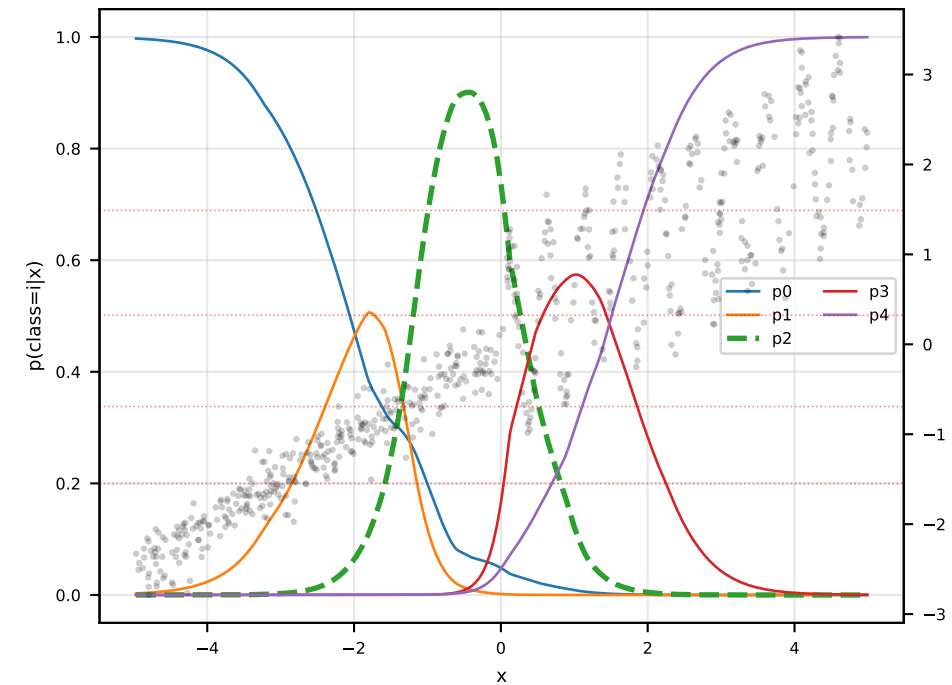


piecewise k=5 — $p_i(x)$ vs x (seed 42)

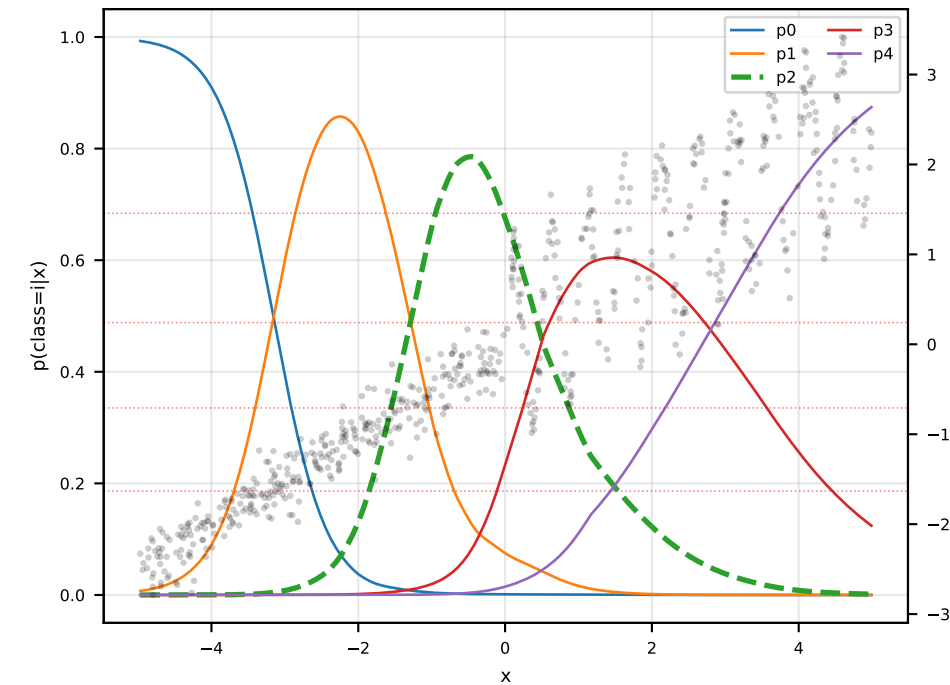
Cell A k=5



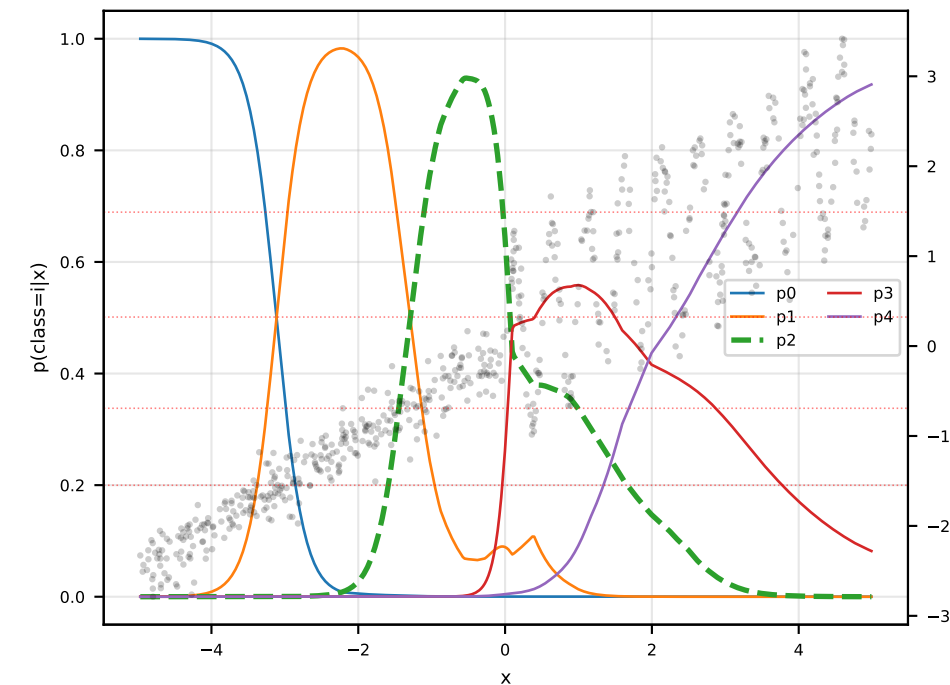
Cell B k=5



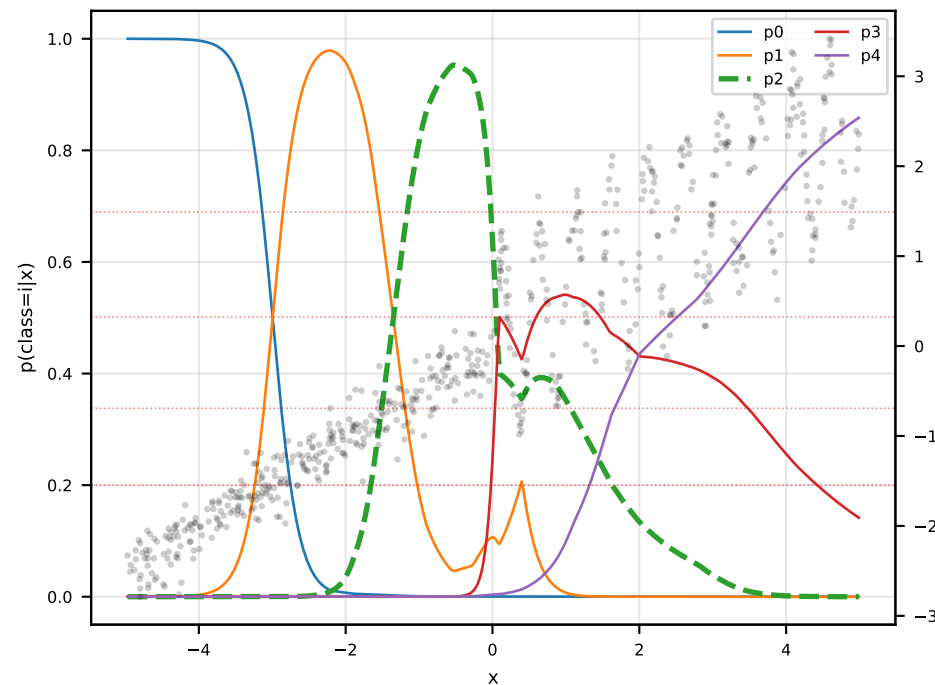
ClassReg k=5



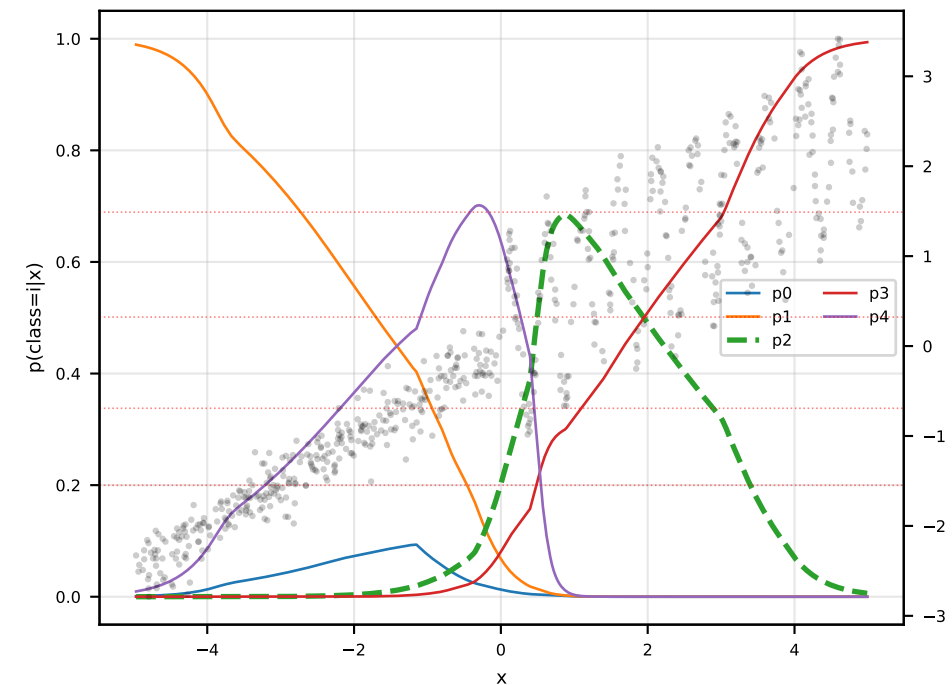
Cell C k=5



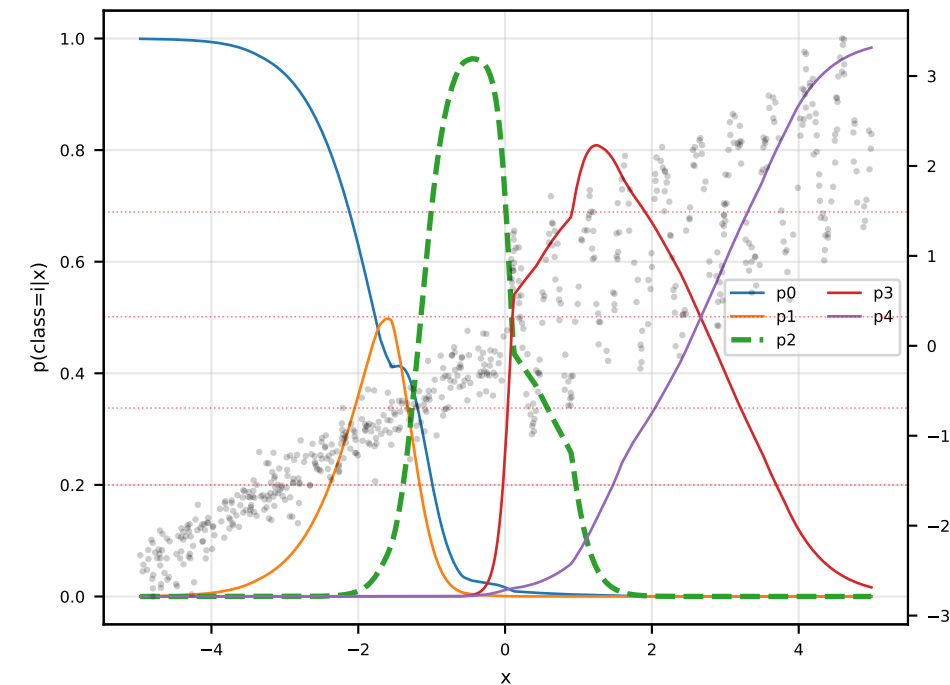
Cell D k=5



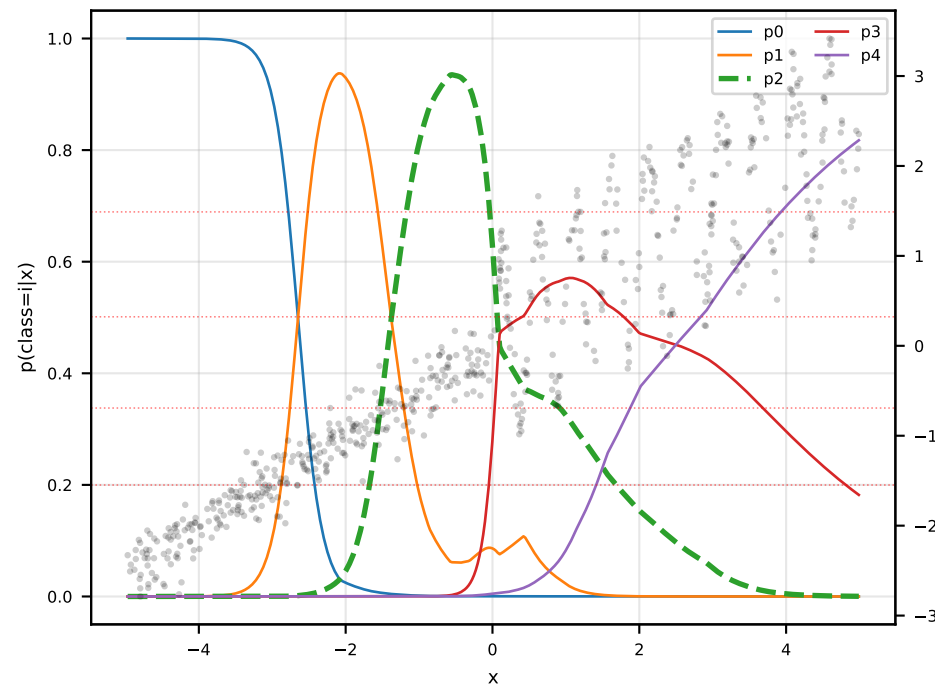
Cell E k=5



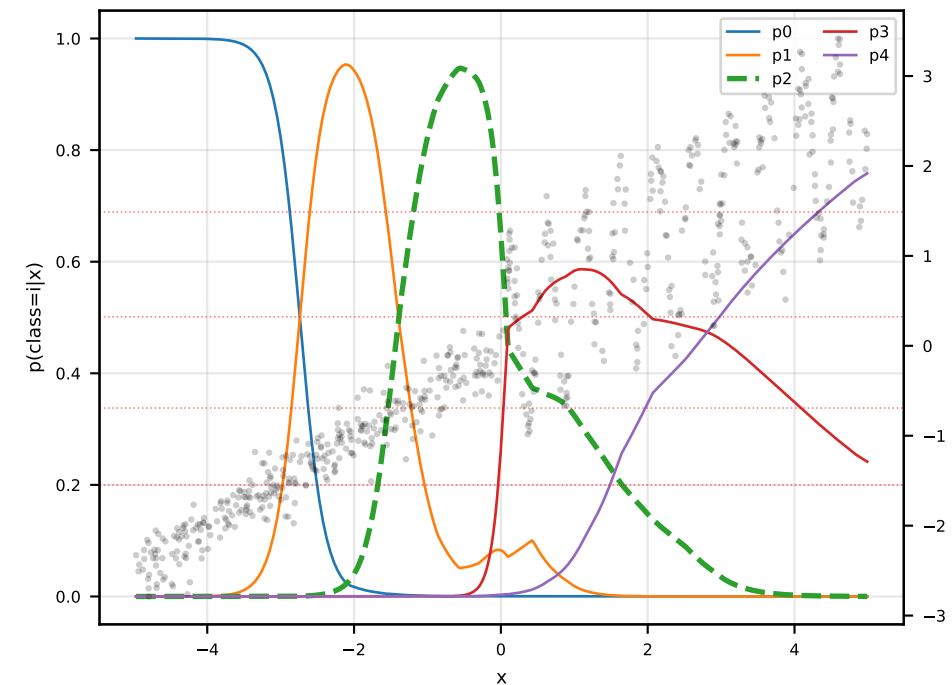
Cell F k=5



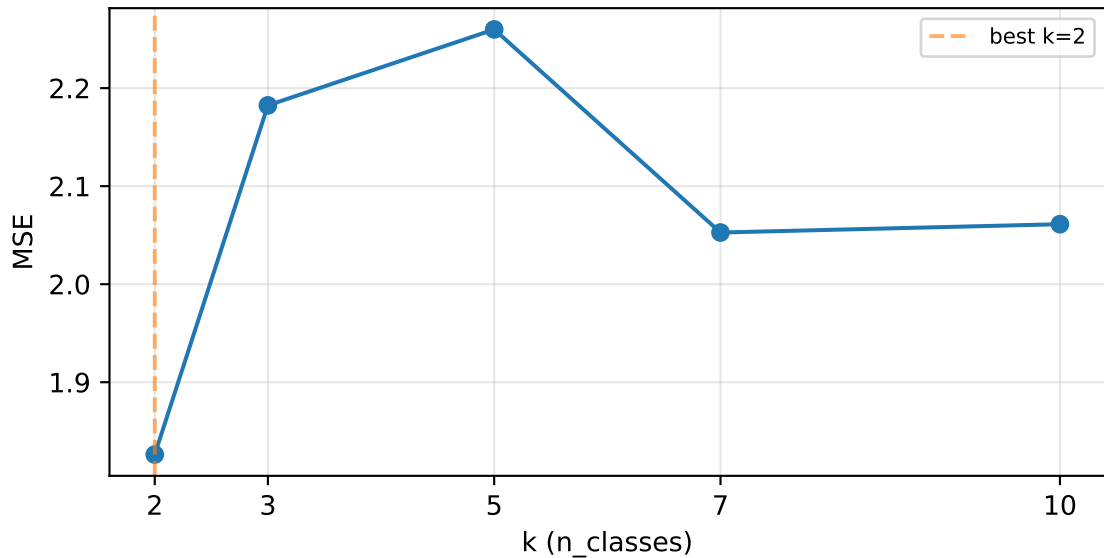
Cell G k=5



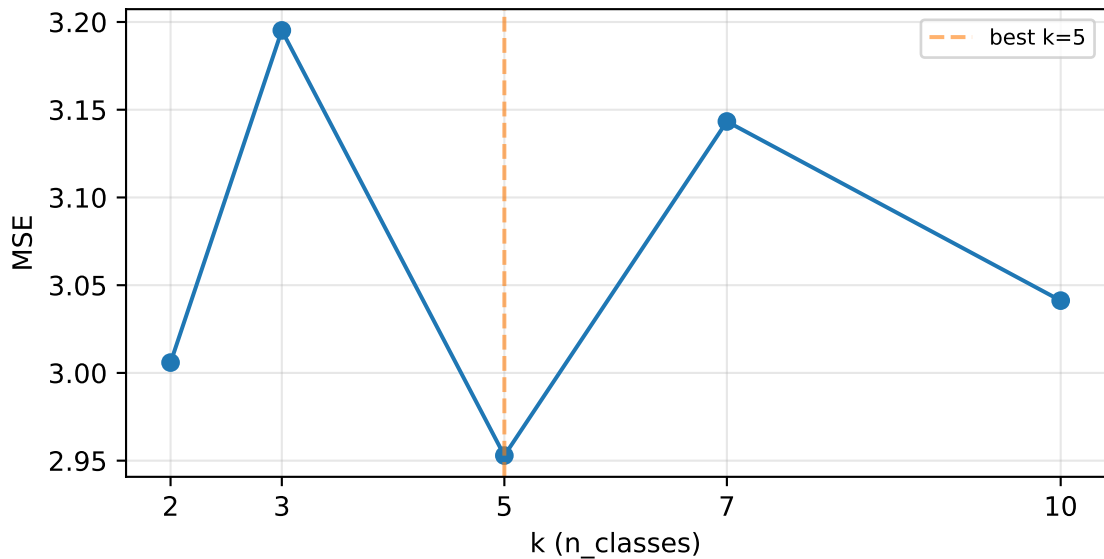
Cell H k=5



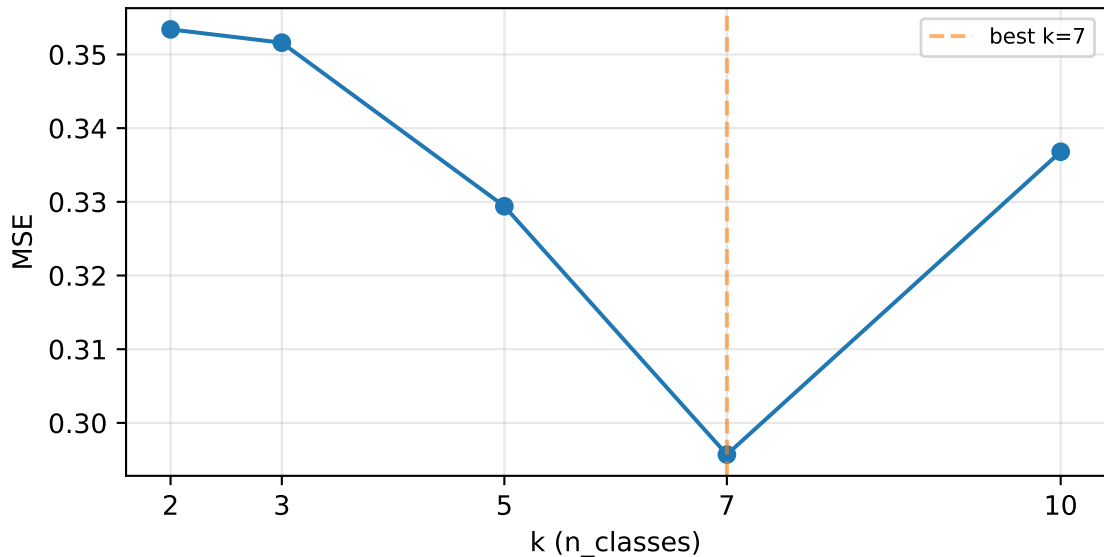
ClassReg k-sweep — heteroscedastic



ClassReg k-sweep — bimodal



ClassReg k-sweep — piecewise



ClassReg k-sweep — exponential

